



Measurement of $R(D^*)$ and $R(D)$ with hadronic FEI tagging method using the 365 fb^{-1} data at Belle II

Lin Wang^{*1}, Taichiro Koga^{†2}, Kodai Matsuoka^{‡2}, Qi-Dong Zhou^{§1}, Koji Hara^{¶2},
Toru Iijima^{||2,3,4}, and Katsuro Nakamura^{**2}

¹Institute of Frontier and Interdisciplinary Science, Shandong University, Qingdao, China

²High Energy Accelerator Research Organization (KEK), Tsukuba, Japan

³Graduate School of Science, Nagoya University, Nagoya, Japan

⁴Kobayashi-Maskawa Institute (KMI), Nagoya University, Nagoya, Japan

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Abstract

1 Measurement of branching ratio $R(D^{(*)})$ of $B \rightarrow D^{(*)}\tau\nu_\tau$ relative to $B \rightarrow D^{(*)}\ell\nu_\ell$
 2 decays with hadronic tagging is performed by using data sample of 365 fb^{-1} collected
 3 at the $\Upsilon(4S)$ resonance with the Belle II detector at the SuperKEKB asymmetric-
 4 energy e^+e^- collider. The analysis uses hadronic reconstruction of the tag-side B
 5 meson based on hadronic FEI and purely leptonic τ decays on signal-side B meson.
 6 We find $R(D^*) = 0.242 \pm 0.019(\text{stat.}) \pm 0.016(\text{syst.})$, $R(D) = 0.439 \pm 0.055(\text{stat.}) \pm$
 7 $0.045(\text{syst.})$, and their correlations $\rho = -0.40(\text{stat.}) - 0.20(\text{syst.})$.

*202217005@mail.sdu.edu.cn

†taichiro@post.kek.jp

‡matsuoka@post.kek.jp

§qzhou@sdu.edu.cn

¶koji.hara@kek.jp

||ijima@hepl.phys.nagoya-u.ac.jp

**katsuro@post.kek.jp

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1.0	Sep.30, 2024	All	Initial note for the internal check.
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1 Introduction

1.1 Introduction

In the Standard Model (SM) of particle physics, semileptonic B decays at the quark level proceed via $b \rightarrow c$ or $b \rightarrow u$ transitions that are mediated by a W boson and produce a charged lepton and its corresponding neutrino. The W boson couples identically to the three lepton generations [1], a symmetry known as Lepton Flavour Universality (LFU) that is a fundamental postulate of the SM. LFU can be tested by measuring $R(D^{(*)}) = \frac{BR(B \rightarrow D^{(*)}\tau\nu)}{BR(B \rightarrow D^{(*)}\ell\nu)}$: where the denominator is referred to as the normalization mode with $\ell = e$ or μ . Semileptonic B decays involving a τ lepton are sensitive to new physics (NP) models [2, 3, 4, 5] and their decays are less constrained by data than are semileptonic decays to electrons and muons. While the couplings to all lepton flavors are the same in the SM, the larger τ mass results in a smaller phase space factor, so $R(D^{*})$ is expected to be 0.254 ± 0.003 [6] by the SM. $R(D^{*})$ has been measured by the BaBar [7, 8], Belle [9, 10, 11], and LHCb [12, 13, 14, 15] collaborations. The combined results of these measurements exceed the SM expectation by approximately 3σ [6].

In 2023, Belle II measured the $R(D^{*})$ with hadronic tag by using 189 fb^{-1} data [16]: $R(D^{*}) = 0.262_{-0.039}^{+0.041}(\text{stat.})_{-0.032}^{+0.035}(\text{syst.})$. Due to low efficiency of hadronic tag (=low branching ratio of hadronic decays), the statistical uncertainty has been the largest error source. In this time, we update the analysis to measure $R(D^{*})$ and $R(D)$ with LS1 dataset. Followings are major changes from the last analysis:

- Data statistics increased from 189 fb^{-1} to 365 fb^{-1}
- MC sample is updated from MC14 run independent to MC15 run dependent
- $B \rightarrow D\tau\nu$ sample is reconstructed and added to fit, in order to measure $R(D)$
- Additional $D^{(*)}$ decay modes are reconstructed to increase statistics
- Selection criteria of daughter particles is modified
- Selection criteria of ROE is modified to remove fake photons and beam background photons with MVA variables
- "isSignal" requirement is removed from definition of signal and background, in order to increase statistics, avoid complex of the MC matching, and avoid difficulty to evaluate data/MC agreement of the MC matching
- Kernel density estimation is not applied to PDF for signal extraction. Instead, binning of PDF is optimised to reduce fitter bias from MC statistics.

This note explains the analysis method in detail. Dataset is described in Sec. 2. Selection criteria and general corrections are in Sec. 3,4. Data/MC comparison at side-band and additional corrections are in Sec. 5. Signal extraction method is in Sec. 6, and systematic uncertainties are in Section 7.

1.2 Analysis over view

In order to measure $R(D^{(*)})$, we reconstruct $B \rightarrow D^{(*)}\ell\nu$ ($\ell = e, \mu$) (normalization mode) and $B \rightarrow D^{(*)}\tau\nu$ (signal mode) simultaneously and take ratio of their branching ratios.

Figure 1.1 shows a schematic view of the signal reconstruction. The $D^{(*)}$ decays are reconstructed by combination of daughter particles ($\pi, K, \pi^0 \rightarrow \gamma\gamma, K_s^0 \rightarrow \pi\pi$) detected by Belle II. In order to increase statistics, several kind of $D^{(*)}$ decay modes are reconstructed. The ℓ is selected by lepton particle identification (PID). The τ is reconstructed by leptonic decays of $\tau \rightarrow e\nu\nu$ and $\tau \rightarrow \mu\nu\nu$, in order to reject hadronic background. Because the reconstructed final state particles of the signal mode are the same as that of the normalization mode, many systematics uncertainties are canceled in this method. On the other hand, identification of signal and normalization modes is difficult due to missing neutrinos. In order to measure the missing neutrinos mass for the signal identification, tag-side B (B_{tag}) is fully reconstructed with hadronic Full Event Interpretation (FEI) algorithm. In the rest of event (ROE), no charged tracks and π^0 candidates are allowed to reject background.

One of the major background in this analysis is $B \rightarrow D^{**}\ell\nu$, where the D^{**} is excited charm-meson cascade decays of D_0^*, D_1, D_1', D_2^* to $D^{(*)}$ ground state. When one or more daughter particles (π, π^0, γ etc.) from D^{**} is mis-reconstructed, it can be identified as $B \rightarrow D^{(*)}\ell\nu$. Because part of the branching ratio of $B \rightarrow D^{**}\ell\nu$ decays have not been measured so far, their systematic uncertainties are estimated conservatively by following the present understanding of the branching ratios. Gap mode, which is the difference of exclusive and sum of inclusive $B \rightarrow D^{**}\ell\nu$ branching ratio measurements, is mainly modeled by $B \rightarrow D^{(*)}\eta\ell$. Another major background is hadronic B decays of $B \rightarrow D^{(*)}D^{(*)}, B \rightarrow D^{(*)}D^{(*)}K$, and $B \rightarrow D^{(*)}n\pi(\pi^0)$ etc. Their uncertainties are estimated based on the past branching ratio measurements on PDG.

The signal mode, normalization modes and backgrounds are extracted by fitting two dimensional distributions of energy sum of ECL clusters in the rest of events ($E_{\text{extra}}^{\text{ECL}}$) and Missing neutrino mass squared (M_{miss}^2):

$$M_{\text{miss}}^2 = (P_{\text{beam}} - P_{B_{\text{tag}}} - P_{D^*} - P_{\ell})^2 \quad (1.1)$$

, where P is four momentum. The signal mode has multiple neutrinos in the final state and tends to have large M_{miss}^2 . On the other hand, the normalization mode has one neutrino and M_{miss}^2 is peaked around zero. The other background with additional daughter particles can be identified by large $E_{\text{extra}}^{\text{ECL}}$, because no remained particles are expected for signal and normalization modes. The fit range of M_{miss}^2 and $E_{\text{extra}}^{\text{ECL}}$ are widely taken to constrain the backgrounds by the fitting. The PDF templates of M_{miss}^2 and $E_{\text{extra}}^{\text{ECL}}$ are obtained by MC. The data/MC agreement of the PDF shape is validated at side-band with low M_{miss}^2 and side-band with additional π^0 candidates in ROE.

The physics sensitivity and fitter performance are evaluated by Asimov data and toy experiments. The systematic uncertainties from detector response, background, signal efficiencies and PDF are evaluated by using toys. Due to low efficiency of hadronic tag, the statistical uncertainty is expected to be the largest error source.

2 Dataset

2.1 Dataset

Below data is used for the analysis:

- FEI skimmed proc13+prompt data, 365 fb^{-1} . The collection is `"/belle/collection/Data/proc13prompt_skim_11180500_v2"`.

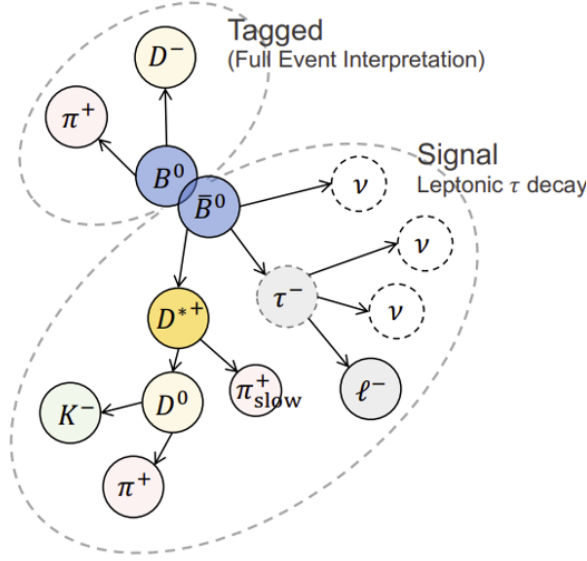


Figure 1.1: Schematic view of the signal reconstruction. The signal side B is reconstructed by combination of ℓ (electron or muon) and D or D^* . The tag side B is reconstructed by hadronic FEI. No additional track is allowed in the rest of events.

Below MC samples are used:

- FEI skimmed MC15rd generic $B\bar{B}$, $c\bar{c}$, $u\bar{u}$, $d\bar{d}$, and $s\bar{s}$ samples equivalent to 1444 fb^{-1} . The collections are
`"/belle/collection/MC/MC15rd_BB_skim_11180500_v2"`,
`"/belle/collection/MC/MC15rd_ccbar_skim_11180500_v2"`,
`"/belle/collection/MC/MC15rd_uds_skim_11180500_v2"`.
- FEI skimmed MC15rd $B \rightarrow D^{**}\ell\nu$ background MC equivalent to 8M events. The collection is
`"/belle/group/physics/SLME/FEI.Dgap_signal"`.
- unskimmed MC15rd generic $B\bar{B}$ sample equivalent to 699 fb^{-1} for correcting D branching ratio in Sec. 4.2.4.

The generic MC15rd is used to simulate both signal and background. Part of $B \rightarrow D^{**}\ell\nu$ decays in the generic MC is replaced with the background MC to improve their modeling. Detail explanation is in Sec. 4.2.

2.2 basf2 release

We used basf2 release of "light-2405-quaxo" for reconstruction and analysis. The analysis code is in "https://gitlab.desy.de/belle2/analyses/wg1_dtaunu_hadtag_p13rel06".

Table 3.1: Selection criteria of event and tag side B .

variable	selection criteria
The number of clean track in an event (nCleanedTrack)	nCleanedTrack > 4
FEI signal probability (p_{FEI})	$p_{\text{FEI}} > 0.01$
Beam constrained mass ($M_{bc} = \sqrt{(E_{\text{beam}})^2 - \vec{p}_{B_{\text{tag}}} ^2}$)	$M_{bc} > 5.27 \text{ GeV}$
Angle between thrust axis of the B_{tag} and others ($\cos\text{TBTO}$)	$\cos\text{TBTO} < 0.9$
Energy difference ($\Delta E = E_{B_{\text{tag}}} - E_{\text{beam}}$)	$-0.15 < \Delta E < 0.1 \text{ GeV}$

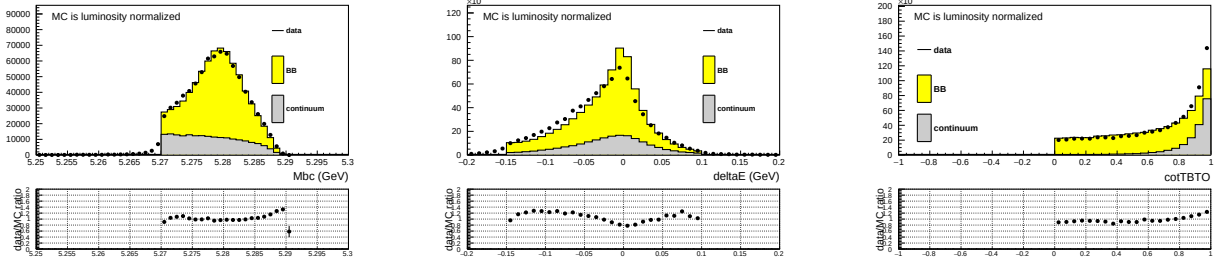


Figure 3.1: B_{tag} variables of M_{bc} (left), ΔE (middle), $\cos\text{TBTO}$ (right) used for the selection. All B_{tag} selections are applied except for $\cos\text{TBTO} < 0.9$. Because track momentum scale correction is applied to data after the selection, there are data points outside the selection criteria in the plots. The same way is used for the FEI calibration analysis.

3 Selection

Signal reconstruction procedure is explained in this section.

3.1 Tag side B

The tag side B (B_{tag}) is reconstructed with hadronic FEI. The standard preselection, which is used for other analysis of FEI efficiency calibration with $B \rightarrow X\ell\nu$ and $B \rightarrow D^{(*)}\pi$ [17], is applied to the B_{tag} candidates. The selections are summarized in Table 3.1 and Fig. 3.1. The signal probability of FEI is required to be more than 0.01. Although data/MC agreement is not perfect, we use the same criteria as the other analysis of FEI calibration, in order to use the calibrated data efficiency provided by them. All B_{tag} candidates in an event are used to increase statistics, although only one B_{tag} candidate with the highest signal probability in an event is used for the FEI calibration analysis.

3.2 Signal side B

After the B_{tag} reconstruction, the remaining tracks and clusters in each event are attributed to the signal B meson candidates of $B \rightarrow D^*\tau\nu$, $B \rightarrow D^*\ell\nu$, $B \rightarrow D\tau\nu$ and $B \rightarrow D\ell\nu$ decays. B_{sig} is reconstructed through the following combinations of a D^* meson and a lepton candidate:

- $B_{\text{sig}}^0 \rightarrow D^{*-}\ell^+$, where $\ell = e$ or μ , $D^{*-} \rightarrow D^0\pi^-$ or $D^{*-} \rightarrow D^-\pi^0$
- $B_{\text{sig}}^+ \rightarrow \bar{D}^{*0}\ell^+$, where $\ell = e$ or μ , $D^{*0} \rightarrow D^0\pi^0$ or $D^{*0} \rightarrow D^0\gamma$
- $B_{\text{sig}}^0 \rightarrow D^-\ell^+$, where $\ell = e$ or μ

Table 3.2: List of reconstructed D^0 decay modes.

Decay	Branching ratio on PDG
$D^0 \rightarrow K^- \pi^+ \pi^0$	$(14.4 \pm 0.50)\%$
$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	$(8.23 \pm 0.14)\%$
$D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$	$(5.20 \pm 0.60)\%$
$D^0 \rightarrow K^- \pi^+$	$(3.95 \pm 0.03)\%$
$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	$(2.80 \pm 0.18)\%$
$D^0 \rightarrow K_S^0 \pi^0$	$(1.24 \pm 0.02)\%$
$D^0 \rightarrow K^- K^+$	$(0.41 \pm 0.01)\%$

Table 3.3: List of reconstructed D^+ decay modes. The D^+ decay modes in gray-shaded lines are newly added.

Decay	Branching ratio on PDG
$D^+ \rightarrow K^- \pi^+ \pi^+$	$(9.38 \pm 0.16)\%$
$D^+ \rightarrow K_S^0 \pi^+ \pi^0$	$(7.36 \pm 0.21)\%$
$D^+ \rightarrow K^- \pi^+ \pi^+ \pi^0$	$(6.25 \pm 0.18)\%$
$D^+ \rightarrow K_S^0 \pi^+ \pi^- \pi^+$	$(3.10 \pm 0.09)\%$
$D^+ \rightarrow K_S^0 \pi^+$	$(1.56 \pm 0.03)\%$
$D^+ \rightarrow K^- K^+ \pi^+$	$(0.968 \pm 0.018)\%$
$D^+ \rightarrow K_S^0 K^+$	$(0.304 \pm 0.01)\%$

126 • $B_{\text{sig}}^+ \rightarrow \bar{D}^0 \ell^+$, where $\ell = e$ or μ

127 the D^0 (D^+) candidates are reconstructed with seven (seven) decay modes, as listed in
128 Table 3.2-3.3. These modes are selected because their branching ratios are relatively large
129 and multiplicity is relatively small. Here, charge conjugation is implied and is considered
130 in all the steps of the analysis.

131 3.2.1 Final state particles

132 The selection criteria are optimized as shown in Table 3.4. On the B_{sig} side, all e, μ, K, π
133 tracks must fulfill the requirements $dr < 2.0$ cm, $|dz| < 4.0$ cm, $p_T > 0.1$ GeV/c. Lepton
134 candidates are required to satisfy global particle-identification requirements for an electron
135 (muon) with a BDT (likelihood) greater than 0.9. $p > 0.4(0.7)$ GeV is required to electron
136 (muon) in order to reach ECL (KLM). Bremsstrahlung correction is applied to electron
137 by using correctBremsBelle module with default arguments. It is required that PID
138 correction is available at region of p and $\cos\theta$ of the lepton candidates. If not, the
139 candidates are removed.

140 Kaon (Pion) candidates are required to satisfy binary PID > 0.1 with likelihood.
141 Neural Network based PID is not used due to smaller coverage of the PID correction
142 table at low momentum, provided by systematic framework. $p > 0.3$ GeV is required to
143 Kaon because fake rate correction is not available below 0.3 GeV. No PID is required to
144 low momentum pion ($p < 0.3$ GeV) to keep statistics. No nCDCHits cut is applied to
145 increase statistics. For the slow charged pion, refers to the π_{slow}^+ daughter produced in the
146 $D^{*+} \rightarrow D^0 \pi_{\text{slow}}^+$ decay, different cut of $p > 0.05$ GeV/c is used without PID.

Table 3.4: Selection criteria on signal side B.

Particles	Selections
good track	$dr < 2 \text{ cm}, dz < 4 \text{ cm}, p_T > 0.1 \text{ GeV}/c$
e^-	good track, $p > 0.4 \text{ GeV}$, $\mathcal{P}_e^{\text{global}} > 0.9$ (BDT), Bremss correction
μ^-	good track, $p > 0.7 \text{ GeV}$, $\mathcal{P}_\mu^{\text{global}} > 0.9$ (likelihood)
π^+	good track, $\mathcal{P}_\pi^{\text{binary}} > 0.1$ or $p < 0.3 \text{ GeV}$
K^+	good track, $\mathcal{P}_K^{\text{binary}} > 0.1$ and $p > 0.3 \text{ GeV}$
π_{slow}^+	$dr < 2 \text{ cm}, dz < 4 \text{ cm}, p > 0.05 \text{ GeV}/c$
π^0	pi0:eff40_May2020, γ : $ \text{clusterTiming} < 200 \text{ ns}$, $122.4 < M_{\pi^0} < 143.0 \text{ MeV}$, Eq:3.1
π_{slow}^0	pi0:eff50_May2020, γ : $ \text{clusterTiming} < 200 \text{ ns}$, $118.3 < M_{\pi^0} < 147.0 \text{ MeV}$, Eq:3.1
K_S^0	ksSelector “standard”, $0.47 < M_{K_S^0} < 0.52 \text{ GeV}$
D	mode dependent M_D cut, Table 3.6
D^*	mode dependent ΔM_{D^*} cut, Table 3.6
B_{sig}	χ^2 of vertex fit > 0 , $q^2 > 4.0 \text{ GeV}^2$

π^0 (slow π^0) is reconstructed with pi0:eff40 (pi0:eff50) selections with a customized mass window of $[122.4, 143.0] \text{ MeV}$ ($[118.3, 147.0] \text{ MeV}$). In addition, $|\text{clusterTiming}| < 200\text{ns}$, minC2TDist and clusterZernikeMVA based cuts are applied:

$$\left(\frac{\text{minC2TDist}}{X \text{ cm}}\right)^2 + \left(\frac{\text{clusterZernikaMVA}}{Y}\right)^2 > 1.0 \quad (3.1)$$

where X and Y are constants summarized in Table 3.5. The same selections as slow π^0 are used for slow photon.

Candidates K_S^0 are reconstructed via $K_S^0 \rightarrow \pi^-\pi^+$, where all candidates pass the “standard” selection in the ksSelector module of basf2. In addition, loose mass selection of $0.47 < M_{K_S^0} < 0.52 \text{ GeV}$ is applied.

3.2.2 D, D^*

Using the final state particles in Sec. 3.2.1, D and D^* candidates are reconstructed as shown in Fig. 3.2–3.4 with treeFit. The window of D mass and mass difference of D^* and D are optimized based on figure of merit as shown in Table 3.6–3.7.

Below is detail of the optimization. First, we evaluated mass resolution (σ) for each M_D and ΔM_{D^*} distributions MC and data. The resolution is defined asymmetrically in the lower side (σ_L) and the higher side (σ_H) from the reconstructed mass peak. They are estimated by fitting the mass distribution. We use following fitting functions: signal of a double-sided Gaussian, peaking background of Gaussian or Crystal ball, and non-peaking background of 1st Chebychev. When MC is fitted, all parameters are floated. The measured σ_L and σ_H of the double Gaussian by MC are summarized in Table 3.7. Data is fitted by the same functions. Normalization parameters of signal and background, and a resolution parameter(R_σ), which scales the sigma of Gaussians of signal and peaking background commonly, are floated for the data fit. Other parameters are fixed with

Table 3.5: Parameter values of the cut function eq. 3.1 for γ selections. $\gamma_{\text{high(low)}}$ denotes either photons of π^0 daughters with higher(lower) energy.

Candidates	cluster region	Parameter values	
		X [cm]	Y
π^0 daughters in D decays	γ_{high}	Forward	85
		Barrel	85
		Backward	85
	γ_{low}	Forward	85
		Barrel	85
		Backward	40
π^0 daughters in D^* decays	γ_{high}	Forward	70
		Barrel	70
		Backward	60
	γ_{low}	Forward	40
		Barrel	40
		Backward	85

the post-fit parameters obtained by MC. The estimated R_σ is shown in Table 3.7. To determine mass window of MC, we define figure of merit as follows: $\text{FoM} = S/\sqrt{(S+N)}$, where S is the number of correctly reconstructed $D^{(*)}$ and B is the number of incorrectly reconstructed $D^{(*)}$. When figure of merit is evaluated, σ_L and σ_H are varied with common fraction and point with maximum FoM is chosen as the selection criteria. For example, 2σ in Table 3.6 represents that lower(higher) mass window of MC is set to $2\sigma_L$ ($2\sigma_H$). Finally, mass window of data is determined by scaling that of MC by the resolution parameter, in order to take account the difference of mass resolution. For example, 2σ in Table 3.6 represents that lower(higher) mass window of data is set to $2R_\sigma\sigma_L$ ($2R_\sigma\sigma_H$).

3.2.3 $B_{\text{sig}} \rightarrow D^{(*)}\tau\nu, \Upsilon(4S)$

Each $D^{(*)}$ is then combined with a lepton candidate to form the B_{sig} . The vertex of the $D^{(*)}\ell$ combination is fitted while constraining all daughters K_S^0 s and π^0 s to the nominal mass, respectively. Any B_{sig} candidate that fails this fit is discarded ($\chi^2 > 0$). The square of momentum transfer to a lepton system in the signal B decays, q^2 , is required to more than $4.0 \text{ (GeV}/c)^2$ in order to enhance signal mode. Finally, the candidate B_{sig} is then combined with the B_{tag} to form the $\Upsilon(4S) \rightarrow B^+B^-$, mixed and unmixed B^0B^0 .

3.3 Rest of events

After the $\Upsilon(4S)$ reconstruction, remaining tracks and clusters are attributed to the rest-of-the-event (ROE). Table 3.8 shows the used ROE mask. Energy threshold of photon at barrel is set to 50 MeV, because data/MC agreement is poor below the 50 MeV. FakephotonSuppressionMVA>0.2 and beamBackgroundSuppressionMVA>0.5 are required to reject fake photon and beam background, as shown in Fig. 3.5. No charged tracks are allowed in the ROE. In addition, no π^0 candidates are required in ROE with pi0:eff30 selection. At the previous version of note, $\eta \rightarrow \gamma\gamma$ veto was applied to ROE. We

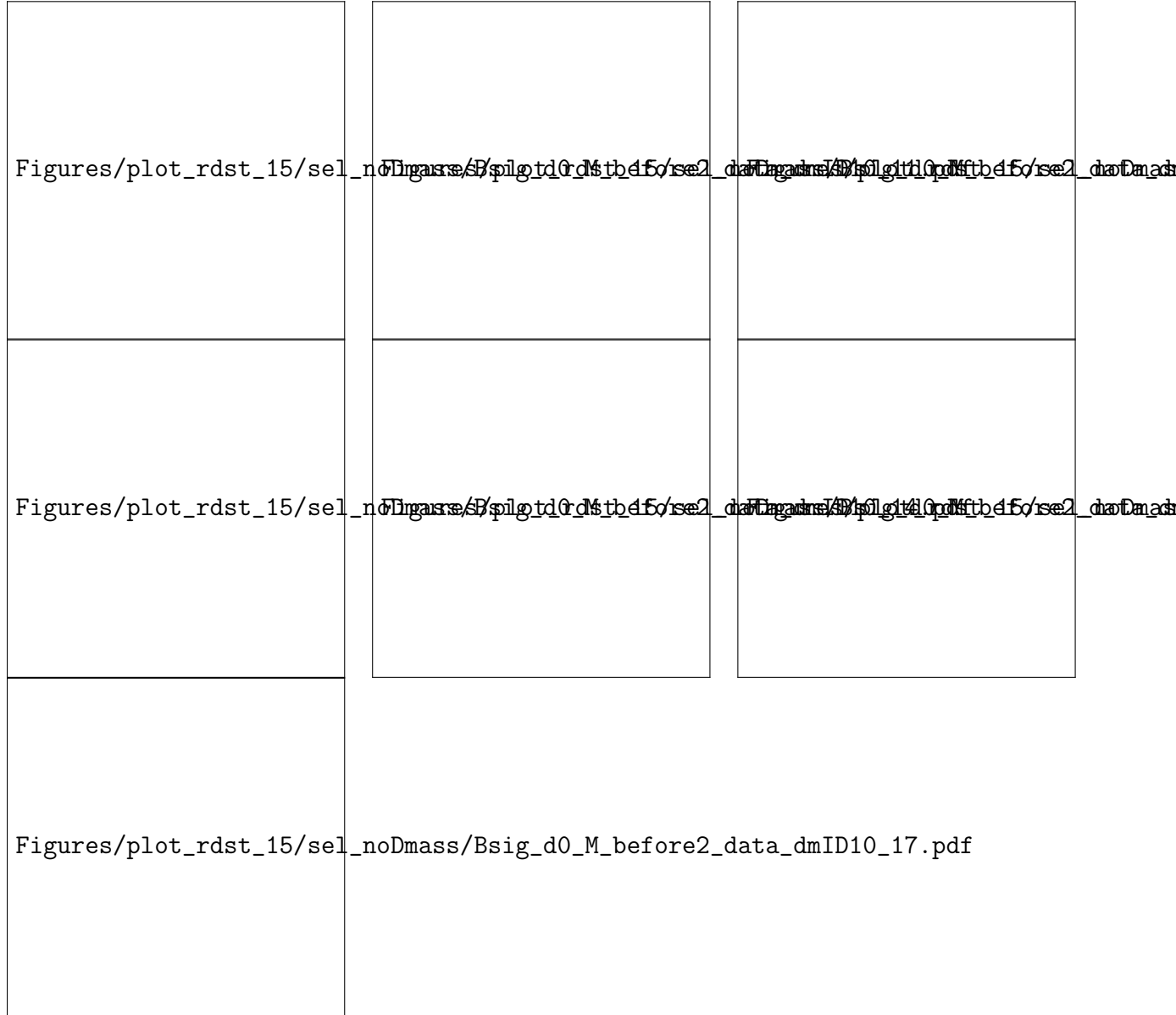


Figure 3.2: Invariant mass of D^0 in each decay mode before applying the mass selection. Mass width and purity are different in each mode due to the difference of daughter particles.

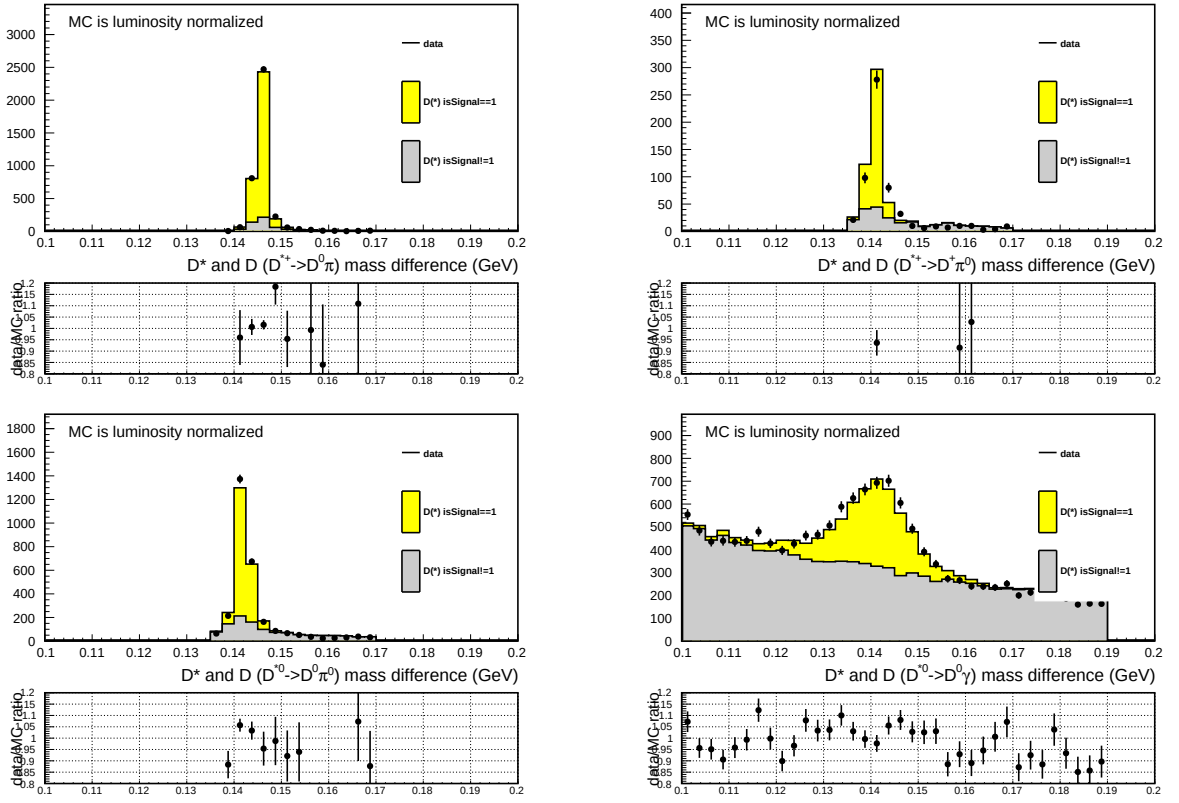


Figure 3.4: Mass differences of D^* and D in each decay mode before applying the mass selection. Mass width and purity are different in each mode due to the difference of charged slow π , slow π^0 , and slow γ .

Table 3.6: Signal regions on M_D and ΔM_{D^*} . The $D^{(*)}$ decay modes in gray-shaded lines are newly added in this analysis.

D^* decays	D decays	M_D	ΔM_{D^*}
$D^{*+} \rightarrow D^0 \pi^+$	$D^0 \rightarrow K^- \pi^+ \pi^0$	2.5σ	3.3σ
	$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	4.1σ	4.9σ
	$D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$	1.8σ	2.6σ
	$D^0 \rightarrow K^- \pi^+$	5.0σ	5.0σ
	$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	4.3σ	4.7σ
	$D^0 \rightarrow K_S^0 \pi^0$	2.4σ	3.6σ
	$D^0 \rightarrow K^- K^+$	4.9σ	3.0σ
$D^{*+} \rightarrow D^+ \pi^0$	$D^+ \rightarrow K^- \pi^+ \pi^+$	3.8σ	3.6σ
	$D^+ \rightarrow K_S^0 \pi^+ \pi^0$	3.3σ	2.5σ
	$D^+ \rightarrow K^- \pi^+ \pi^+ \pi^0$	1.6σ	2.1σ
	$D^+ \rightarrow K_S^0 \pi^+ \pi^- \pi^+$	1.6σ	2.3σ
	$D^+ \rightarrow K_S^0 \pi^+$	4.6σ	1.5σ
	$D^+ \rightarrow K^- K^+ \pi^+$	2.4σ	1.5σ
	$D^+ \rightarrow K_S^0 K^+$	1.2σ	2.7σ
$D^{*0} \rightarrow D^0 \pi^0$	$D^0 \rightarrow K^- \pi^+ \pi^0$	1.4σ	2.1σ
	$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	2.6σ	2.3σ
	$D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$	1.4σ	1.4σ
	$D^0 \rightarrow K^- \pi^+$	5.0σ	2.5σ
	$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	3.8σ	2.2σ
	$D^0 \rightarrow K_S^0 \pi^0$	1.8σ	2.5σ
	$D^0 \rightarrow K^- K^+$	3.9σ	2.0σ
$D^{*0} \rightarrow D^0 \gamma$	$D^0 \rightarrow K^- \pi^+ \pi^0$	1.4σ	2.1σ
	$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	3.4σ	4.4σ
	$D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$	1.4σ	1.7σ
	$D^0 \rightarrow K^- \pi^+$	4.6σ	5.0σ
	$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	2.5σ	3.4σ
	$D^0 \rightarrow K_S^0 \pi^0$	1.9σ	4.3σ
	$D^0 \rightarrow K^- K^+$	1.5σ	3.9σ
D^0	$D^0 \rightarrow K^- \pi^+ \pi^0$	2.1σ	
	$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	4.4σ	
	$D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$	1.7σ	
	$D^0 \rightarrow K^- \pi^+$	5.0σ	
	$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	3.4σ	
	$D^0 \rightarrow K_S^0 \pi^0$	4.3σ	
	$D^0 \rightarrow K^- K^+$	3.9σ	
D^+	$D^+ \rightarrow K^- \pi^+ \pi^+$	4.1σ	
	$D^+ \rightarrow K_S^0 \pi^+ \pi^0$	1.6σ	
	$D^+ \rightarrow K^- \pi^+ \pi^+ \pi^0$	1.5σ	
	$D^+ \rightarrow K_S^0 \pi^+ \pi^- \pi^+$	2.2σ	
	$D^+ \rightarrow K_S^0 \pi^+$	3.1σ	
	$D^+ \rightarrow K^- K^+ \pi^+$	2.5σ	
	$D^+ \rightarrow K_S^0 K^+$	2.5σ	

Table 3.7: M_D and ΔM_{D^*} resolutions.

$D^{(*)}$ decays	σ_L [MeV/ c^2] (MC)	σ_H [MeV/ c^2](MC)	data/MC ratio of σ (R_σ)
$D^0 \rightarrow K^- \pi^+ \pi^0$	15.85	10.47	0.954
$D^0 \rightarrow K^- \pi^+ \pi^- \pi^+$	4.58	3.48	1.087
$D^0 \rightarrow K_S^0 \pi^+ \pi^- \pi^0$	11.60	8.33	1.052
$D^0 \rightarrow K^- \pi^+$	4.24	3.92	1.127
$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	4.02	3.43	0.984
$D^0 \rightarrow K_S^0 \pi^0$	22.46	14.70	0.725
$D^0 \rightarrow K^- K^+$	4.31	3.60	0.862
$D^+ \rightarrow K^- \pi^+ \pi^+$	4.18	3.65	0.925
$D^+ \rightarrow K_S^0 \pi^+ \pi^0$	16.30	10.30	1.079
$D^+ \rightarrow K^- \pi^+ \pi^+ \pi^0$	12.36	8.56	1.051
$D^+ \rightarrow K_S^0 \pi^+ \pi^- \pi^+$	3.89	3.23	1.004
$D^+ \rightarrow K_S^0 \pi^+$	3.89	3.63	1.232
$D^+ \rightarrow K^- K^+ \pi^+$	3.57	3.05	0.814
$D^+ \rightarrow K_S^0 K^+$	4.57	4.42	1.250
$D^{*+} \rightarrow D^0 \pi^+$	0.701	0.694	1.141
$D^{*+} \rightarrow D^+ \pi^0$	1.080	1.185	0.819
$D^{*0} \rightarrow D^0 \pi^0$	1.230	1.156	1.125
$D^{*0} \rightarrow D^0 \gamma$	9.909	6.817	0.987

removed the η veto in order to constrain gap mode by fitting.

3.4 Best candidate selection

There are multiple $Y(4S)$ candidates in an event as shown in Figure 3.6 and Table 3.10. A best candidate selection is thus applied following five steps in chronological order:

1. B_{tag} probability: the candidates with the highest p_{FEI} are selected. B_{tag}^0 and B_{tag}^+ are selected independently.
2. D^* mode selection: $D^{*+} \rightarrow D^0 \pi^+$ candidates are selected at the first priority, $D^{*+} \rightarrow D^+ \pi^0$ at the second priority, $D^{*0} \rightarrow D^0 \pi^0$ at the third priority, $D^{*0} \rightarrow D^0 \gamma$ at the fourth priority, D^0 at the fifth priority, and D^+ at the sixth priority.
3. D^* -mode-dependent selections: the candidates in Table 3.9 are selected.
4. D mode selection: the candidates with D mode with highest branching ratio is selected.
5. Random: Finally, one of the remaining multiple candidates are selected randomly.

Main reasons of the high multiplicity are multiple B_{tag} candidates found by FEI, and D^* candidates which are double counted to D^* sample and D sample. After step2., the multiplicity is small as shown in Figure 3.6.

Table 3.11 shows the simulated events after passing all selections including the best candidate selections. The number of selected $B \rightarrow D^* \tau \nu$, $B \rightarrow D \tau \nu$, $B \rightarrow D^* \ell \nu$, $B \rightarrow D \ell \nu$ are about 600, 300, 15000, and 4000, respectively. The main backgrounds are $B \rightarrow D^{**} \ell \nu$

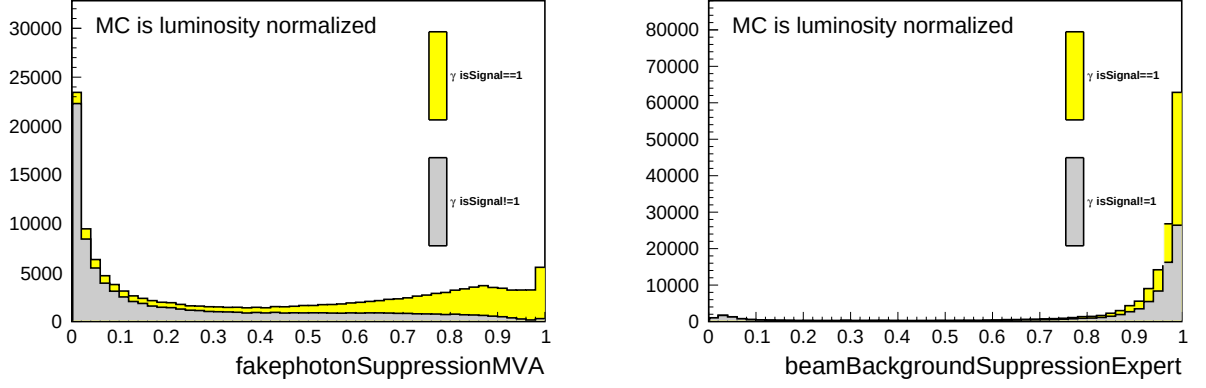


Figure 3.5: FakephotonSuppressionMVA (left) and beambackgroundSuppressionMVA (right) of the photons in the rest of the events.

Table 3.8: Selection criteria for the rest of events.

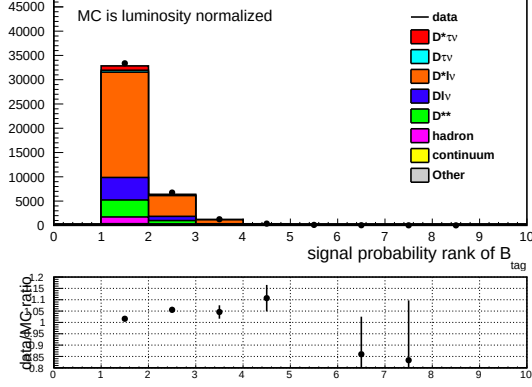
Particles		Selections
ROE mask	Tracks	$dr < 5$ cm, $ dz < 10$ cm, $p_T > 0.1$ GeV/ c , <code>nCDCHits</code> > 0
	Clusters	energy > 0.10 (FW), 0.05 (BR), 0.10 GeV(BW), clusterTiming < 200 ns, minC2TDist > 15 cm, fakePhotonSuppressionMVA > 0.2 , beambackgroundPhotonSuppressionMVA > 0.5
Charged track		<code>roeCharge</code> == 0, <code>nROE_Tracks</code> < 1
Neutral cluster		$N(\pi^0) < 1$ with eff30

Table 3.9: D^* mode dependent best candidate selection for step 3.

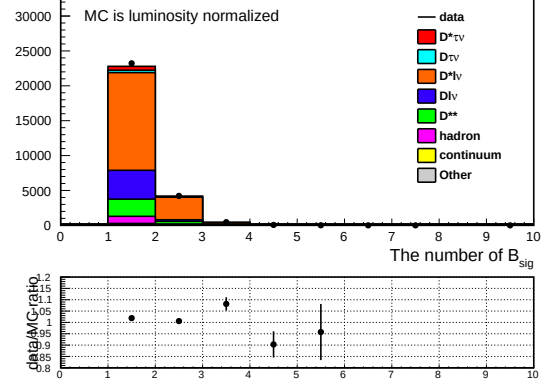
D^* mode	Selection
$D^{*+} \rightarrow D^0 \pi^+$	The maximum χ^2 probability of the second B_{sig} vertex fit
$D^{*+} \rightarrow D^+ \pi^0$	The minimum $\chi^2(M_{\pi^0})$
$D^{*0} \rightarrow D^0 \pi^0$	The minimum $\chi^2(M_{\pi^0})$
$D^{*0} \rightarrow D^0 \gamma$	The most energetic γ_{low}

Table 3.10: The averaged number of event multiplicity of data. Values in parentheses show that of MC.

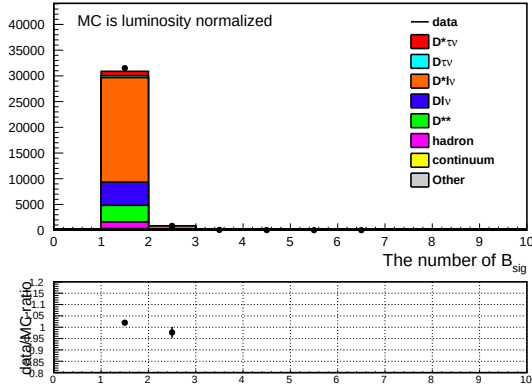
	$D^{*+} \rightarrow D^0 \pi^+$	$D^{*+} \rightarrow D^+ \pi^0$	$D^{*0} \rightarrow D^0 \pi^0$	$D^{*0} \rightarrow D^0 \gamma$	D^0	D^+
before step 1.			1.26 (1.26) (B_{tag})			
after step 1.			1.19 (1.21)			
after step 2.	1.01(1.01)	1.03(1.04)	1.04(1.04)	1.07(1.08)	1.03(1.03)	1.01 (1.02)
after step 3.	1.00(1.00)	1.01(1.01)	1.00(1.01)	1.00(1.00)	1.03(1.03)	1.01 (1.02)
after step 4.	1.00(1.00)	1.01(1.01)	1.00(1.01)	1.00(1.00)	1.02(1.02)	1.01 (1.01)
after step 5.	1.00 (1.00)	1.00 (1.00)	1.00 (1.00)	1.00 (1.00)	1.00 (1.00)	1.00 (1.00)



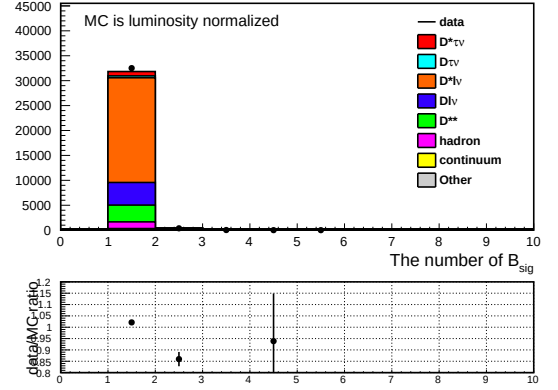
(a) before step 1.



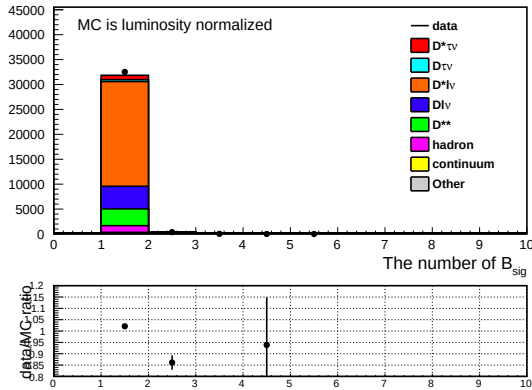
(b) after step 1.



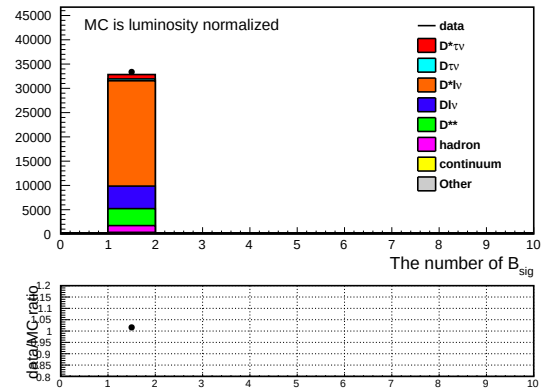
(c) after step 2.



(d) after step 3.



(e) after step 4.



(f) after step 5.

Figure 3.6: Multiplicity of $Y(4S)$ candidates in each event after the selection. The rank of B_{tag} ordered by FEI signal probability before the best candidate selection (top left) and the number of B_{sig} candidates in each step (other five plots) are shown.

Table 3.11: Expected yields of candidate categories and D^* modes at 365 fb^{-1} in MC after all selections. The world average $R(D^{(*)})$ values are assumed for yields of $\bar{B} \rightarrow D^* \tau^- \bar{\nu}_\tau$.

Category	Expected yields in MC					
	$D^{*+} \rightarrow D^0 \pi^+$	$D^{*+} \rightarrow D^+ \pi^0$	$D^{*0} \rightarrow D^0 \pi^0$	$D^{*0} \rightarrow D^0 \gamma$	D^0	D^+
$B \rightarrow D^* \tau \nu$	107	14	63	104	372	61
$B \rightarrow D \tau \nu$	0	0	1.4	21	240	101
$B \rightarrow D^* \ell \nu$	2366	294	1306	2379	9788	1478
$B \rightarrow D \ell \nu$	0.9	2.9	6.9	216	2920	1229
$B \rightarrow D^{**} \ell \nu, D^{**} \tau \nu$ and gap	196	37	120	442	1583	557
hadronic B decay	86	14	44	162	605	279
continuum	2.1	0.8	2.2	33	181	84
others	0.2	0	0	0.1	0.8	4.2
Total	2758	360	1543	3358	15686	3793

213 and hadronic B decays. In $D^{*+} \rightarrow D^0 \pi^+$, $D^{*+} \rightarrow D^+ \pi^0$ and $D^{*0} \rightarrow D^0 \pi^0$ samples,
 214 $B \rightarrow D^* \tau \nu$ and $B \rightarrow D^* \ell \nu$ are selected with high purity. In $D^{*0} \rightarrow D^0 \gamma$ sample, $B \rightarrow D \tau \nu$
 215 and $B \rightarrow D \ell \nu$ are contaminated due to low purity of slow γ . In D samples, the D^* and
 216 D are contaminated due to missed slow pions and slow gamma. Especially, in D^0 sample,
 217 feed down from $D^{*+} \rightarrow D^0 \pi^+$, $D^{*0} \rightarrow D^0 \pi^0$, and $D^{*0} \rightarrow D^0 \gamma$ are large. Figure 3.7–3.9
 218 show distributions of the M_{miss}^2 and $E_{\text{extra}}^{\text{ECL}}$. The signal mode has multiple neutrinos in the
 219 final state and tends to have large M_{miss}^2 . On the other hand, the normalization mode has
 220 one neutrino and M_{miss}^2 is peaked around zero. The background with additional daughter
 221 particles can be identified by large $E_{\text{extra}}^{\text{ECL}}$, because no remaining particles are expected for
 222 signal and normalization modes peaked at $E_{\text{extra}}^{\text{ECL}} = 0$. The signal mode can be extracted
 223 from the small $E_{\text{extra}}^{\text{ECL}}$ with high M_{miss}^2 as shown in Fig. 3.9.

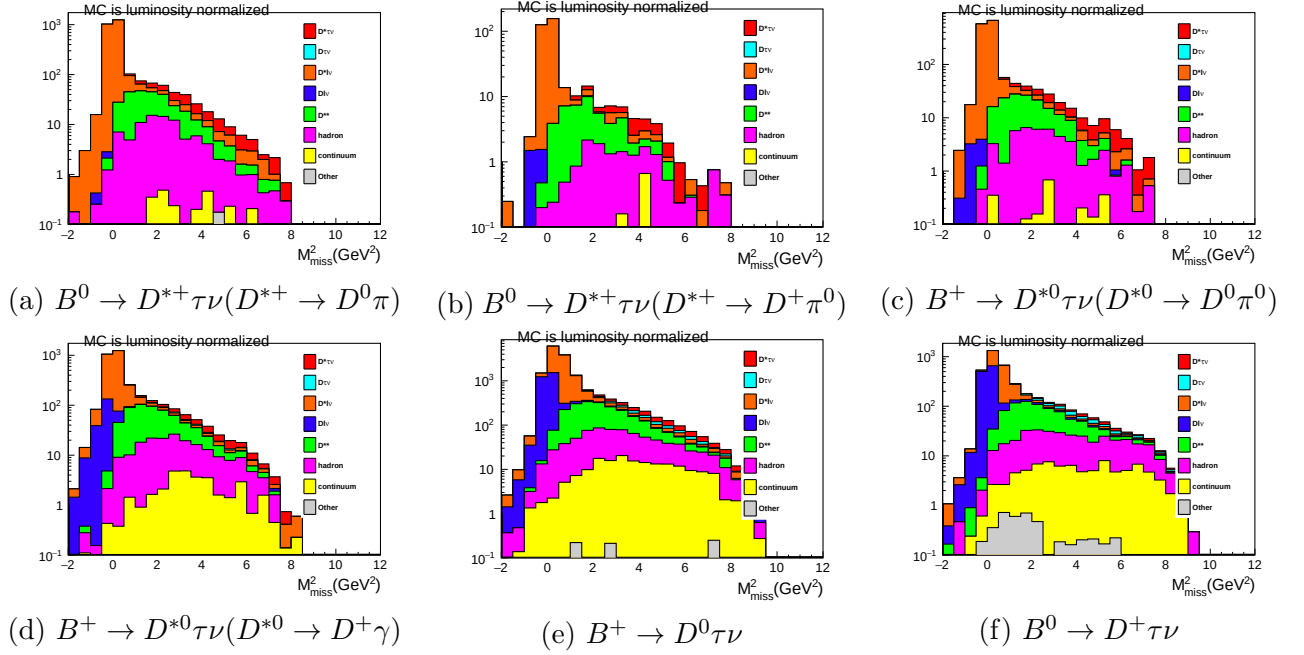


Figure 3.7: MC distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode. The signal mode (red and light blue) has multiple neutrinos in the final state and tends to have large M_{miss}^2 . On the other hand, the normalization mode (orange and blue) has one neutrino and M_{miss}^2 is peaked around zero.

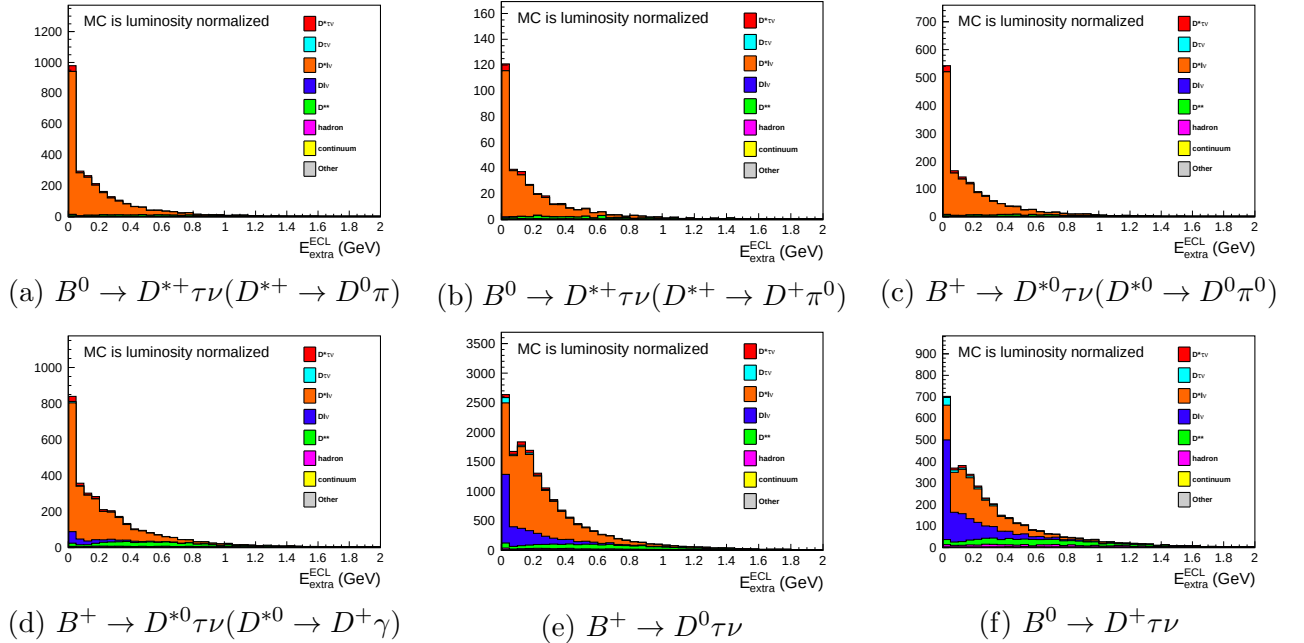


Figure 3.8: MC distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode. The y-axis is linear scale and only normalization mode (orange and blue) are visible. Because no remained particles are expected for normalization modes, they are peaked at $E_{\text{extra}}^{\text{ECL}} = 0$. In D^0 (e) and D^+ (f) samples, $D^* \ell \nu$ is peaked around 0.2 GeV due to γ from slow π^0 .

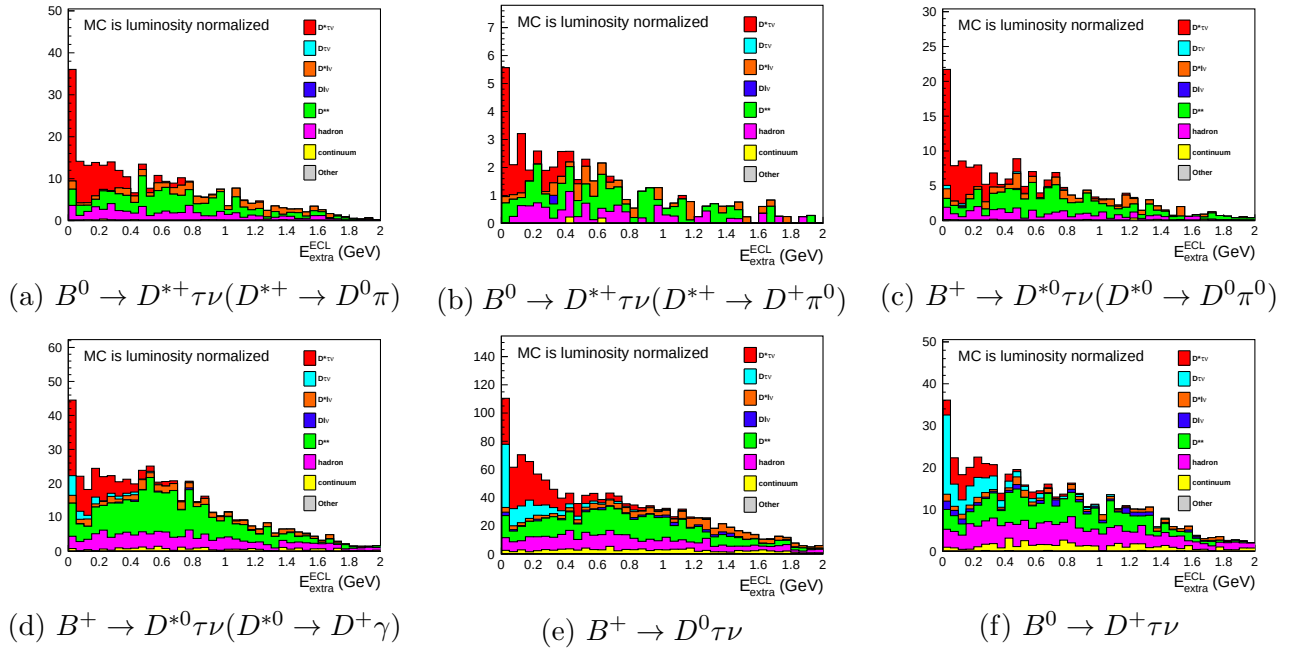


Figure 3.9: MC distributions of the $E_{\text{extra}}^{\text{ECL}}$ with high Missing mass square cut ($1.5 < M_{\text{miss}}^2 < 6.0 \text{ GeV}^2$ for D^* , $3.0 < M_{\text{miss}}^2 < 8.0 \text{ GeV}^2$ for D) in each $D^{(*)}$ decay mode. Because no remained particles are expected for signal modes (red and light blue), they are peaked at $E_{\text{extra}}^{\text{ECL}} = 0$. Distribution is not smooth due to MC statistics uncertainty. For fitting, coarser binning is used.

4 Correction

Several corrections are applied to data and MC, in order to improve data/MC agreement. In this section, general corrections following recommendations from physics performance group, WG1, and PDG are explained. In this note, all corrections are applied to all plots and tables, if nothing mentioned.

4.1 Correction of detector performance

Several corrections are applied based on the recommendations from physics performance group. Detail explanations are in <https://xwiki.desy.de/xwiki/rest/p/3f4b5>. Below corrections are applied to data:

- track momentum scaling (tracking_data_Moriond23_v1)
- photon energy bias (PhotonEnergyBiasCorrection_MC15rd_June2023)

Below corrections are applied to MC:

- hadron PID correction (systematic framework)
- lepton PID correction (leptonid_official_rel6_mc15rd, chargedpidmva_rel6_v5, systematic framework)
- slow $\pi^\pm$ efficiency (provided by tracking group on the above xwiki)
- γ efficiency (PhotonEfficiencyDataMCRatio_Run1MC15rd_April2024)
- π^0 efficiency (provided by neutral group locally by using our π^0 selection)
- mode-dependent FEI efficiency provided by [git@gitlab.desy.de:belle2/performance/correction-tables.git](https://gitlab.desy.de/belle2/performance/correction-tables.git)

The weights are applied to each event. Total weight of an event is multiplication of all weights of the listed corrections.

Track momentum scaling is applied after B_{tag} selection, because track momentum scaling is not applied in the FEI efficiency calibration analysis of $B \rightarrow X\ell\nu$ and $B \rightarrow D^{(*)}\pi$.

The slow π , π^0 , and γ efficiencies are applied to not only D^* samples but also D samples, because if the slow particles are missed in an event, only D is reconstructed and selected by the best candidate selection. They are fully anti-correlated between D^* and D samples. Below is detail of the correction method: we checked events in which both D^* and D candidates with additional slow π or γ are reconstructed. Usually, a D^* candidate is selected in an event by the ROE and best candidate selection. When we apply slow π or γ correction, we assume that D^* candidate is missed and D candidate is selected with the corrected weight probability.

π^0 efficiency corrections are applied to correctly reconstructed candidates, following the same way as the measurement of π^0 efficiency performed by neutral group.

4.2 Correction of branching ratios

Values of the branching ratios of signal and background are different between the Belle II simulation and PDG. The differences are evaluated and corrections are applied to MC.

4.2.1 Branching ratio of $B \rightarrow D^{(*)}\ell\nu$

Branching ratio of $B \rightarrow D^{(*)}\ell\nu$ decays in the Belle II simulation (decfiles/dec/DECAY_BELLE2.DEC) are listed in Table 4.1. During WG review, the values are updated following PDG live and the latest $\text{BR}(B \rightarrow D\ell\nu)$ measurement by Belle II.

Table 4.1: Branching ratio of $B \rightarrow D^{(*)}\ell\nu$ decays.

Decay	MC	PDG and BelleII
$B^0 \rightarrow D^{*+}e\nu$	0.0511	0.0487 ± 0.0009
$B^0 \rightarrow D^{*+}\mu\nu$	0.0511	0.0487 ± 0.0009
$B^0 \rightarrow D^+e\nu$	0.0214	$0.0206 \pm 0.0005 \pm 0.0010$
$B^0 \rightarrow D^+\mu\nu$	0.0214	$0.0206 \pm 0.0005 \pm 0.0010$
$B^- \rightarrow D^{*0}e\nu$	0.0549	0.0526 ± 0.0010
$B^- \rightarrow D^{*0}\mu\nu$	0.0549	0.0526 ± 0.0010
$B^- \rightarrow D^0e\nu$	0.0231	$0.0231 \pm 0.0004 \pm 0.0009$
$B^- \rightarrow D^0\mu\nu$	0.0231	$0.0231 \pm 0.0004 \pm 0.0009$

4.2.2 Branching ratio of $\tau \rightarrow \ell\nu\nu$

Branching ratio of $\tau \rightarrow \ell\nu\nu$ decays in the simulation and PDG 2025 are listed in Table 4.2. They are identical and no corrections are applied.

Table 4.2: Branching ratio of τ leptonic decays.

Decay	MC	PDG
$\tau \rightarrow e\nu\nu$	0.178175	0.1782 ± 0.004
$\tau \rightarrow \mu\nu\nu$	0.173937	0.1739 ± 0.0004

4.2.3 Branching ratio of D^*

Branching ratio of D^* decays in the simulation and PDG 2025 are listed in Table 4.3. They are consistent and no corrections are applied.

Table 4.3: Branching ratios of D^* decays.

Decay	MC	PDG
$D^{*+} \rightarrow D^0\pi^+$	0.677	0.677 ± 0.005
$D^{*+} \rightarrow D^+\pi^0$	0.307	0.307 ± 0.005
$D^{*0} \rightarrow D^0\pi^0$	0.647	0.647 ± 0.009
$D^{*0} \rightarrow D^0\gamma$	0.353	0.353 ± 0.009

4.2.4 Branching ratio of D

The branching ratios of D on PDG are listed in Table 3.2, 3.3. We counted the number of produced D decays in $B \rightarrow D^{(*)}\ell\nu$ in generic MC sample by using MC generator information, and compared them with the expectations from PDG 2025 as shown in Table 4.4. The ratio is applied to MC as an event by event weight.

Table 4.4: The number of simulated D decays in $B \rightarrow D^{(*)}\ell\nu$ MC and corrections of branching ratios. $699fb^{-1}$ equivalent generic MC is used without FEI skim. **Old values were copied from the old version of note wrongly. They are fixed in v3.0.**

D decays	MC	PDG	Ratio(correction factor)
$D^0 \rightarrow K^-\pi^+\pi^0$	7522524	7103608	1.0589
$D^0 \rightarrow K^-\pi^+\pi^-\pi^+$	4775228	4059910	1.1762
$D^0 \rightarrow K_S^0\pi^+\pi^-\pi^0$	1841432	1772545	1.0388
$D^0 \rightarrow K^-\pi^+$	1929107	1948560	0.9900
$D^0 \rightarrow K_S^0\pi^+\pi^-$	946747	954449	0.9919
$D^0 \rightarrow K_S^0\pi^0$	410943	422678	0.9722
$D^0 \rightarrow K^-K^+$	199526	201263	0.9913
$D^+ \rightarrow K^-\pi^+\pi^+$	2054800	2098307	0.9792
$D^+ \rightarrow K_S^0\pi^+\pi^0$	1097158	1137685	0.9643
$D^+ \rightarrow K^-\pi^+\pi^+\pi^0$	1933500	1398125	1.3829
$D^+ \rightarrow K_S^0\pi^+\pi^-\pi^+$	421483	479185	0.8795
$D^+ \rightarrow K_S^0\pi^+$	235347	241448	0.9743
$D^+ \rightarrow K^-K^+\pi^+$	217910	216543	1.0063
$D^+ \rightarrow K_S^0K^+$	45927	46986	0.9774

4.2.5 Branching ratio of hadronic B decay

We categorized the hadronic B decays, reconstructed as B_{sig} , based on MC generator information with GenMC TagTool [18], as shown in Fig. 4.1 and 4.2. We listed up the major hadronic B decays, by selecting the number of expected events more than 2(4) for $B^0(B^+)$ as shown in Table 4.5. Most of the decay modes can be categorized to $B \rightarrow D^{(*)}D^{(*)}$, $B \rightarrow D^{(*)}D^{(*)}K$, and $B \rightarrow D^{(*)}n\pi(\pi^0)$. Their branching ratios on PDG 2025 are compared with decay file configuration used for the Belle II simulation. In the case that the measured branching ratios are listed on PDG, they are consistent with each other and no correction is applied. If the decays are not listed on PDG, no correction is applied and the values on Belle II simulation is used with 100% uncertainty. The treatment of uncertainty is explained in Sec. 7.

4.2.6 Branching ratio of $B \rightarrow D^{**}\ell\nu$ decay

The D^{**} represents the $D_0^{*(0,+)}(2400)$, $D_1^{(0,+)}(2420)$, $D_1^{'(0,+)}(2430)$, $D_2^{*(0,+)}(2460)$ states, and the non-resonant contributions ($D^{**} \rightarrow \eta D^{(*)}, \pi D^{(*)}$). They are excited charm-meson cascade decays to D ground state. Table 4.6 shows the branching fractions of $B \rightarrow D^{**}\ell\nu$ decays. The values are taken from the HFLAV2023 and recommendation from WG1 group. It is difficult to measure each decay mode independently, and a few branching

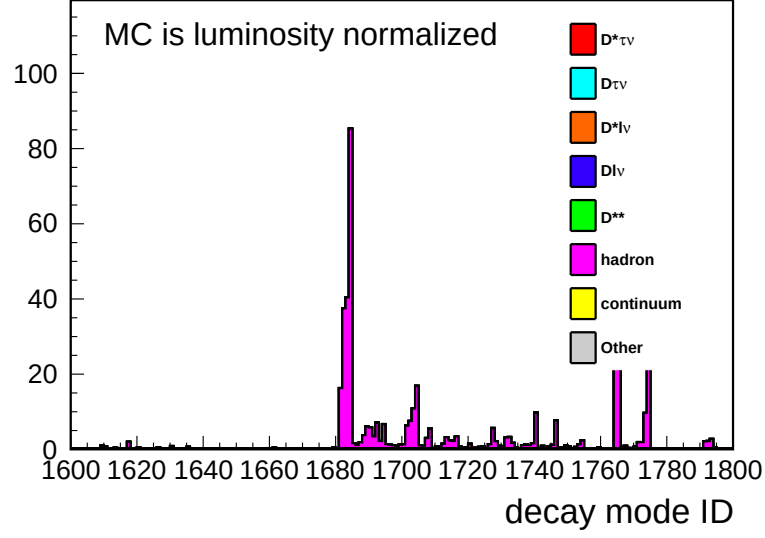


Figure 4.1: Decay mode ID of hadronic B^- decays after event selection.

1681-1684, 1725-1728: $D_{(s)}^{(*)} D^{(*)}$

1685-1690: $D_s^{(*)} D_{1,2}^{(*)}$

1691-1700: $D_s^{(*)} D n \pi$

1701-1724: $D^{(*)} D^{(*)} K^{(*)}$

1729-1748: $D^* n \pi$

1764-1767, 1773-1774: $D_{s0,s1}^{(*)} D^*$

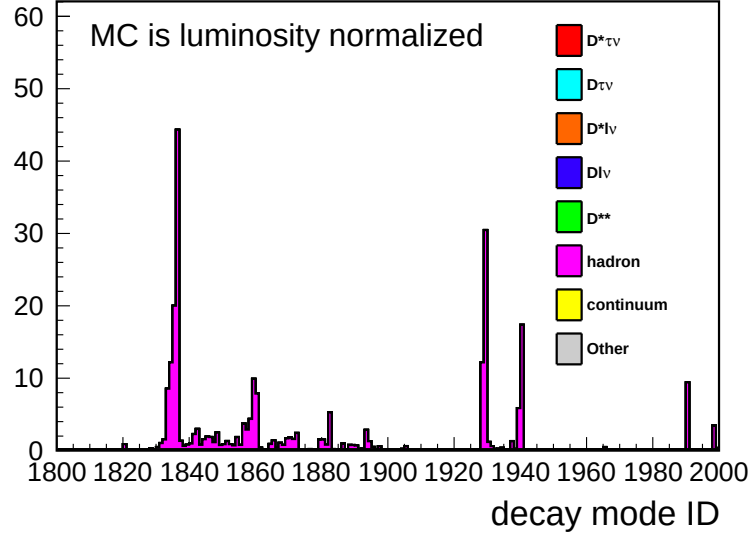


Figure 4.2: Decay mode ID of hadronic B^0 decays after event selection.

1829-1836: $D_{(s)}^{(*)} D^{(*)}$

1837-1842: $D_s^{(*)} D_{1,2}^{(*)}$

1843-1852: $D_s^{(*)} D n \pi$

1853-1876: $D^{(*)} D^{(*)} K^{(*)}$

1877-1893, 1998: $D^* n \pi$

1928-1931: $D_{s0,s1}^{(*)} D^*$

1939-1940: $D'_{s1} D^*$

1990: $D^* p \pi$

Table 4.5: Branching ratios of hadronic B^+ decays.

Decay		decayModelID	$\mathcal{B}$ in MC [10^{-3}]	$\mathcal{B}$ in PDG [10^{-3}]
$D^{(*)}D^{(*)}$	$B^+ \rightarrow D_s^+ \bar{D}$	1681	9.0	9.3 ± 0.6
	$B^+ \rightarrow D_s^+ \bar{D}^*$	1682	8.2	7.0 ± 1.0
	$B^+ \rightarrow D_s^{*+} \bar{D}$	1683	7.6	7.6 ± 1.6
	$B^+ \rightarrow D_s^{*+} \bar{D}^*$	1684	17.1	17.1 ± 2.4
	$B^+ \rightarrow \bar{D}_2^{*0} D_s^+$	1689	4.2	–
	$B^+ \rightarrow \bar{D}_2^{*0} D_s^{*+}$	1690	4.0	–
	$B^+ \rightarrow \bar{D}^{*0} D^+$	1727	0.63	0.63 ± 0.17
	$B^+ \rightarrow \bar{D}^0 D_{s0}^{*+}$	1764	7.6	–
	$B^+ \rightarrow \bar{D}^0 D_{s1}^{*+}$	1773	0.45	0.4 ± 0.1
	$B^+ \rightarrow \bar{D}^{*0} D_{s1}^{*+}$	1774	0.94	0.94 ± 0.42
	$B^0 \rightarrow D_s^+ D^-$	1833	7.2	8.1 ± 0.6
	$B^0 \rightarrow D_s^+ D^{*-}$	1834	8.0	8.2 ± 0.8
	$B^0 \rightarrow D_s^{*+} D^-$	1835	7.4	7.4 ± 1.6
	$B^0 \rightarrow D_s^{*+} D^{*-}$	1836	17.7	17.7 ± 1.4
	$B^0 \rightarrow D_2^{*-} D_s^+$	1841	4.2	–
	$B^0 \rightarrow D_2^{*-} D_s^{*+}$	1842	4.0	–
	$B^0 \rightarrow D_{s0}^{*+} D^-$	1928	7.4	–
	$B^0 \rightarrow D_{s0}^{*+} D^{*-}$	1929	17.7	–
	$B^0 \rightarrow D_{s1}^{*+} D^-$	1939	0.43	0.43 ± 0.17
	$B^0 \rightarrow D_{s1}^{*+} D^{*-}$	1940	0.83	0.83 ± 0.28
$D^{(*)}D^{(*)}K$	$B^+ \rightarrow \bar{D}^0 D^+ K^0$	1701	1.55	1.55 ± 0.21
	$B^+ \rightarrow \bar{D}^0 D^{*+} K^0$	1702	3.8	3.8 ± 0.4
	$B^+ \rightarrow \bar{D}^{*0} D^+ K^0$	1703	2.1	2.1 ± 0.5
	$B^+ \rightarrow \bar{D}^{*0} D^{*+} K^0$	1704	9.2	9.2 ± 1.2
	$B^+ \rightarrow \bar{D}^0 D^{*0} K^+$	1707	6.3	6.3 ± 0.5
	$B^+ \rightarrow \bar{D}^{*0} D^{*0} K^+$	1708	11.2	11.2 ± 1.3
	$B^0 \rightarrow D^{*-} D^{*0} K^+$	1856	10.6	10.6 ± 0.9
	$B^0 \rightarrow D^- D^+ K^0$	1857	0.75	0.75 ± 0.17
	$B^0 \rightarrow D^{*-} D^+ K^0$	1858	1.7	1.7 ± 0.1
	$B^0 \rightarrow D^- D^{*+} K^0$	1859	4.8	4.8 ± 0.4
	$B^0 \rightarrow D^{*-} D^{*+} K^0$	1860	8.1	8.1 ± 0.7
	$B^0 \rightarrow D^- D^+ K^{*0}$	1869	2.5	–
	$B^0 \rightarrow D^{*-} D^{*+} K^{*0}$	1872	5.0	–
$D^* n \pi(\pi^0)$ etc.	$B^+ \rightarrow D_s^+ \bar{D}^0 \pi^0$	1692	1.8	–
	$B^+ \rightarrow D_s^{*+} \bar{D}^0 \pi^0$	1694	1.8	–
	$B^+ \rightarrow \bar{D}^{*-} \pi^0 \pi^+ \pi^+$	1746	15	15 ± 7
	$B^+ \rightarrow \rho^+ \bar{D}^0$	1731	13.4	9.7 ± 1.1
	$B^+ \rightarrow \rho^+ \bar{D}^{*0}$	1732	9.8	9.8 ± 1.7
	$B^+ \rightarrow \bar{D}^{*0} a_1^+$	1740	19	19 ± 5
	$B^0 \rightarrow D_s^+ D^- \pi^0$	1845	1.8	–
	$B^0 \rightarrow D_s^{*+} \bar{D}^0 \pi^-$	1846	3.7	–
	$B^0 \rightarrow D_s^+ D^- \pi^0 \pi^0$	1848	2.2	–
	$B^0 \rightarrow \rho^+ D^-$	1879	7.6	7.6 ± 1.2
	$B^0 \rightarrow \rho^+ D^{*-}$	1880	6.8	6.8 ± 0.9
	$B^0 \rightarrow D^{*-} \pi^+ \pi^0$	1882	8.35	8.2 ± 0.5
	$B^0 \rightarrow a_1^+ D^-$	1886	6.0	6.0 ± 3.3
	$B^0 \rightarrow a_1^+ D^{*-}$	1893	13.0	13.0 ± 0.27
	$B^0 \rightarrow D^* \pi^+ \pi^+ \pi^- \pi^0$	1998	17.6	17.6 ± 2.7
	$B^0 \rightarrow D^{*-} p^+ \bar{n}^0$	1990	1.5	–

ratios are estimated by subtraction of multiple measurements. Detail explanations are in [19] and <https://xwiki.desy.de/xwiki/rest/p/ab7ff>.

Part of $B \rightarrow D^{**}\ell\nu$ decays in the generic MC is replaced with the background MC as shown in Table 4.7, in order to improve modeling of gap mode, which is the difference of the sum of exclusive and of inclusive $B \rightarrow X\ell\nu$ branching ratio measurements. In the generic MC, the decays are generated based on phase space with multi-body decays. On the other hand, in the background MC, they are simulated through D_0^* or D_1' resonances. Detail explanations are in <https://xwiki.desy.de/xwiki/rest/p/ab7ff>. The composition of gap mode is not understood well and large uncertainty is assigned for $D^{**} \rightarrow \eta D^{(*)}$ and $D^{**} \rightarrow D^{(*)}\pi\pi$ decay.

4.3 Correction of form factors

Table 4.8 shows the form factors used in the Belle II simulation for $B \rightarrow D^{(*)}\tau\nu$, $B \rightarrow D^{(*)}\ell\nu$, and $B \rightarrow D^{**}\ell\nu$. The form factors and their correlations are updated based on the angular measurements of $B \rightarrow D^*\ell\nu$ listed in Table 4.9, by using a tool named Hammer [20].

Table 4.6: Branching ratios of $\bar{B} \rightarrow D^{**}\ell^-\bar{\nu}_\ell$ decays.

Decay	BR in MC ($\times 10^{-3}$)	BR from HFLAV2023 ($\times 10^{-3}$)
$\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(0.7 \pm 0.8)/2$
$\bar{B}^0 \rightarrow D_1^{'+}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(0.7 \pm 0.8)/2$
$\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(2.0 \pm 1.0)/2$
$\bar{B}^0 \rightarrow D_1^{'+}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(2.0 \pm 1.0)/2$
$\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell$	– (8M events)	8.8 ± 8.8
$\bar{B}^0 \rightarrow D_1^{'+}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell$	– (8M events)	8.8 ± 8.8
$\bar{B}^0 \rightarrow D^{*0}\pi^+\ell^-\bar{\nu}_\ell$	0.92	0
$\bar{B}^0 \rightarrow D^0\pi^+\ell^-\bar{\nu}_\ell$	0.92	0
$\bar{B}^0 \rightarrow D^{*+}\pi^0\ell^-\bar{\nu}_\ell$	0.46	0
$\bar{B}^0 \rightarrow D^+\pi^0\ell^-\bar{\nu}_\ell$	0.46	0
$\bar{B}^0 \rightarrow D_1^+\ell^-\bar{\nu}_\ell$	7.04	5.9 ± 1.0
$\bar{B}^0 \rightarrow D_0^{*+}\ell^-\bar{\nu}_\ell$	3.62	1.2 ± 1.8
$\bar{B}^0 \rightarrow D_1^{'+}\ell^-\bar{\nu}_\ell$	4.01	2.6 ± 0.4
$\bar{B}^0 \rightarrow D_2^{*+}\ell^-\bar{\nu}_\ell$	3.47	2.99 ± 0.27
$\bar{B}^0 \rightarrow D_1^+\tau^-\bar{\nu}_\tau$	1.3	0.59 ± 0.59
$\bar{B}^0 \rightarrow D_0^{*+}\tau^-\bar{\nu}_\tau$	1.3	0.96 ± 0.96
$\bar{B}^0 \rightarrow D_1^{'+}\tau^-\bar{\nu}_\tau$	2.0	0.13 ± 0.13
$\bar{B}^0 \rightarrow D_2^{*+}\tau^-\bar{\nu}_\tau$	2.0	0.21 ± 0.21
$B^- \rightarrow D_0^{*0}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(0.7 \pm 0.9)/2$
$B^- \rightarrow D_1^{'0}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(0.7 \pm 0.9)/2$
$B^- \rightarrow D_0^{*0}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(2.2 \pm 1.0)/2$
$B^- \rightarrow D_1^{'0}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	– (8M events)	$(2.2 \pm 1.0)/2$
$B^- \rightarrow D_0^{*0}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell$	– (8M events)	9.2 ± 9.2
$B^- \rightarrow D_1^{'0}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell$	– (8M events)	9.2 ± 9.2
$B^- \rightarrow D^{*0}\pi^0\ell^-\bar{\nu}_\ell$	0.50	0
$B^- \rightarrow D^0\pi^0\ell^-\bar{\nu}_\ell$	0.50	0
$B^- \rightarrow D^{*+}\pi^-\ell^-\bar{\nu}_\ell$	1.00	0
$B^- \rightarrow D^+\pi^-\ell^-\bar{\nu}_\ell$	1.00	0
$B^- \rightarrow D_s^{*+}K^-\ell^-\bar{\nu}_\ell$	0.30	0.29 ± 0.19
$B^- \rightarrow D_s^+K^-\ell^-\bar{\nu}_\ell$	0.30	0.30 ± 0.13
$B^- \rightarrow D_1^0\ell^-\bar{\nu}_\ell$	7.57	6.4 ± 1.0
$B^- \rightarrow D_0^{*0}\ell^-\bar{\nu}_\ell$	3.89	1.3 ± 1.9
$B^- \rightarrow D_1^{'0}\ell^-\bar{\nu}_\ell$	4.31	2.8 ± 0.4
$B^- \rightarrow D_2^{*0}\ell^-\bar{\nu}_\ell$	3.73	3.21 ± 0.29
$B^- \rightarrow D_1^0\tau^-\bar{\nu}_\tau$	1.3	0.64 ± 0.64
$B^- \rightarrow D_0^{*0}\tau^-\bar{\nu}_\tau$	1.3	0.10 ± 0.10
$B^- \rightarrow D_1^{'0}\tau^-\bar{\nu}_\tau$	2.0	0.14 ± 0.14
$B^- \rightarrow D_2^{*0}\tau^-\bar{\nu}_\tau$	2.0	0.22 ± 0.22

Table 4.7: Simulated $\bar{B} \rightarrow D^{**}\ell^-\bar{\nu}_\ell$ decays.

Event type	Decay	MC
1196700003	$\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1196700001	$\bar{B}^0 \rightarrow D_1^{'+}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1196700004	$\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1196700002	$\bar{B}^0 \rightarrow D_1^{'+}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1196708001	$\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell$	background MC
1196708000	$\bar{B}^0 \rightarrow D_1^{'+}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell$	background MC
	$\bar{B}^0 \rightarrow D^{*0}\pi^+\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D^0\pi^+\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D^{*+}\pi^0\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D^+\pi^0\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D_1^+\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D_0^{*+}\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D_1^{'+}\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D_2^{*+}\ell^-\bar{\nu}_\ell$	generic MC
	$\bar{B}^0 \rightarrow D_1^+\tau^-\bar{\nu}_\tau$	generic MC
	$\bar{B}^0 \rightarrow D_0^{*+}\tau^-\bar{\nu}_\tau$	generic MC
	$\bar{B}^0 \rightarrow D_1^{'+}\tau^-\bar{\nu}_\tau$	generic MC
	$\bar{B}^0 \rightarrow D_2^{*+}\tau^-\bar{\nu}_\tau$	generic MC
1296700003	$B^- \rightarrow D_0^{*0}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1296700001	$B^- \rightarrow D_1^{'0}(\rightarrow D\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1296700004	$B^- \rightarrow D_0^{*0}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1296700002	$B^- \rightarrow D_1^{'0}(\rightarrow D^*\pi\pi)\ell^-\bar{\nu}_\ell$	background MC
1296708001	$B^- \rightarrow D_0^{*0}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell$	background MC
1296708000	$B^- \rightarrow D_1^{'0}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell$	background MC
	$B^- \rightarrow D^{*0}\pi^0\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D^0\pi^0\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D^{*+}\pi^-\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D^+\pi^-\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_s^{*+}K^-\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_s^+K^-\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_1^0\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_0^{*0}\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_1^{'0}\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_2^{*0}\ell^-\bar{\nu}_\ell$	generic MC
	$B^- \rightarrow D_1^0\tau^-\bar{\nu}_\tau$	generic MC
	$B^- \rightarrow D_0^{*0}\tau^-\bar{\nu}_\tau$	generic MC
	$B^- \rightarrow D_1^{'0}\tau^-\bar{\nu}_\tau$	generic MC
	$B^- \rightarrow D_2^{*0}\tau^-\bar{\nu}_\tau$	generic MC

Table 4.8: Summary of form factor used in Belle II simulation

Decay	configuration in dec file
$B \rightarrow D^* \tau \nu$	HQET3 0.912 1.205 1.15 1.404 0.854
$B \rightarrow D \tau \nu$	HQET3 1.0541 1.128 1.0
$B \rightarrow D_0^*, D_1' \tau \nu$	LLSW 0.68 -0.2 0.3
$B \rightarrow D_1, D_2^* \tau \nu$	LLSW 0.71 -1.6 -0.5 2.9
$B \rightarrow D^* \ell \nu$	BGL 0.02596 -0.06049 0.01311 0.01713 0.00753 -0.09346
$B \rightarrow D \ell \nu$	BGL 0.0126 -0.094 0.34 -0.1 0.0115 -0.057 0.12 0.4
$B \rightarrow D_0^*, D_1' \ell \nu$	LLSW 0.68 -0.2 0.3
$B \rightarrow D_1, D_2^* \ell \nu$	LLSW 0.71 -1.6 -0.5 2.9

Table 4.9: Summary of paper used for form factor correction by Hammer

Decay	model	paper
$B \rightarrow D^{(*)} \tau \nu$	BLPRXP	2206.11281
$B \rightarrow D^{(*)} \ell \nu$	BLPRXP	2206.11281
$B \rightarrow D^{**} \ell \nu$	BLR	1711.03110, 1606.09300
$B \rightarrow D^{**} \tau \nu$	BLR	1711.03110, 1606.09300

5 Data/MC comparison in side-band

The data-MC difference is studied with side-band regions, as summarized in Table 5.1. In the low q^2 side-band, the normalization mode is enhanced which allows to validate the data-MC differences of the ECL energy sum in ROE ($E_{\text{extra}}^{\text{ECL}}$) and Missing mass squared (M_{miss}^2), which are used for signal extraction in Sec. 6. Because the final states of the normalization and signal modes are very similar, the validation is effective for the signal mode too. In addition, background yield and PDF shapes are studied in π^0 veto side-band. PDF shapes of wrongly reconstructed $D^{(*)}$ are studied in $D^{(*)}$ mass sideband.

Table 5.1: Summary of side-band regions

Side-band	Selection	Enhanced mode	Check items
q^2 side-band	$q^2 < 3.5 \text{ (GeV/c)}^2$	$\bar{B} \rightarrow D^* \ell^- \bar{\nu}_\ell$	FEI efficiency signal PDF shapes
π^0 veto side-band	$N_{\pi^0_{\text{ROE}}} > 0,$ $E_{\text{ECL}}^{\text{extra}} > 1.0 \text{ GeV},$ $M_{\text{miss}}^2 > 2 \text{ GeV}^2$	$\bar{B} \rightarrow D^{**} \ell^- \bar{\nu}_\ell,$ hadron	BG yield, PDF
$D^{(*)}$ mass side-band	outside of Table 3.6	wrongly reconstructed D^*	PDF

5.1 q^2 side-band

5.1.1 Data yield and efficiency

Although the absolute number of normalization mode and signal mode are canceled for $R(D^{(*)})$ measurement, we checked absolute data yield of normalization mode, in order to check how well it is modeled by MC. Table 5.2 shows the number of selected events in the q^2 side-band. The data yield is consistent with MC expectation within a few percents. These are reasonable, because a few percent systematic errors are expected from FEI efficiency and other detector performances. Data yield of $D^{*+} \rightarrow D^+ \pi^0$ is lower than MC expectation by about 2σ of statistical error, which can be statistical fluctuation. Figure 5.1–5.3 show the D mass and mass difference of D^* and D . The data yield of $D^+ \rightarrow K\pi\pi\pi^0$, $D^0 \rightarrow K\pi\pi\pi$, $D^0 \rightarrow KK$, $D^+ \rightarrow K_S\pi\pi^0$ are about 20-30% different from MC (more than 2σ of statistical error). The difference is taken as systematic error in Sec. 7.

Figure 5.4 shows decay mode ID of B_{tag} . Although FEI efficiency is calibrated in each decay mode by using $B \rightarrow X\ell\nu$ and $B \rightarrow D^{(*)}\pi$, the data yield does not agree with MC expectation in some decay modes. In order to cover these data/MC difference, systematic error is estimated in Sec. 7. **Figure 5.5 shows lepton momentum in each decay mode. The data agree with MC within statistical error.**

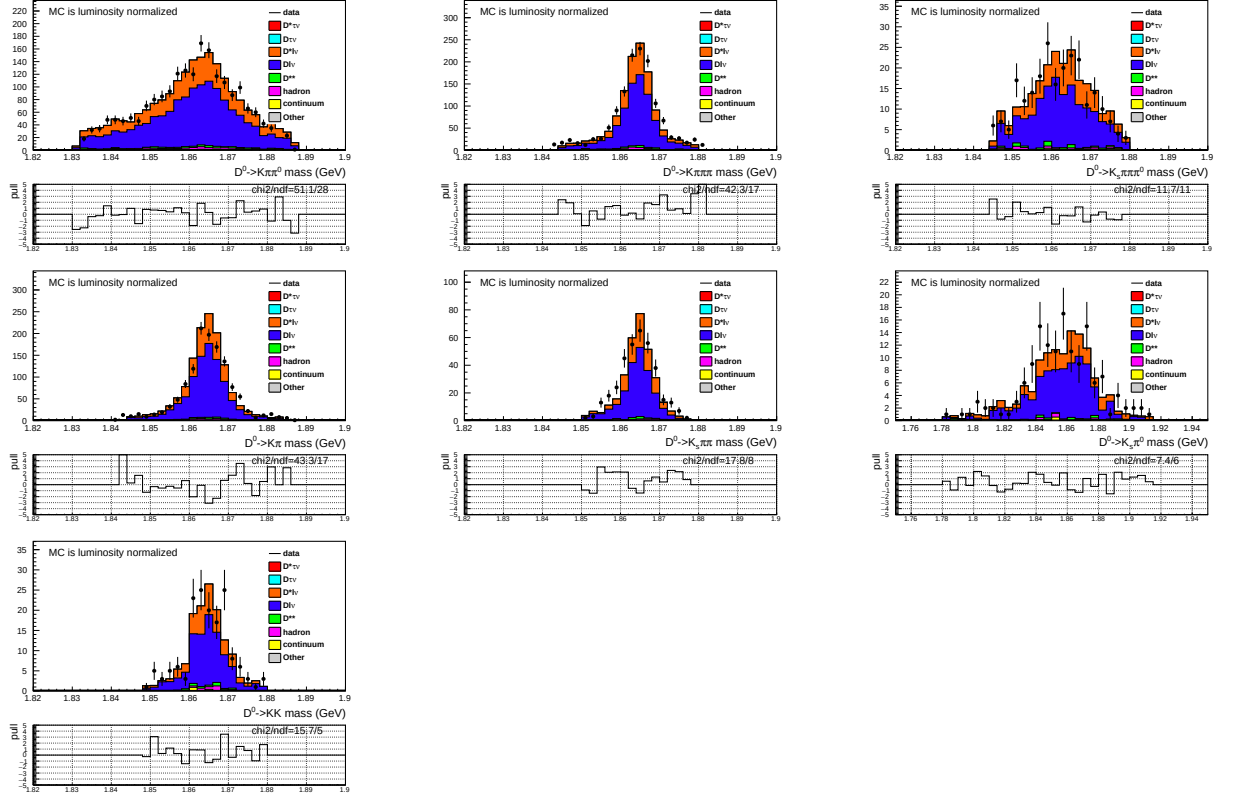


Figure 5.1: Invariant mass of D^0 in each decay mode in the q^2 sideband with all selections including the mass selection.

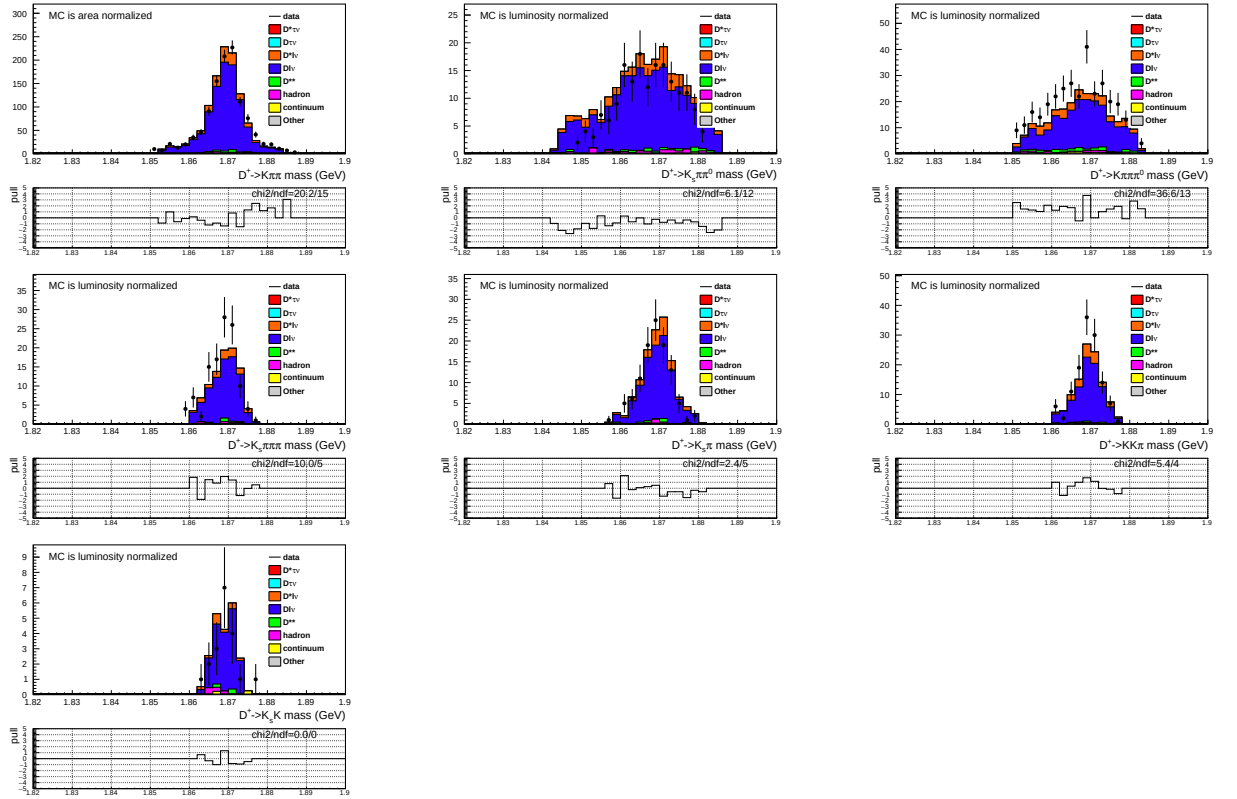


Figure 5.2: Invariant mass of D^+ in each decay mode in the q^2 sideband with all selections including the mass selection.

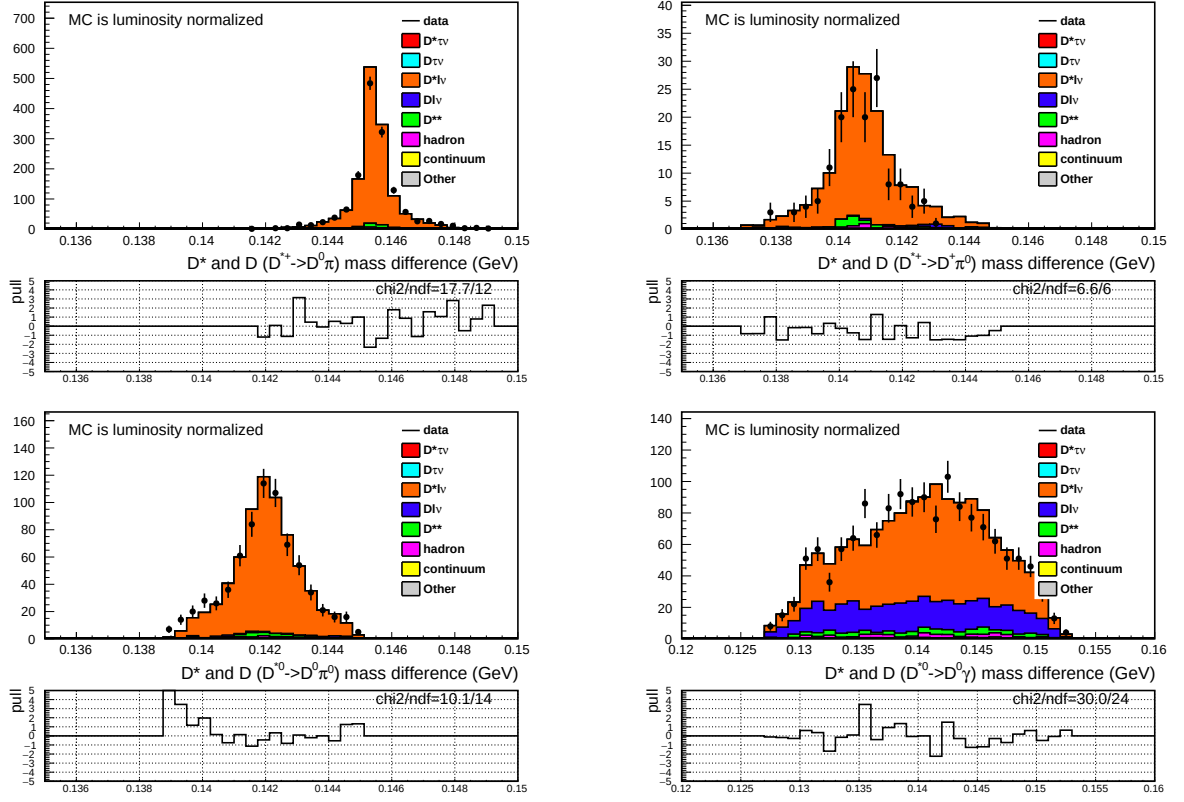


Figure 5.3: Mass differences of D^* and D in each decay mode in the q^2 sideband with all selections including the mass selection.

Table 5.2: Data/MC comparison of the number of selected events in the q^2 side-band. The error shows statistical error of data. systematic error is not included in the error.

Category	The number of selected events)					
	$D^{*+} \rightarrow D^0 \pi^+$	$D^{*+} \rightarrow D^+ \pi^0$	$D^{*0} \rightarrow D^0 \pi^0$	$D^{*0} \rightarrow D^0 \gamma$	D^0	D^+
Data	1422	144	712	1481	5585	1988
Total MC	1432	170	707	1476	5443	1928
Data/MC ratio	0.993	0.848	1.007	1.004	1.026	1.032
(stat. error)	± 0.026	± 0.071	± 0.038	± 0.026	± 0.014	± 0.023

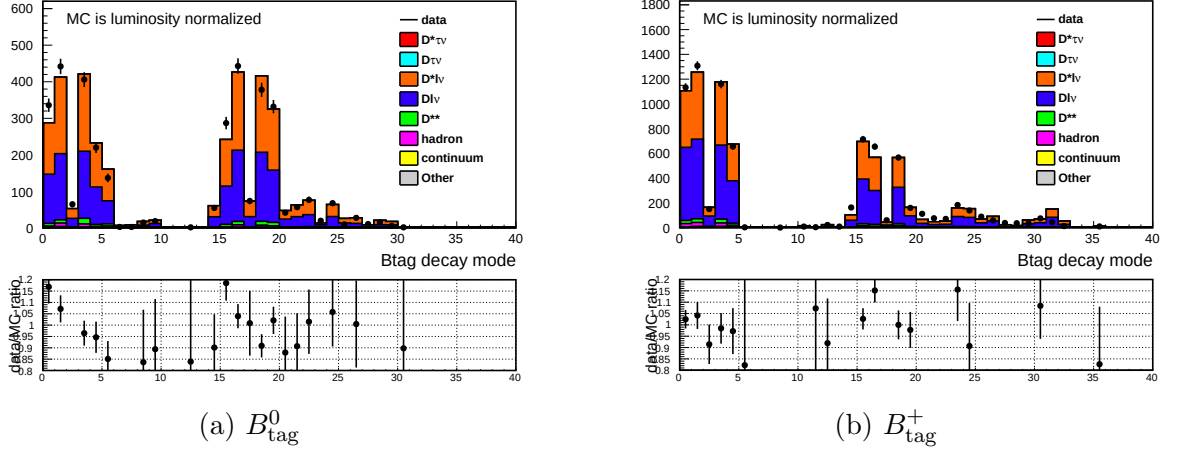


Figure 5.4: B_{tag} decay mode ID in the q^2 side-band. Major decay modes are $B^0 \rightarrow D^-\pi^+(0)$, $D^-\pi^+\pi^0(1)$, $D^-\pi^+\pi^+\pi^-(3)$, $D^-\pi^+\pi^+\pi^-\pi^0(4)$, $\bar{D}^0\pi^+\pi^-(5)$, $D^{*-}\pi^+(15)$, $\bar{D}^{*-}\pi^+\pi^0(16)$, $\bar{D}^{*-}\pi^+\pi^+\pi^-(18)$, $\bar{D}^{*-}\pi^+\pi^+\pi^-\pi^0(19)$, $\Lambda_c^- p\pi^+\pi^-(26)$, $B^+ \rightarrow \bar{D}^0\pi^+(0)$, $\bar{D}^0\pi^+\pi^0(1)$, $\bar{D}^0\pi^+\pi^+\pi^-(3)$, $\bar{D}^0\pi^+\pi^+\pi^-\pi^0(4)$, $\bar{D}^{*0}\pi^+(15)$, $\bar{D}^{*0}\pi^+\pi^0(16)$, $\bar{D}^{*0}\pi^+\pi^+\pi^-(18)$, $\bar{D}^{*0}\pi^+\pi^+\pi^-\pi^0(19)$, $D^-\pi^+\pi^+(23)$, $D^-\pi^+\pi^+\pi^0(24)$, $\Lambda_c^- p\pi^+\pi^-\pi^+(30)$.

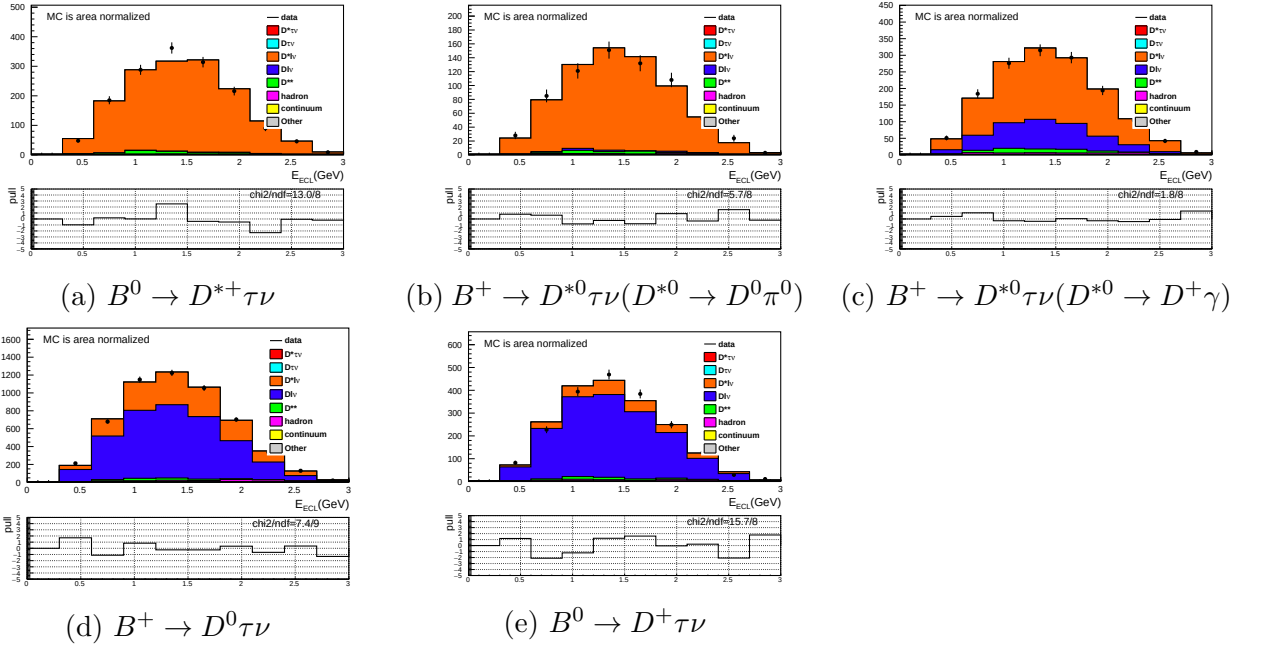
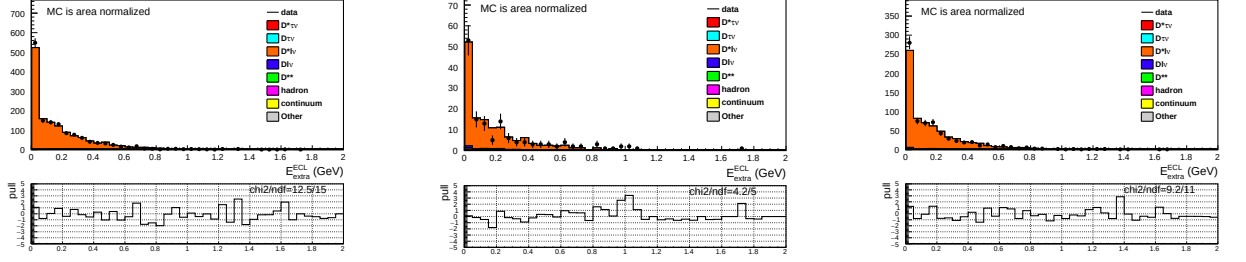
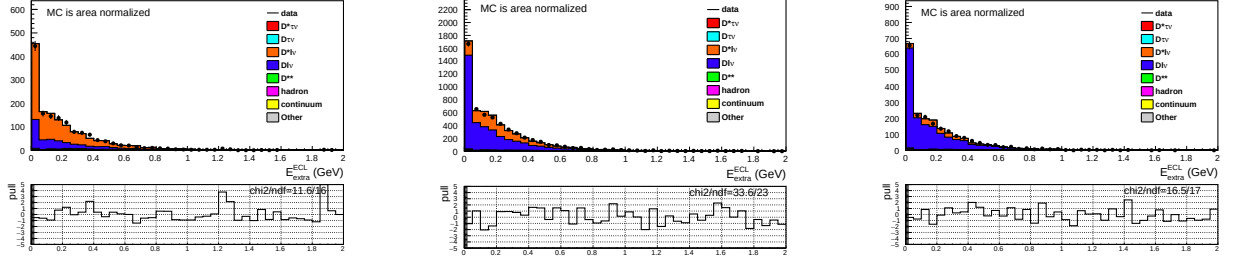


Figure 5.5: Distributions of the lepton momentum in each $D^{(*)}$ decay mode in the q^2 side-band with fine binning.



(a) $B^0 \rightarrow D^{*+} \tau \nu (D^{*+} \rightarrow D^0 \pi)$ (b) $B^0 \rightarrow D^{*+} \tau \nu (D^{*+} \rightarrow D^+ \pi^0)$ (c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(d) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^+ \gamma)$ (e) $B^+ \rightarrow D^0 \tau \nu$ (f) $B^0 \rightarrow D^+ \tau \nu$

Figure 5.6: Distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode in the q^2 side-band. The data agree with MC within statistical error.

5.1.2 $E_{\text{extra}}^{\text{ECL}}$ PDF shape

Figure 5.6 shows the $E_{\text{extra}}^{\text{ECL}}$ distribution. In order to compare shape, MC is area normalized. The data agree with MC within statistical error.

5.1.3 M_{miss}^2 PDF shape

Figure 5.7 shows the M_{miss}^2 distribution. In order to compare shape, MC is area normalized. The data agree with MC within statistical error. A coarse binning is chosen because the same binning is used for the fitting.

Figure 5.8 shows the same distribution with fine binning. Small discrepancies in peak widths are observed between data and MC simulations. We apply a MC correction to cover this difference as follows. To evaluate the M_{miss}^2 width, the M_{miss}^2 distributions in data and MC are fitted by combinations of Gaussian and Crystal Ball functions, as shown in Table 5.3. When data is fitted, shape parameters of these functions are determined from MC, with a floating normalization parameter and a single floating shape parameter (R_{res}) which scales all of the sigma of Gaussians commonly. Table 5.4 shows the fitted R_{res} and a weighted average of fitted Gaussians's sigma in MC (σ^{MC}). Due to the limited statistics of the $D^{*+} \rightarrow D^+ \pi^0$ decay channel, it is combined with the $D^{*+} \rightarrow D^0 \pi^+$ channel. In order to correct the M_{miss}^2 of MC to data, the M_{miss}^2 of MC is smeared in each event by

$$M_{\text{miss}}^2 = M_{\text{miss}}^2 + \text{Gauss}(0, \Delta\sigma) \quad (5.1)$$

where Gauss is a random number generated by Gaussian with mean of 0 and width of $\Delta\sigma = \sigma^{\text{MC}} \sqrt{(R_{\text{res}}^2 - 1)}$. The uncertainty of the fit is taken into account as systematic error as described in Sec. 7.

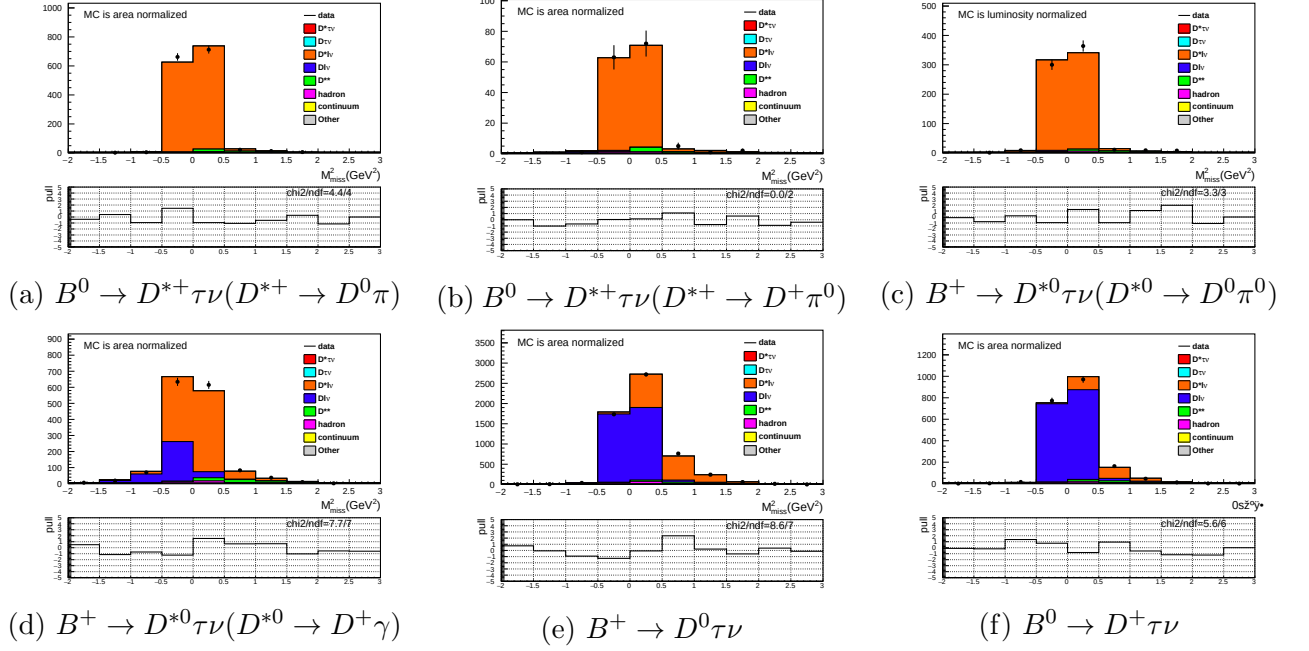


Figure 5.7: Distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode in the q^2 side-band. The data agree with MC within statistical error.

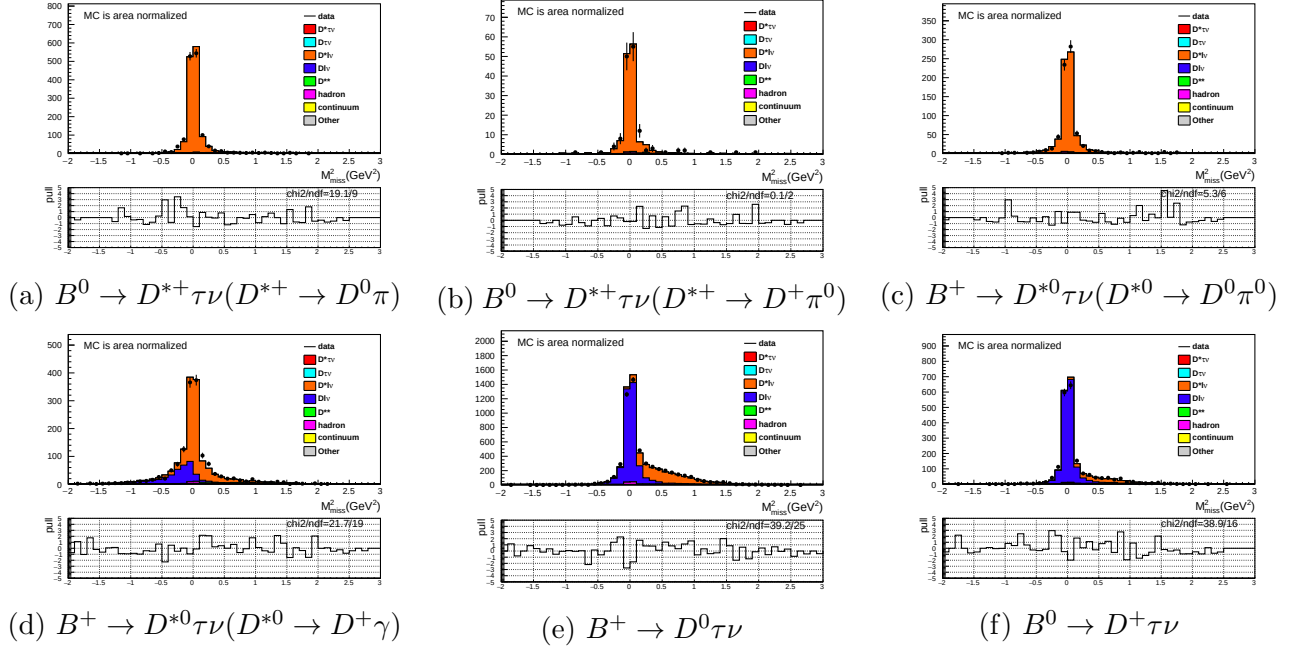


Figure 5.8: Distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode in the q^2 side-band with fine binning.

Table 5.3: Modeling functions on M_{miss}^2

D decay	signal	background
D^{*+}	triple gaussian	double gaussian + crystall ball
$D^{*0} \rightarrow D^0 \pi^0$	triple gaussian	double gaussian + crystall ball
$D^{*0} \rightarrow D^0 \gamma$	triple gaussian	triple gaussian
D^0	triple gaussian	double gaussian + crystall ball
D^+	triple gaussian	double gaussian + crystall ball

Table 5.4: Resolution and smearing factors in the M_{miss}^2 distributions in the q^2 side-band region

D decay	R_{res}	$\sigma^{\text{MC}}(\text{GeV}^2)$	$\Delta\sigma$ (GeV^2)
D^{*+}	1.132 ± 0.009	0.0750 ± 0.0011	0.0398 ± 0.0037
$D^{*0} \rightarrow D^0 \pi^0$	1.039 ± 0.003	0.0883 ± 0.0025	0.0249 ± 0.0129
$D^{*0} \rightarrow D^0 \gamma$	1.076 ± 0.006	0.1187 ± 0.0018	0.0472 ± 0.0070
D^0	1.083 ± 0.007	0.0960 ± 0.0026	0.0399 ± 0.0098
D^+	1.144 ± 0.008	0.0811 ± 0.0023	0.0450 ± 0.0070

5.1.4 Fraction of correctly reconstructed $D^{(*)}$

Figure 5.9-5.10 show $E_{\text{extra}}^{\text{ECL}}$ and M_{miss}^2 distributions, categorized by correctly and incorrectly $D^{(*)}$ distribution. Their PDF shape is different at low $E_{\text{extra}}^{\text{ECL}}$ region mainly. These distribution will be used to estimate a fraction of correctly reconstructed D^* candidates in Sec. 7.

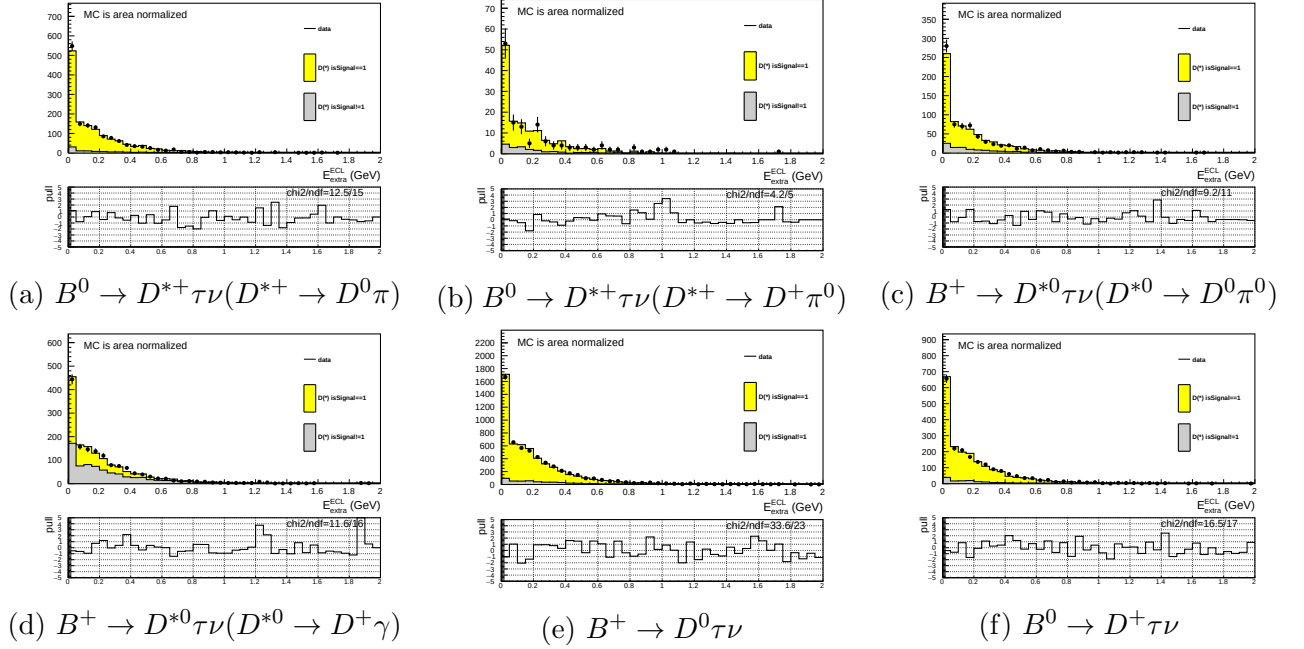


Figure 5.9: Distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode in the q^2 side-band. MC is categorized by correctly and incorrectly reconstructed $D^{(*)}$.

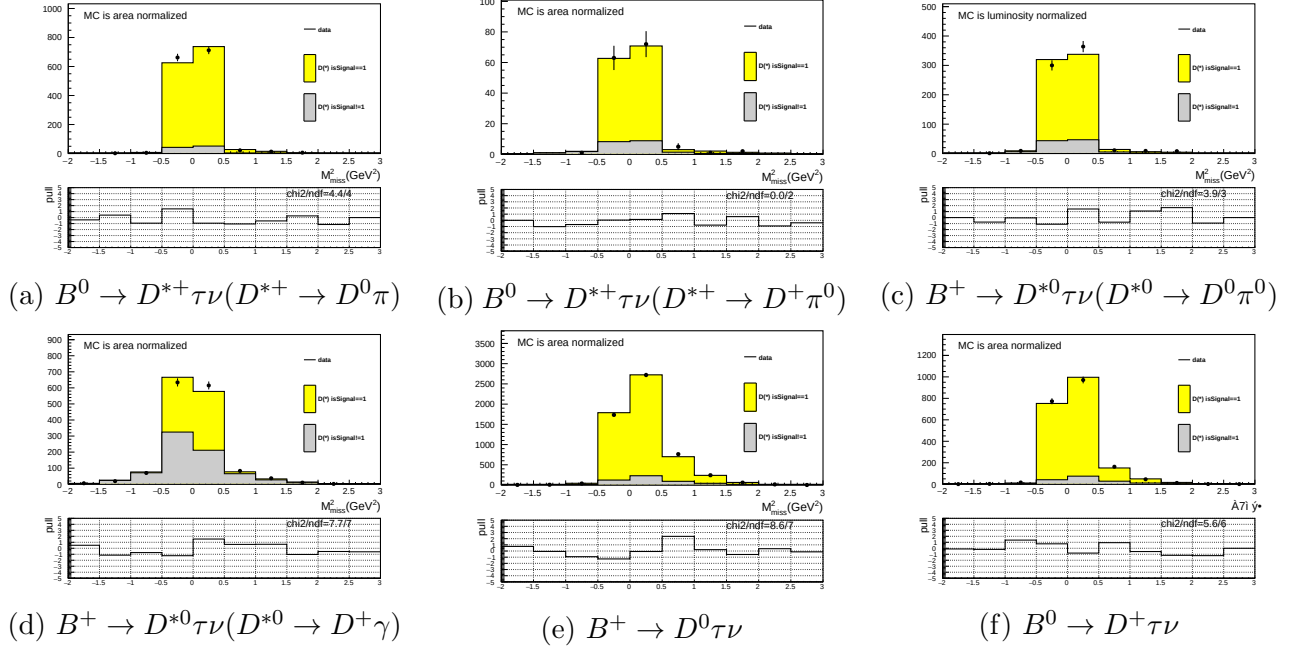


Figure 5.10: Distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode in the q^2 side-band. MC is categorized by correctly and incorrectly reconstructed $D^{(*)}$.

Table 5.5: Data/MC comparison of the number of selected events in the π^0 veto side-band. The error shows statistical error of data.

Category	The number of selected events					
	$D^{*+} \rightarrow D^0 \pi^+$	$D^{*+} \rightarrow D^+ \pi^0$	$D^{*0} \rightarrow D^0 \pi^0$	$D^{*0} \rightarrow D^0 \gamma$	D^0	D^+
Data ($\pm$ stat.)	85 ± 9	24 ± 5	49 ± 7	397 ± 20	1421 ± 38	764 ± 28
MC $B \rightarrow D^{**} \ell \nu$, gap mode	22	3.1	19	79	354	131
MC hadronic B decay, unmeasured BR	28	11	17	90	287	293
MC $B \rightarrow D^{**} \ell \nu$, non-gap mode	13	1.4	9.9	34	191	78
MC hadronic B decay, measured BR	34	11	31	124	420	215
MC continuum	5.5	3.6	5.5	67	348	199
MC others, $B \rightarrow D^{(*)} \ell \nu$	10.4	1.5	4.4	15	118	30
MC Total ($\pm$ 100% error of unmeasured BR)	116 $\pm 22 \pm 28$	31.7 $\pm 3.1 \pm 11$	86.9 $\pm 19 \pm 17$	408.8 $\pm 79 \pm 90$	1718 $\pm 354 \pm 287$	946 $\pm 131 \pm 293$
Data/MC (stat.)	0.75 ± 0.08	0.76 ± 0.15	0.56 ± 0.08	0.97 ± 0.05	0.83 ± 0.02	0.81 ± 0.03
($\pm$ 100% error of unmeasured BR)	$\pm 0.17 \pm 0.25$	$\pm 0.09 \pm 0.44$	$\pm 0.15 \pm 0.15$	$\pm 0.24 \pm 0.28$	$\pm 0.22 \pm 0.17$	$\pm 0.13 \pm 0.36$

5.2 π^0 veto side-band

Table 5.5 shows the number of selected events in the π^0 veto side-band. The data yield is 10-40% smaller than MC expectation in each sample. In order to investigate this issue, new MC category of "un-measured hadronic B decay" and " $D^{**} \ell \nu$ gap mode", where their branching ratios are not measured so far on PDG, are added to the table. Because their branching ratio uncertainty is large ($\pm 100\%$), the MC expectation has large uncertainty and it covers the data/MC difference. Then, systematic uncertainties from their branching ratios is estimated in Sec. 7, and no additional systematic uncertainties are added. The other analysis of $B \rightarrow D^{(*)} \ell \nu$ [21] reported the similar observation at their control region: the data suggest a configuration with a substantially smaller fraction of gap modes.

Figure 5.11 and 5.12 show the $E_{\text{extra}}^{\text{ECL}}$ and M_{miss}^2 distributions. The data agree with MC within statistical error and the branching ratio error. Then we do not apply additional corrections and systematics by using this side-band. The distribution suggests a configuration with a substantially smaller fraction of gap modes too. In order to constrain gap mode from data, gap mode is independently categorized from other $D^{**} \ell \nu$ and floated for the fitting, as described in Sec. 6.

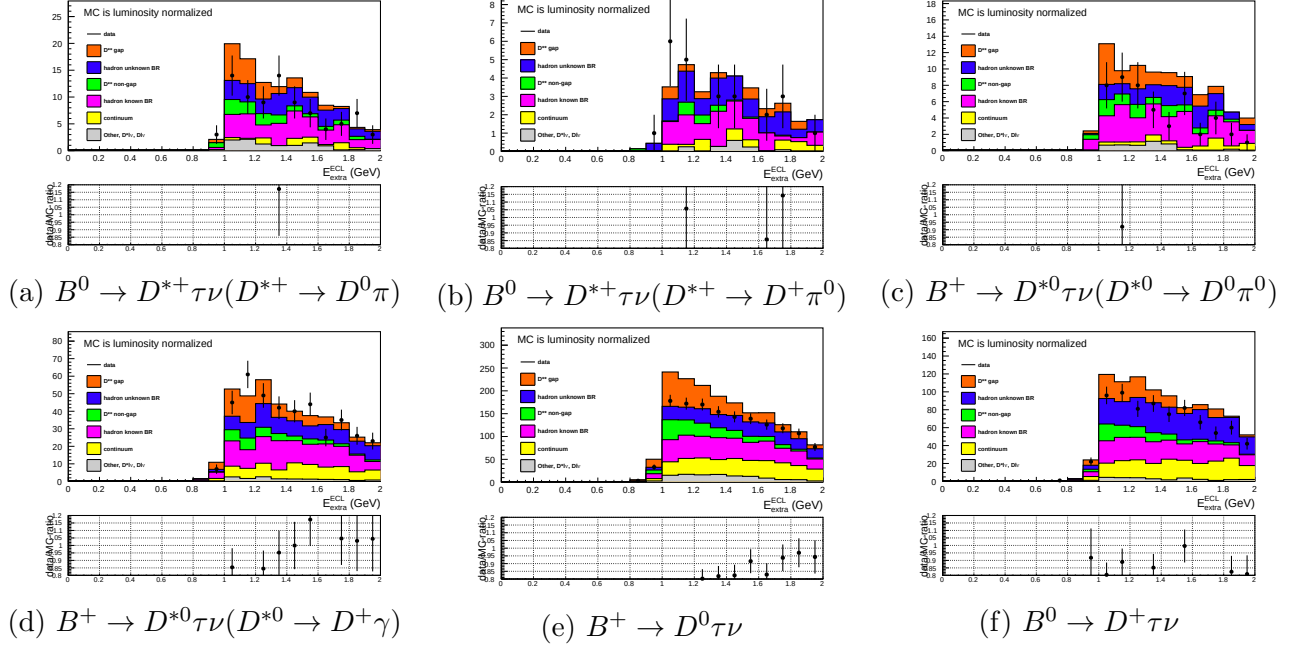


Figure 5.11: Distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode in the π^0 veto side-band. The "D** gap" and "hadron unmeasured BR" have about 100% error of their branching ratio.

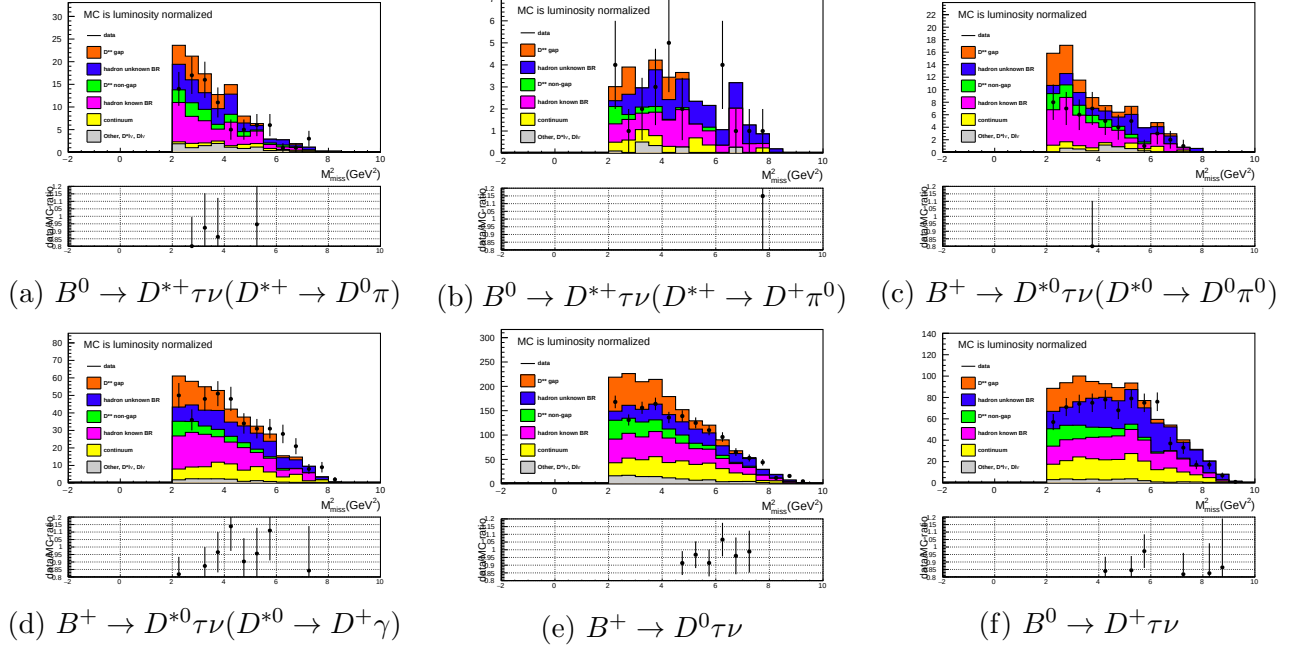


Figure 5.12: Distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode in the π^0 veto side-band. The "D** gap" and "hadron unmeasured BR" have about 100% error of their branching ratio.

Table 5.6: Data/MC comparison of the number of selected events in the $D^{(*)}$ mass sideband with $M_{\text{miss}}^2 > 2\text{GeV}^2$ and $E_{\text{extra}}^{\text{ROE}} > 1 \text{ GeV}$. The error shows statistical error of data. Systematic error is not included in the error.

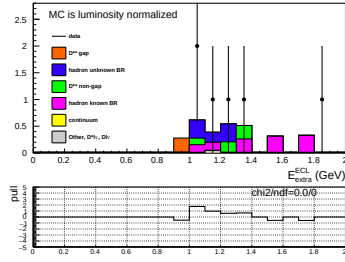
Category	The number of selected events					
	$D^{*+} \rightarrow D^0\pi^+$	$D^{*+} \rightarrow D^+\pi^0$	$D^{*0} \rightarrow D^0\pi^0$	$D^{*0} \rightarrow D^0\gamma$	D^0	D^+
Data	6	2	15	89	58	68
($\pm$ stat.)	± 2	± 1	± 4	± 9	± 8	± 8
MC $B \rightarrow D^{**}\ell\nu$, gap mode	0.3	0.7	4.2	32	6.0	6.6
MC hadronic B decay, unmeasured BR	0.9	1.7	2.2	14	7.0	19
MC $B \rightarrow D^{**}\ell\nu$, non-gap mode	0.6	0.5	1.6	9.9	4.6	4.7
MC hadronic B decay, measured BR	0.9	1.2	4.6	23	8.4	15
MC continuum	0	1.5	3.6	18	18	13
MC others, $B \rightarrow D^{(*)}\ell\nu$	0	0.3	3.6	19	7.9	4.7
MC Total	2.7	5.9	19.8	117	51.5	63.3
Data/MC (stat.)	2.3 ± 0.92	0.34 ± 0.24	0.76 ± 0.20	0.76 ± 0.08	1.13 ± 0.15	1.07 ± 0.13

Table 5.7: Data/MC comparison of the number of selected events in the $D^{(*)}$ mass sideband with $M_{\text{miss}}^2 < 1\text{GeV}^2$. The error shows statistical error of data. Systematic error is not included in the error.

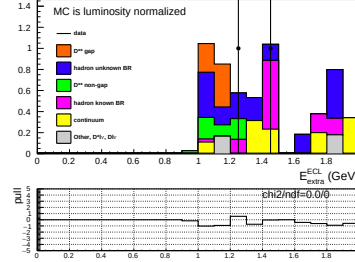
Category	The number of selected events)					
	$D^{*+} \rightarrow D^0\pi^+$	$D^{*+} \rightarrow D^+\pi^0$	$D^{*0} \rightarrow D^0\pi^0$	$D^{*0} \rightarrow D^0\gamma$	D^0	D^+
Data	48	46	206	1953	383	305
Total MC	44.0	42.2	237	1836	387	289
Data/MC ratio	1.09	1.09	0.870	1.06	0.99	1.06
(stat. error)	± 0.16	± 0.16	± 0.06	± 0.02	± 0.05	± 0.06

5.3 $D^{(*)}$ mass sideband

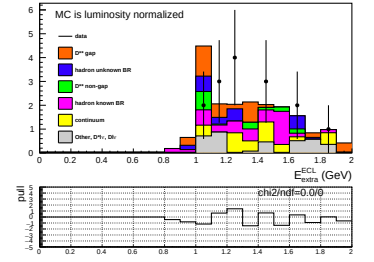
In order to check PDF shape of wrongly reconstructed $D^{(*)}$, events outside of $D^{(*)}$ mass window ($D^{(*)}$ mass sideband) are used. The sideband is separated into two regions: one requires $M_{\text{miss}}^2 > 2\text{GeV}^2$ and $E_{\text{extra}}^{\text{ROE}} > 1 \text{ GeV}$ to enhance background, another requires $M_{\text{miss}}^2 < 1\text{GeV}^2$ to enhance normalization mode. Table 5.6, Figure 5.13-5.14 show the $E_{\text{extra}}^{\text{ROE}}$ and M_{miss}^2 distributions in the background enhanced region. The same tendency is seen as the π^0 veto side-band and data suggest a configuration with a substantially smaller fraction of gap modes. Table 5.7, Figure 5.15-5.16 show the $E_{\text{extra}}^{\text{ROE}}$ and M_{miss}^2 in the signal enhanced region. The same tendency is seen in the q^2 sideband and data/MC agrees within their errors.



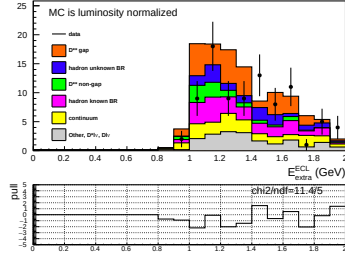
(a) $B^0 \rightarrow D^{*+} \tau \nu (D^{*+} \rightarrow D^0 \pi^0)$



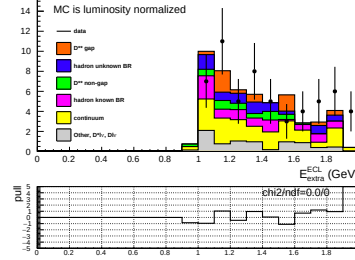
(b) $B^0 \rightarrow D^{*+} \tau \nu (D^{*+} \rightarrow D^+ \pi^0)$



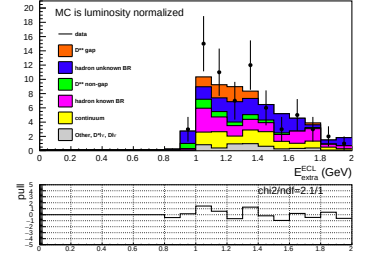
(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(d) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^+ \gamma)$

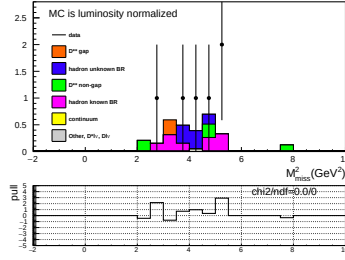


(e) $B^+ \rightarrow D^0 \tau \nu$

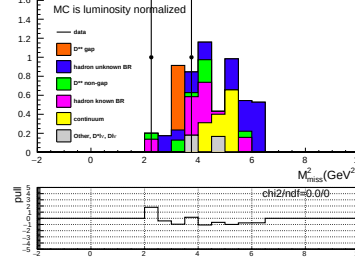


(f) $B^0 \rightarrow D^+ \tau \nu$

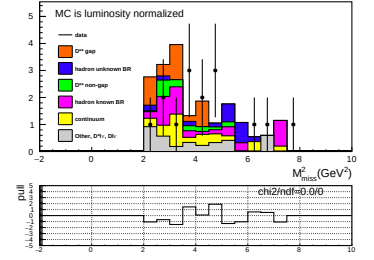
Figure 5.13: Distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode in the $D^{(*)}$ mass side-band with $M_{\text{miss}}^2 > 2 \text{ GeV}^2$, $E_{\text{extra}}^{\text{ROE}} > 1 \text{ GeV}$.



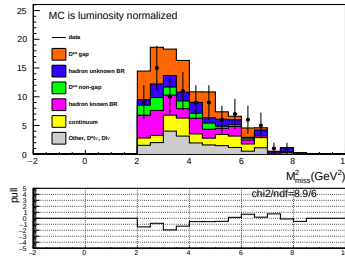
(a) $B^0 \rightarrow D^{*+} \tau \nu (D^{*+} \rightarrow D^0 \pi^0)$



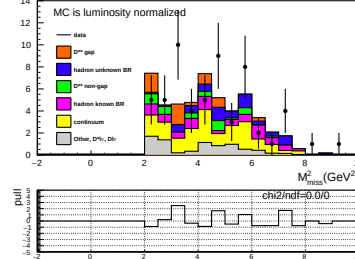
(b) $B^0 \rightarrow D^{*+} \tau \nu (D^{*+} \rightarrow D^+ \pi^0)$



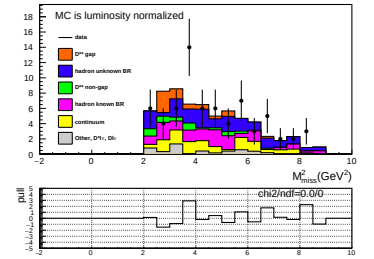
(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(d) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^+ \gamma)$



(e) $B^+ \rightarrow D^0 \tau \nu$



(f) $B^0 \rightarrow D^+ \tau \nu$

Figure 5.14: Distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode in the $D^{(*)}$ mass side-band with $M_{\text{miss}}^2 > 2 \text{ GeV}^2$, $E_{\text{extra}}^{\text{ROE}} > 1 \text{ GeV}$.

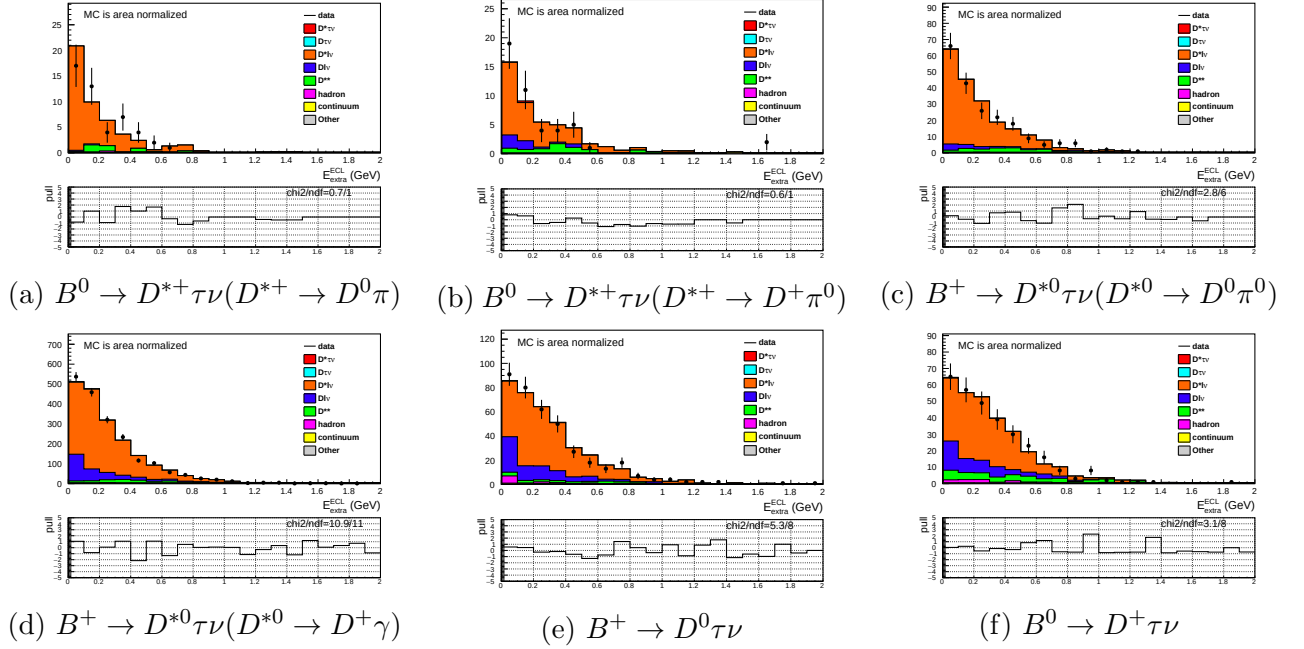


Figure 5.15: Distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode in the $D^{(*)}$ mass side-band with $M_{\text{miss}}^2 < 1 \text{ GeV}^2$.

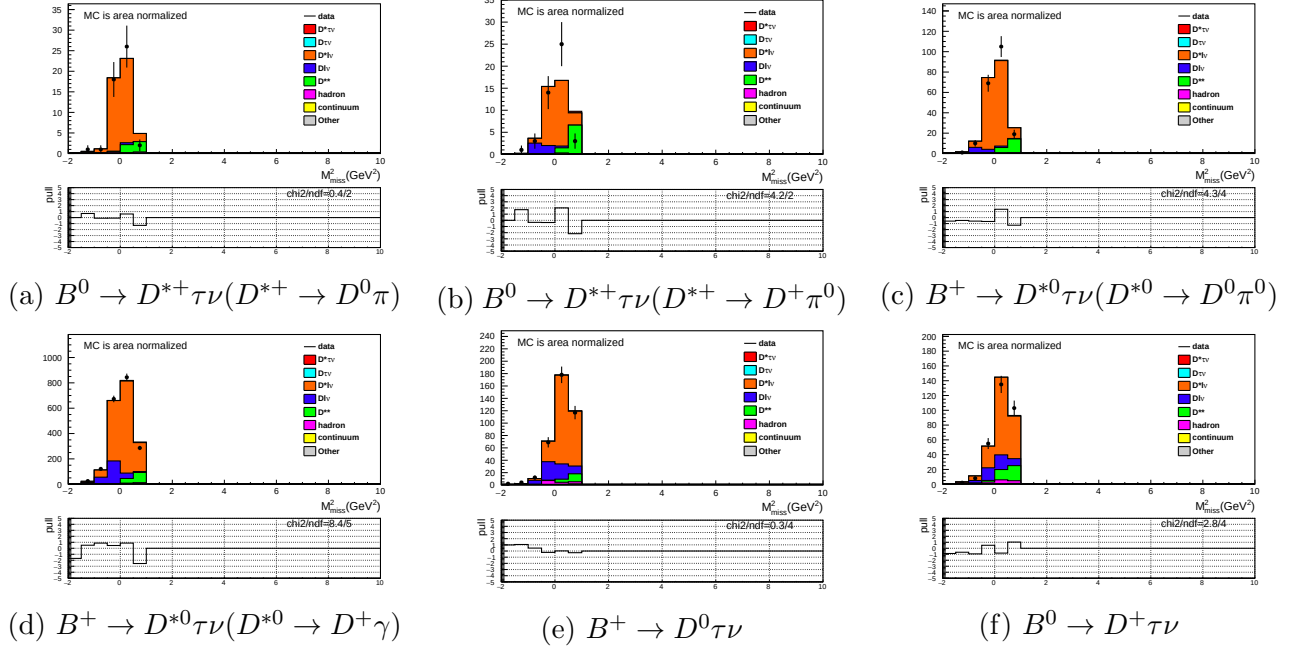


Figure 5.16: Distributions of the M_{miss}^2 in each $D^{(*)}$ decay mode in the $D^{(*)}$ mass side-band with $M_{\text{miss}}^2 < 1 \text{ GeV}^2$.

6 Signal extraction

6.1 Sample and PDF category

$R(D^*)$ and $R(D)$ are extracted through a two-dimensional extended binned maximum likelihood fit. The two-dimensional probability density functions (PDFs) are constructed independently in each of five fit samples:

- $B_{\text{sig}}^0 \rightarrow D^{*-}\ell^+$, where $\ell = e$ or μ , $D^{*-} \rightarrow D^0\pi^-$ or $D^{*-} \rightarrow D^-\pi^0$
- $B_{\text{sig}}^+ \rightarrow \bar{D}^{*0}\ell^+$, where $\ell = e$ or μ , $D^{*0} \rightarrow D^0\pi^0$
- $B_{\text{sig}}^+ \rightarrow \bar{D}^{*0}\ell^+$, where $\ell = e$ or μ , $D^{*0} \rightarrow D^0\gamma$
- $B_{\text{sig}}^0 \rightarrow D^-\ell^+$, where $\ell = e$ or μ
- $B_{\text{sig}}^+ \rightarrow \bar{D}^0\ell^+$, where $\ell = e$ or μ .

The latter three samples are newly added in this analysis. The $D^{*-} \rightarrow D^0\pi^-$ and $D^{*-} \rightarrow D^-\pi^0$ are combined due to low statistics of the $D^{*-} \rightarrow D^-\pi^0$. A simultaneous fit is performed across the five samples. The variables used in the fit are $E_{\text{extra}}^{\text{ECL}}$ and M_{miss}^2 . Their binnings are:

- $E_{\text{extra}}^{\text{ECL}}$:
[0.00,0.05,0.10,0.20,0.3,0.4,0.6,0.8,1.0,2.0] GeV
- M_{miss}^2 :
[-2,-0.5,0.0,0.5,1.0,1.5,2.0,3.0,4.0,5.0,6.0,7.0,8.0,10.0] GeV^2 .

The number of bins in each sample is $9 \times 13 = 117$. The binnings are optimized to improve $R(D^{*})$ sensitivity, while keeping small, less than 2%, fitter bias from MC statistics uncertainty. The binning is updated again in version 2.8 of the note, because MC statistics error is increased by separating gap mode template from D^{**} modes. Narrow bin width are set around the peak of signal mode, and coarse bin width are set low statistic region. In each sample, nine PDF categories are used:

- $B \rightarrow D^*\tau\nu$
- $B \rightarrow D\tau\nu$
- $B \rightarrow D^*\ell\nu$
- $B \rightarrow D\ell\nu$
- $D^{**}\ell\nu$ gap mode
- $B \rightarrow D^{**}\tau, \ell\nu$
- Hadronic B decay
- Continuum

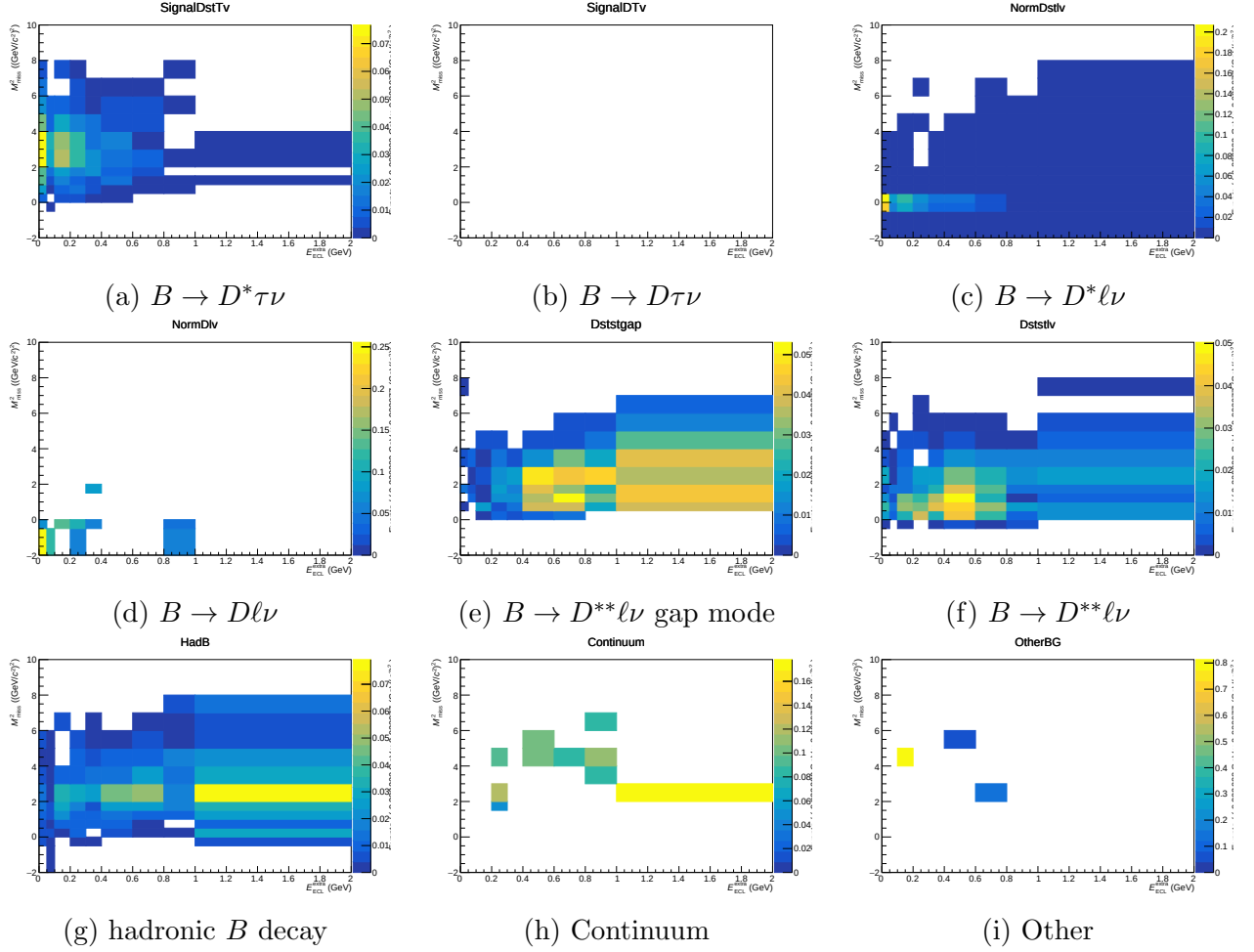


Figure 6.1: PDF of each category in $B^0 \rightarrow D^{*-} \ell^+$ sample. z-axis is probability.

423

• Other

424 they are categorized based on the generated decay mode id, obtained from GenMCSTag-
 425 Tool. "isSignal" and any MCmatching information are not used for the categorization.

426 Figure 6.1 and 1.1–1.4 in appendix A show the two dimensional PDF in each sample
 427 and category. The $B \rightarrow D^* \tau \nu$ and $B \rightarrow D \tau \nu$ signals are peaked at low E_{ECL} and high
 428 M_{miss}^2 in D^* and D samples. The D^* samples are sensitive to the $B \rightarrow D^* \tau \nu$. The D
 429 samples are sensitive to sum of $B \rightarrow D \tau \nu$ and $B \rightarrow D^* \tau \nu$, because there is no large
 430 PDF difference between them. The normalization modes of $B \rightarrow D^* \ell \nu$ and $B \rightarrow D \ell \nu$
 431 are peaked at low E_{ECL} and low M_{miss}^2 with large yield and high purity. The other
 432 backgrounds of D^{**} , hadronic B decays, continuum are distributed at high E_{ECL} and
 433 high M_{miss}^2 regions. Amount of other backgrounds are small and negligible.

Table 6.1: Parameters used for the fit

parameters	floated/fixed	note
$R(D^*)$	floated	one parameter, common for all samples
$R(D)$	floated	one parameter, common for all samples
$BR(B \rightarrow D^* \ell \nu)$	floated	one parameter, common for all samples
$BR(B \rightarrow D \ell \nu)$	floated	one parameter, common for all samples
$N(B \rightarrow D^{**} \tau, \ell \nu)$	floated	independent five parameters in five samples
$N(\text{gap mode})$	floated	independent five parameters in five samples
$N(\text{hadronic } B)$	floated	independent five parameters in five samples
$N(\text{continuum})$	fixed	independent five parameters in five samples
$N(\text{other})$	fixed	independent five parameters in five samples
$\varepsilon_{D^* \tau \nu}$	fixed	independent five parameters in five samples
$\varepsilon_{D \tau \nu}$	fixed	independent five parameters in five samples
$\varepsilon_{D^* \ell \nu}$	fixed	independent five parameters in five samples
$\varepsilon_{D \ell \nu}$	fixed	independent five parameters in five samples
f_{00}	floated	one parameter, common for all samples
f_{+-}	floated	one parameter, common for all samples
$N_{B\bar{B}}$	floated	one parameter, common for all samples

6.2 Fit parameter

Table 6.1 summarizes parameters used for the fit. $R(D^*)$ is described using the number of reconstructed signal and normalization events and reconstruction efficiencies,

$$R(D^{(*)}) = \frac{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \tau^- \nu)}{\mathcal{B}(\bar{B} \rightarrow D^{(*)} \ell^- \nu)}, \quad (\ell = e, \mu) \quad (6.1)$$

$$= \frac{N_{D^{(*)} \tau \nu}}{(N_{D^{(*)} \ell \nu} / 2)} \cdot \frac{\varepsilon_{D^{(*)} \ell \nu}}{\varepsilon_{D^{(*)} \tau \nu}}, \quad (6.2)$$

where $N_{D^{(*)} \tau(\ell) \nu}$ is the observed number of $\bar{B} \rightarrow D^{(*)} \tau^- \bar{\nu}_\tau$ ($D^{(*)} \ell^- \bar{\nu}_\ell$) candidates in the data and $\varepsilon_{D^{(*)} \tau(\ell) \nu}$ is reconstruction efficiency of reconstructed $\bar{B} \rightarrow D^{(*)} \tau^- \bar{\nu}_\tau$ ($D^{(*)} \ell^- \bar{\nu}_\ell$) candidates. The factor 2 in the denominator averages the summed yield from two light leptons, $\ell \in \{e, \mu\}$. Here, the reconstruction efficiencies are defined as follows,

$$\varepsilon_{D^{(*)} \tau(\ell) \nu} = \frac{N_{D^{(*)} \tau(\ell) \nu}^{\text{rec}}}{N_{D^{(*)} \tau(\ell) \nu}^{\text{gen}}}, \quad (6.3)$$

where $N_{D^{(*)} \tau(\ell) \nu}^{\text{rec}}$ and $N_{D^{(*)} \tau(\ell) \nu}^{\text{gen}}$ are the number of reconstructed and generated $\bar{B} \rightarrow D^{(*)} \tau^- \bar{\nu}_\tau$ ($D^{(*)} \ell^- \bar{\nu}_\ell$) decays, respectively. $N_{D^{(*)} \tau(\ell) \nu}^{\text{gen}}$ is derived by Eq. (6.4) and Eq. (6.5),

$$N_{D^{(*)} \tau \nu}^{\text{gen}} = N_{B\bar{B}}^{\text{gen}} \cdot 2 \mathcal{B}(\bar{B} \rightarrow D^{(*)} \tau^- \bar{\nu}_{\text{tau}})^{\text{gen}} \quad (6.4)$$

$$N_{D^{(*)} \ell \nu}^{\text{gen}} = N_{B\bar{B}}^{\text{gen}} \cdot 2 [\mathcal{B}(\bar{B} \rightarrow D^{(*)} e^- \bar{\nu}_e)^{\text{gen}} + \mathcal{B}(\bar{B} \rightarrow D^{(*)} \mu^- \bar{\nu}_\mu)^{\text{gen}}]. \quad (6.5)$$

Here $N_{B\bar{B}}^{\text{gen}}$ is the number of $B\bar{B}$ pairs generated in the simulation, and $\mathcal{B}(B \rightarrow X)^{\text{gen}}$ refers to a simulated branching fraction for $B \rightarrow X$ decay. The number of generated $\bar{B} \rightarrow D^{(*)} \ell^- \bar{\nu}_\ell$ decays is twice the individual counts from one of the $B\bar{B}$ pair, which is represented by the factor of 2 in Eq. (6.5). We assume $R(D^{(*)})$ holds a consistent value

Table 6.2: The fixed or constrained input parameters for the fitting.

Parameter		Values				
f_{00}		0.4861 ± 0.0080				
f_{+-}		0.5113 ± 0.0108				
$N_{B\bar{B}}$	$[10^6]$	387.1 ± 5.6				
		D^{*+}	$D^{*0} \rightarrow D^0\pi^0$	$D^{*0} \rightarrow D^0\gamma$	D^0	D^+
$\varepsilon_{D^*\tau\nu}$	$[10^{-5}]$	1.208	0.623	1.034	3.691	0.616
$\varepsilon_{D\tau\nu}$	$[10^{-5}]$	0	0.028	0.430	4.766	2.018
$\varepsilon_{D^*\ell\nu}$	$[10^{-5}]$	3.408	1.666	3.054	12.506	1.910
$\varepsilon_{D\ell\nu}$	$[10^{-5}]$	0.011	0.020	0.644	8.647	3.677

between B^0 and B^+ decays under the isospin symmetry, setting $R(D^{(*)}) = R(D^{(*)+}) = R(D^{(*)0})$. All five samples take the common value during the fit for the signal extraction. Furthermore, $\mathcal{B}(\bar{B}^0 \rightarrow D^{(*)+}\ell^-\bar{\nu}_\ell)$ is required to be the same in all five samples. The efficiencies and the number of $B\bar{B}$ pairs are estimated from MC and fixed in the fit, as shown in Table 6.2. Effect from their uncertainties are evaluated as systematic errors.

The number of reconstructed backgrounds, $B \rightarrow D^{**}\tau, \ell\nu$ and hadronic B decays, are floated independently in each $D^{(*)}$ mode. The number of continuum and other backgrounds are fixed in the fit and their fluctuation is estimated as systematic error in Sec. 7. Therefore, the fit involves a total of 21 floated parameters.

Systematic parameters are not included in the fit, and nominal PDF and input values are used to determine the best fit point. Effect from their uncertainties are evaluated as systematic error by using toy ensemble and re-fitting method as described in Sec. 7.

6.3 Likelihood function

We perform an extended binned maximum likelihood. The likelihood(L) is defined as

$$L(\vec{n}|\vec{\lambda}) \equiv \prod_{i=1}^{N_{\text{sample}}} \prod_{j=1}^{N_{\text{bin}}} \left[\frac{(\mathcal{P}_{ij})^{n_{ij}}}{n_{ij}!} \cdot e^{-\mathcal{P}_{ij}} \right], \quad (6.6)$$

where n_{ij} represents the observed number of events in the j -th bin for the i -th sample, and $\vec{n}$ represents the collection of these observed counts. $\vec{\lambda}$ consists of 16 fit parameters floated in the fit, which includes $R(D^{(*)})$. The indices i and j run fit samples and bins in two dimensions, up to $N_{\text{sample}} = 5$ and $N_{\text{bin}} = 260$, respectively. The PDF $\mathcal{P}_{ij}$ is constructed as a sum of the PDF components from nine categories:

$$\mathcal{P}_{ij}(\vec{\lambda}) = \sum_{k=1}^{N_{\text{category}}} \left[\nu_i^k(\vec{\lambda}) f_{ij}^k \right]. \quad (6.7)$$

Here, ν_i^k is the expected yield of the k -th category in the i -th sample, and f_{ij}^k corresponds to the PDF component in the j -th bin for this candidate category. Empty bins, where the expected number of MC events is zero, are removed from likelihood calculation.

6.4 Asimov fit sensitivity

Fit is performed to Asimov data assuming $R(D^*) = 0.254$ and $R(D) = 0.300$. When Asimov data is generated, the branching ratios of $B \rightarrow D^{(*)}\ell\nu$ is fixed following Table 4.1, and branching ratios of $B \rightarrow D^{(*)}\tau\nu$ is weighted following the $R(D^{(*)})$ assumptions. The fit results are

$$R(D^*) = 0.254 \pm 0.022(\text{stat.}), \quad (6.8)$$

$$R(D) = 0.300 \pm 0.051(\text{stat.}), \quad (6.9)$$

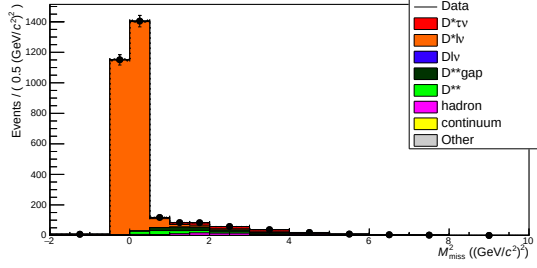
$$\rho = -0.42(\text{stat.}). \quad (6.10)$$

Figure 6.2–6.5 show the projection of the fitted distributions.

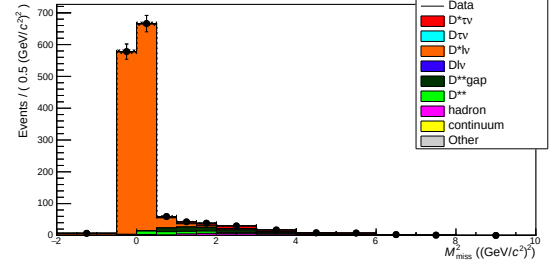
6.5 Fitter validation

Fitter performance is validated by using toy ensemble. 1000 set of toy is generated by fluctuating statistic in each bin assuming Poisson distribution. $R(D^{(*)})$ is fixed to 0.254 and 0.300 in this study. They are fitted and their pull is estimated as shown in Fig. 6.6–6.10. They are consistent with Gaussian of mean 0, sigma 1. In Figure 6.9, the pull of $N(\text{gap})$ differs from 1 about 3σ in D^0 mode. This is newly found after separating gap mode templates from others. The reason is not clearly understood, but one possible explanation is that the templates are similar between gap mode and hadronic B decay in the D^0 mode, as shown in Fig. 1.3. The deviation of the pull is small, 2.3%, and impact to $R(D^{(*)})$ is negligible.

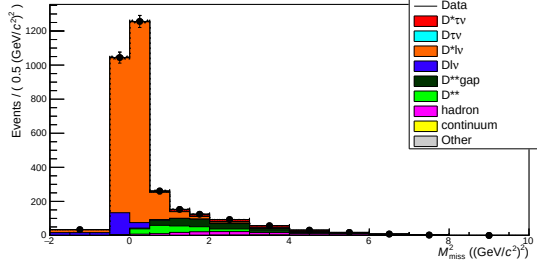
In addition, linearity test is performed by varying $R(D^*) = 0.1 - 0.4$ and $R(D) = 0.1 - 0.5$ with 0.003 and 0.004 intervals. Depending on value of $R(D^*)$ and $R(D)$, small fitter bias is seen, as shown in Fig. 6.11. This will be taken into systematic error in Sec. 7.



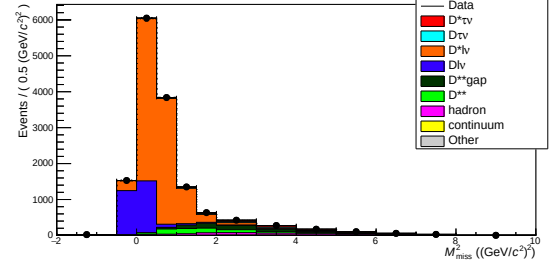
(a) $B^0 \rightarrow D^{*+} \tau \nu$



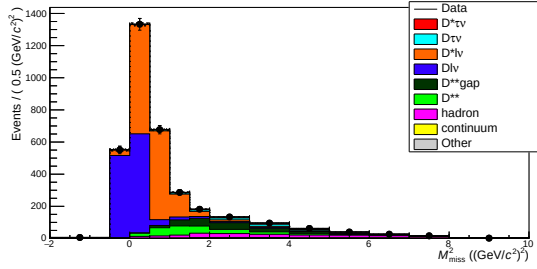
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

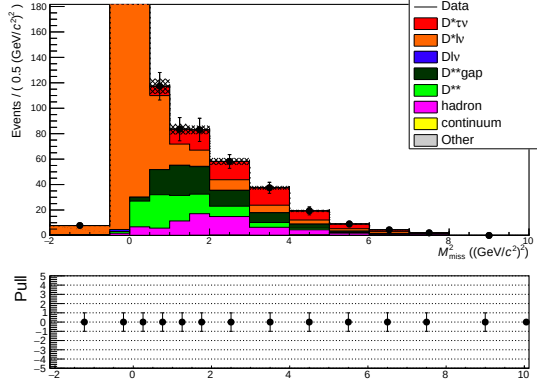


(d) $B^+ \rightarrow D^0 \tau \nu$

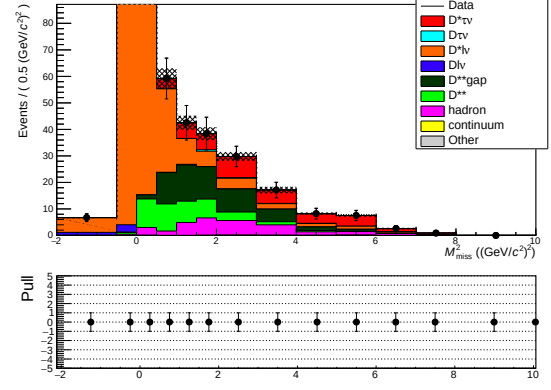


(e) $B^0 \rightarrow D^+ \tau \nu$

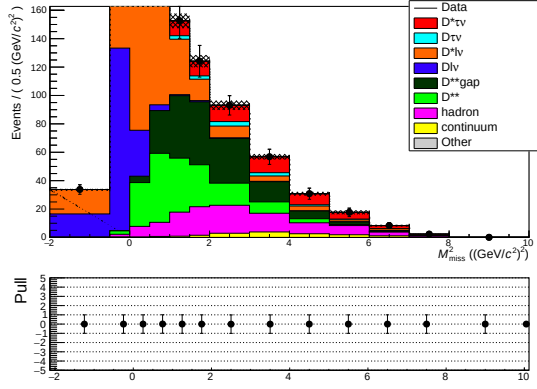
Figure 6.2: Asimov fit projection to the M_{miss}^2 in each sample. Data point shows Asimov data assuming $R(D^*) = 0.254$ and $R(D) = 0.300$.



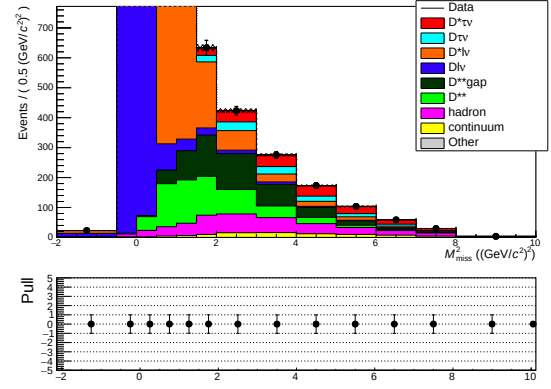
(a) $B^0 \rightarrow D^{*+} \tau \nu$



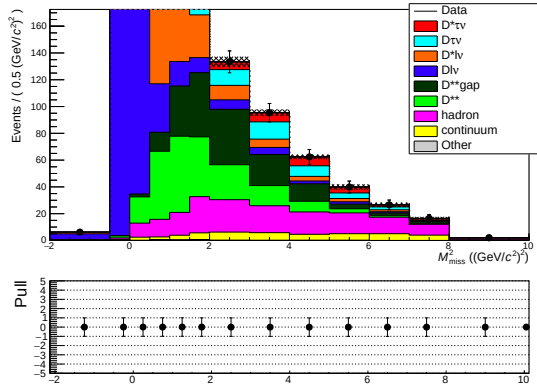
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

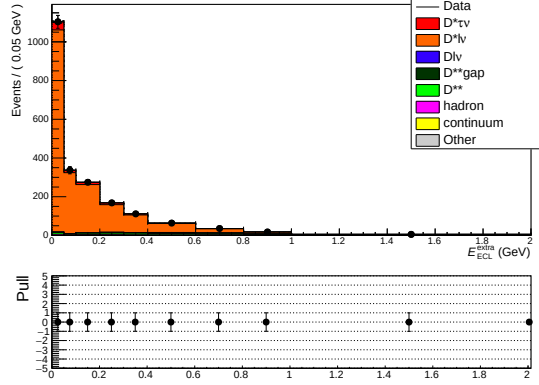


(d) $B^+ \rightarrow D^0 \tau \nu$

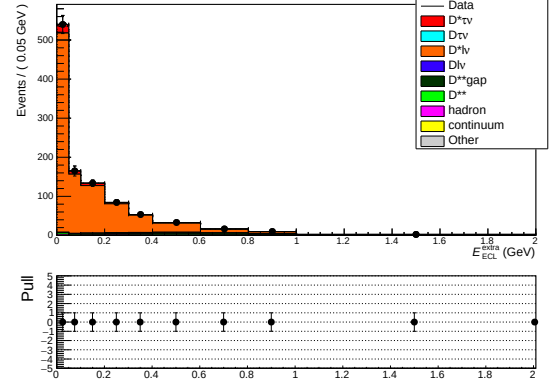


(e) $B^0 \rightarrow D^+ \tau \nu$

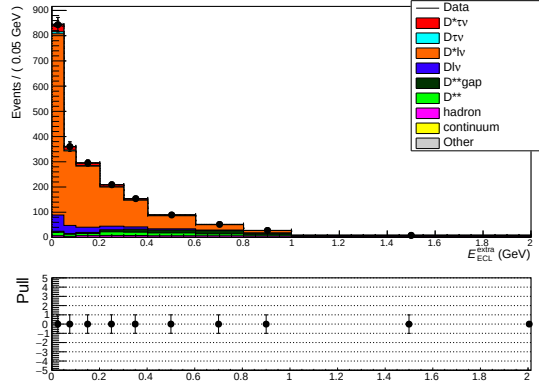
Figure 6.3: Asimov fit projection to the M_{miss}^2 in each sample. Data point shows Asimov data assuming $R(D^*) = 0.254$ and $R(D) = 0.300$. y axis is zoomed.



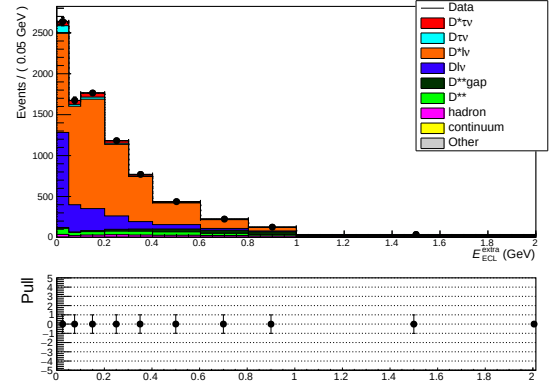
(a) $B^0 \rightarrow D^{*+} \tau \nu$



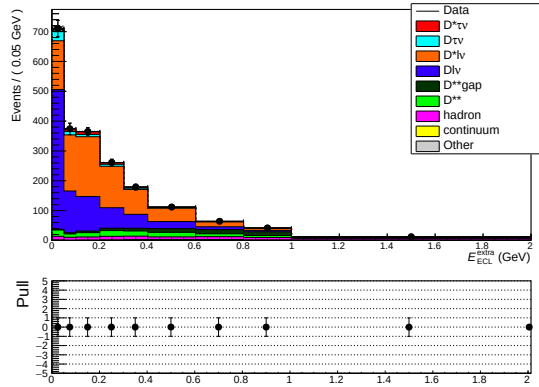
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

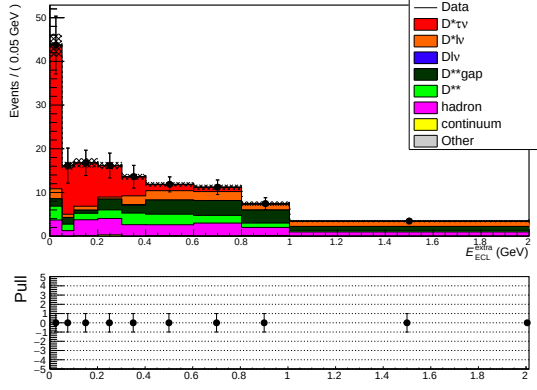


(d) $B^+ \rightarrow D^0 \tau \nu$

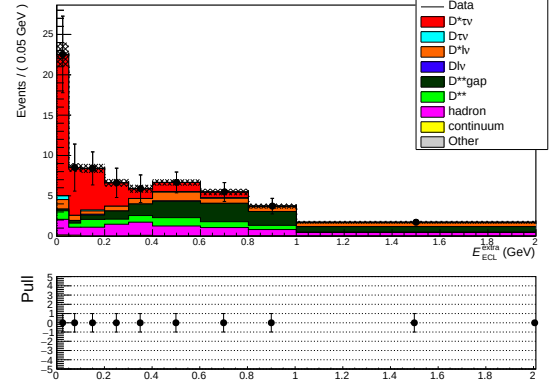


(e) $B^0 \rightarrow D^+ \tau \nu$

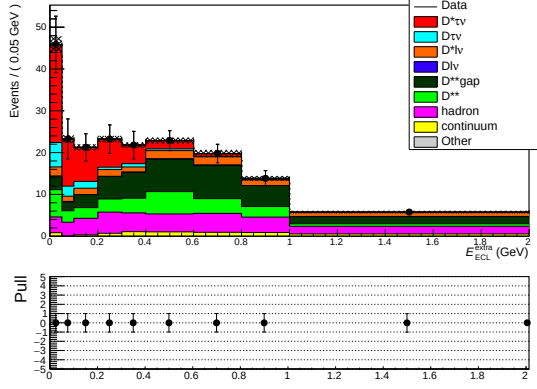
Figure 6.4: Asimov fit projection to the $E_{\text{extra}}^{\text{ECL}}$ in each sample. Data point shows Asimov data assuming $R(D^*) = 0.254$ and $R(D) = 0.300$.



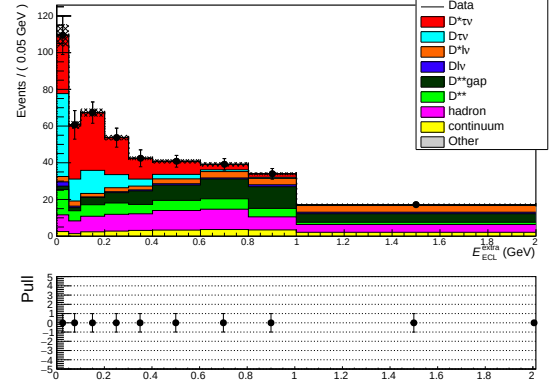
(a) $B^0 \rightarrow D^{*+} \tau \nu$



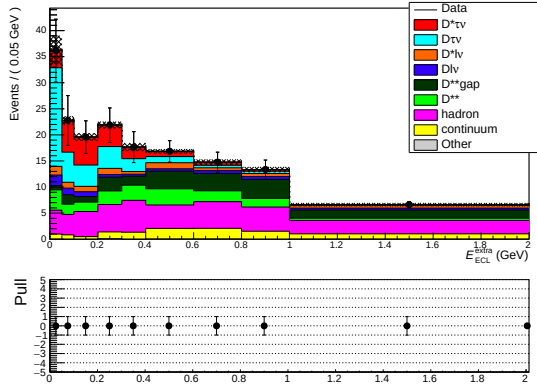
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$



(d) $B^+ \rightarrow D^0 \tau \nu$



(e) $B^0 \rightarrow D^+ \tau \nu$

Figure 6.5: Asimov fit projection to the $E_{\text{extra}}^{\text{ECL}}$ at $1.5 \text{ GeV}^2 < M_{\text{miss}}^2$ ($3.0 \text{ GeV}^2 < M_{\text{miss}}^2$) for $D^*(D)$ sample in each sample. Data point shows Asimov data assuming $R(D^*) = 0.254$ and $R(D) = 0.300$.

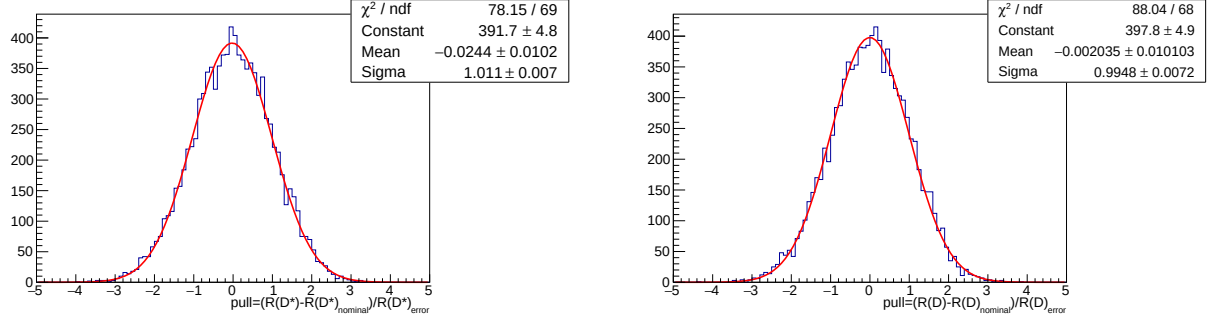


Figure 6.6: Pull of $R(D^*)$ (left) and $R(D)$ (right) with 10000 toys. Only statistical error is fluctuated and other parameters are fixed for toy generation. f_{00} and N_{BB} are fixed for the fitting.

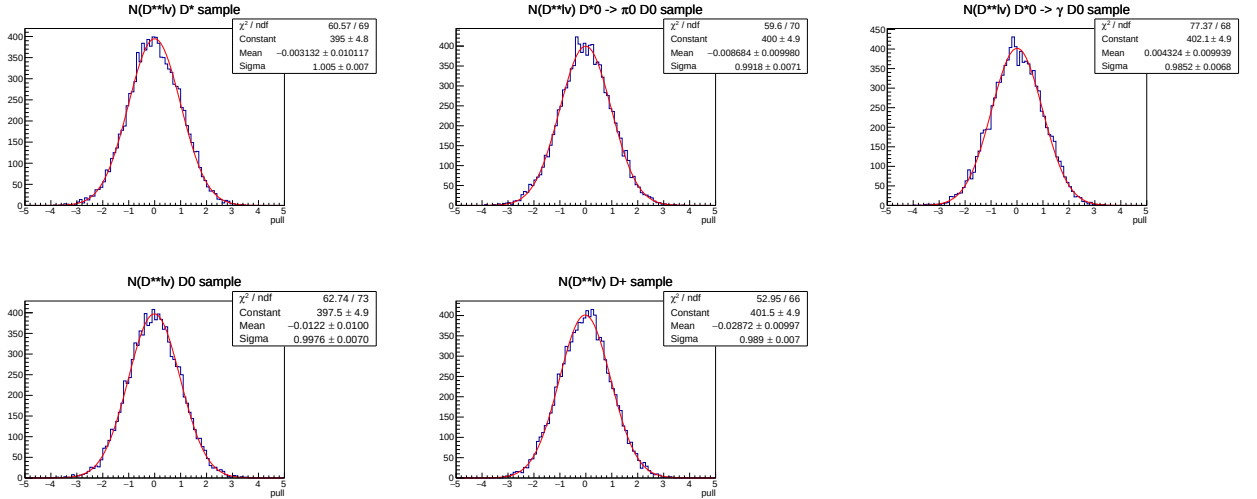


Figure 6.7: Pull of nuisance parameters, $N(B \rightarrow D^{**} \ell \nu)$, with 10000 toys. Only statistical error is fluctuated and other parameters are fixed for toy generation. f_{00} and N_{BB} are fixed for the fitting.

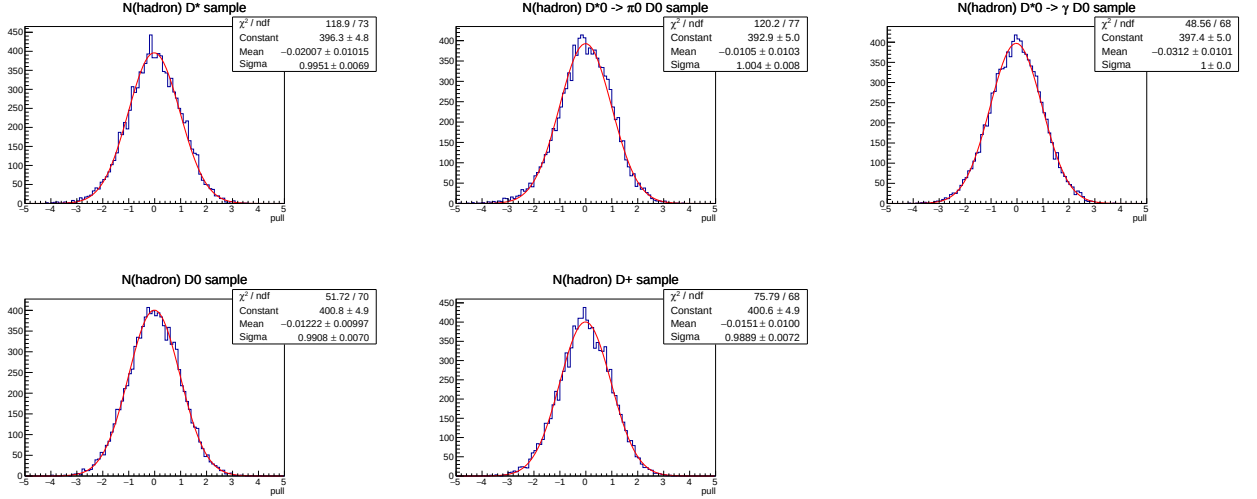


Figure 6.8: Pull of nuisance parameters, $N(\text{hadron})$, with 10000 toys. Only statistical error is fluctuated and other parameters are fixed for toy generation. f_{00} and N_{BB} are fixed for the fitting.

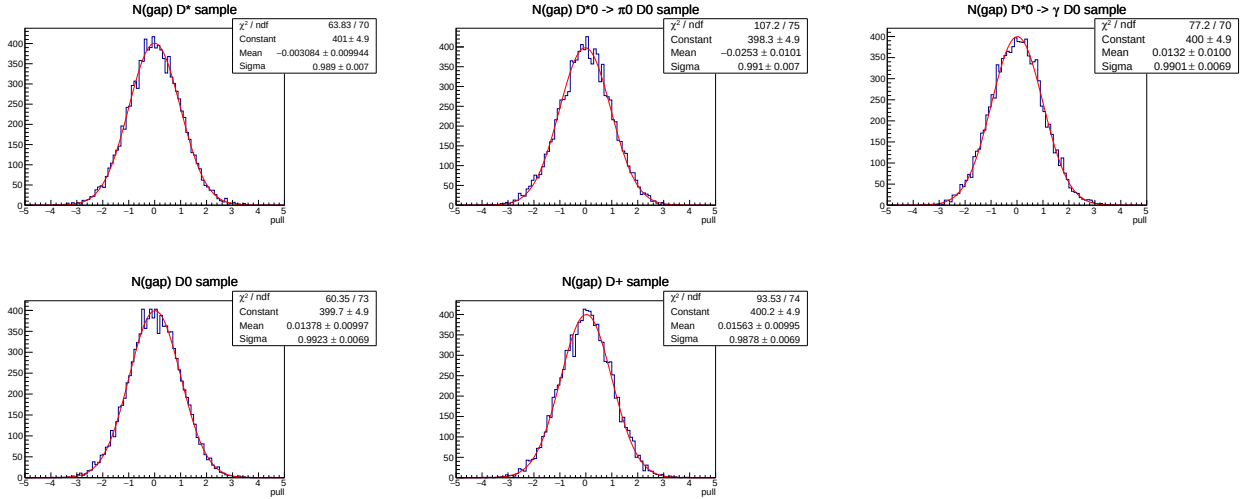


Figure 6.9: Pull of nuisance parameters, $N(\text{gap})$, with 10000 toys. Only statistical error is fluctuated and other parameters are fixed for toy generation. f_{00} and N_{BB} are fixed for the fitting.

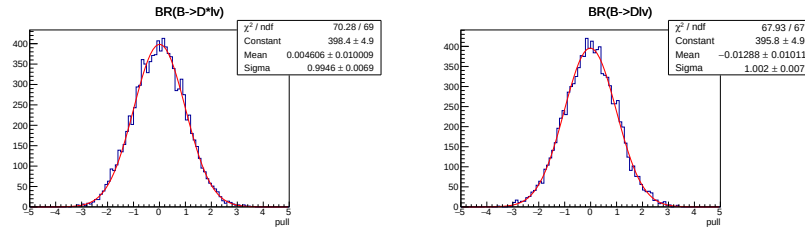


Figure 6.10: Pull of nuisance parameters, $BR(B \rightarrow D^{(*)} \ell \nu)$, with 10000 toys. Only statistical error is fluctuated and other parameters are fixed for toy generation. f_{00} and N_{BB} are fixed for the fitting.

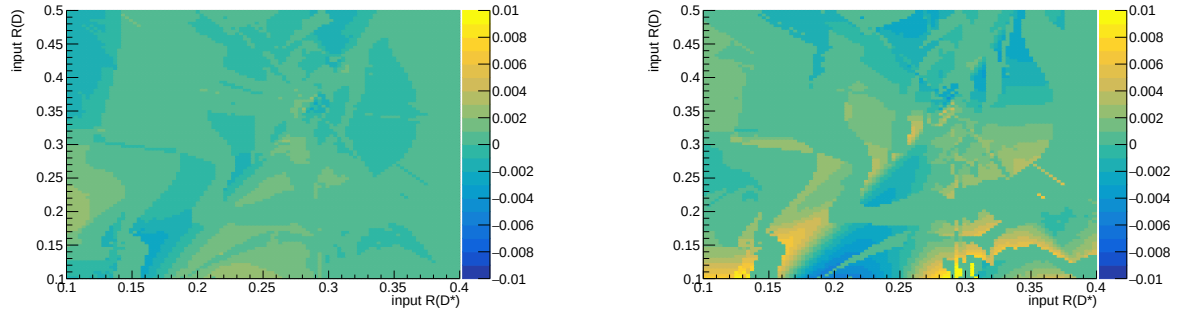


Figure 6.11: Fitter bias of $R(D^*)$ (left) and $R(D)$ (right) with Asimov fit. x and y axis are input value of $R(D^*)$ and $R(D)$, and z axis is the bias defined as $(R(D^{(*)}) - R(D^{(*)})_{\text{input}})/R(D^{(*)})_{\text{input}}$.

Table 7.1: Summary of the systematic uncertainties.

Error source	$R(D^*)$	$R(D)$	ρ
$\bar{B} \rightarrow D^{**}\ell^-\bar{\nu}_\ell$ branching fractions	0.6%	1.3%	0.02
MC statistics	4.2%	8.2%	-0.40
gap mode branching fractions $\pm 100\%$	0.1-6.5%	0.1-6.3%	0.0
gap mode branching fractions	1.6%	2.1%	-0.49
Hadronic B decay branching fractions	1.7%	2.9%	0.06
Continuum background	1.4%	1.9%	0.95
Slow π^0, γ efficiency	1.5%	3.0%	0.99
Other efficiency corrections	1.0%	1.0%	0.80
FEI efficiency of data	0.3%	1.1%	-1.0
FEI efficiency of $B \rightarrow D\tau\nu$	0.1%	2.4%	1.0
M_{miss}^2 resolution	0.3%	0.9%	0.55
Fraction of fake $D^{(*)}$	0.7%	1.4%	0.08
Fitter bias	0.3%	1.4%	0.00
Form factors correction	0.4%	1.0%	-0.34
Total systematic uncertainty	5.1-8.3%	10.6-12.4%	-0.20

Table 7.2: Break down of other efficiency corrections.

Error source	$R(D^*)$	$R(D)$	ρ
FEI efficiency calibration	0.9%	1.0%	0.80
Tracking	<0.1%	<0.1%	-0.12
Lepton PID	0.3%	0.5%	0.64
Hadron PID	0.8%	0.8%	-0.87
Slow π	0.3%	0.8%	0.97
Slow π^0, γ	1.5%	3.0%	0.99
π^0 efficiency	0.7%	0.3%	0.84

7 Systematics uncertainties

Table 7.1 shows a summary of systematic uncertainties. We account for uncertainties in detector responses, underlying physics processes, and discrepancies between the data and MC simulation. Details of each error source will be explained in each sub-section. The small errors ($< 0.2\%$) are not listed in the table but explained in the sub-sections.

We generated a Asimov data, assuming $R(D^*) = 0.254$ and $R(D) = 0.300$ without systematic uncertainties. For systematic evaluation, we fixed the Asimov data to the nominal expectation and refit it with new PDF templates and efficiency, which are fluctuated according to the systematic uncertainties. After data is unblinded, the Asimov data will be replaced with data.

7.1 $\bar{B} \rightarrow D^{**}\ell^-\bar{\nu}_\ell$ and gap mode branching fractions

The branching fractions of $\bar{B} \rightarrow D^{**}\ell^-\bar{\nu}_\ell$ decays and gap modes listed in Table 4.6 are fluctuated following their errors to generate 1000 toys. Branching ratio of each decay

Table 7.3: Change of $R(D^{(*)})$ by varying branching ratio of $B \rightarrow D^{(*)}\eta\ell\nu$ by $\pm 100\%$.

Error source	$R(D^*)$	$R(D)$
$\text{BR}(\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.5%	0.6%
$\text{BR}(\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) - 100\%$	6.5%	4.0%
$\text{BR}(\bar{B}^0 \rightarrow D_1^{' +}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.4%	0.1%
$\text{BR}(\bar{B}^0 \rightarrow D_1^{' +}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) - 100\%$	0.7%	6.3%
$\text{BR}(B^- \rightarrow D_0^{*0}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.6%	0.1%
$\text{BR}(B^- \rightarrow D_0^{*0}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) - 100\%$	4.7%	0.2%
$\text{BR}(B^- \rightarrow D_1^{' 0}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.4%	0.2%
$\text{BR}(B^- \rightarrow D_1^{' 0}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) - 100\%$	1.5%	3.2%

Table 7.4: Change of $R(D^{(*)})$ by varying branching ratio of $B \rightarrow D^{(*)}\eta\ell\nu$ by $\pm 100\%$. We used another Asimov data without gap mode. PDF template is nominal.

Error source	$R(D^*)$	$R(D)$
$\text{BR}(\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.1%	0.1%
$\text{BR}(\bar{B}^0 \rightarrow D_0^{*+}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) - 100\%$	0.1%	0.1%
$\text{BR}(\bar{B}^0 \rightarrow D_1^{' +}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.1%	0.1%
$\text{BR}(\bar{B}^0 \rightarrow D_1^{' +}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) - 100\%$	0.1%	0.1%
$\text{BR}(B^- \rightarrow D_0^{*0}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.1%	0.1%
$\text{BR}(B^- \rightarrow D_0^{*0}(\rightarrow D\eta)\ell^-\bar{\nu}_\ell) - 100\%$	0.1%	0.1%
$\text{BR}(B^- \rightarrow D_1^{' 0}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) + 100\%$	0.1%	0.1%
$\text{BR}(B^- \rightarrow D_1^{' 0}(\rightarrow D^*\eta)\ell^-\bar{\nu}_\ell) - 100\%$	0.1%	0.1%

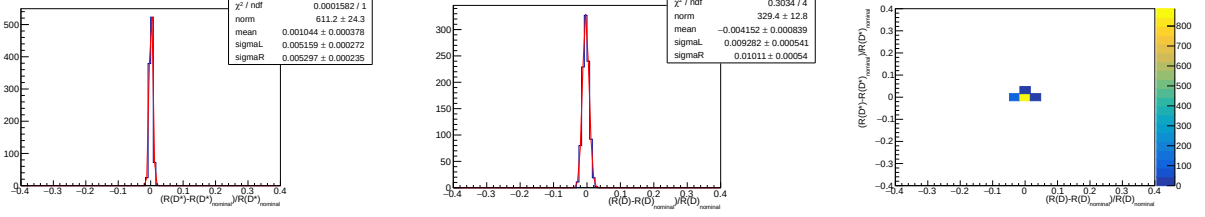


Figure 7.1: Comparison of Asimov fit results with nominal PDF and varied PDF with $D^{**}\ell\nu$ branching fractions uncertainty, except for $B \rightarrow D^{(*)}\eta\ell\nu$.

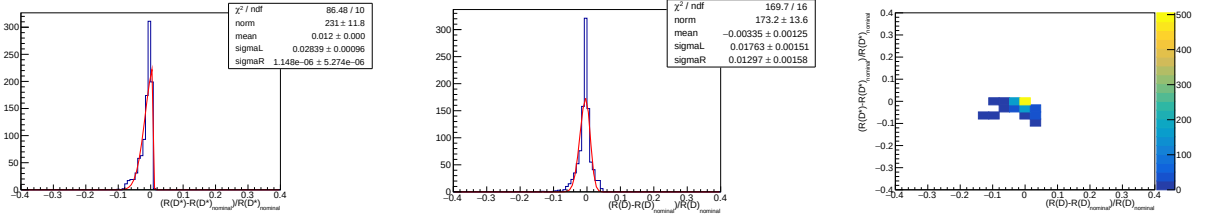


Figure 7.2: Comparison of Asimov fit results with nominal PDF and varied PDF with $D^{(*)}\eta\ell\nu$ branching fractions uncertainty.

mode is randomly fluctuated independently following Gaussian function. If the measured branching ratio is not available, it is fluctuated $\pm 100\%$ by uniform function. For each toy, PDF is regenerated and Asimov fit is performed as shown in Fig. 7.1-7.2. Because of large uncertainty, $B \rightarrow D^{(*)}\eta\ell\nu$ is plotted separately. The $R(D)$ ($R(D^{*})$) fit results of 1000 toys are fitted by double-side Gaussian. When mean of the double-side gaussian is 0, the larger sigma of the double-side gaussians is assigned as a systematic error of $R(D^{*})$. When the mean is different from 0 more than 1.5σ of fitting error, the larger one of $\text{abs}(\text{mean} - \text{lower sigma})$ or $\text{abs}(\text{mean} + \text{higher sigma})$ is assigned as a systematic error. Pearson's correlation coefficient (ρ) of $R(D^{*})$ and $R(D)$ is estimated by using the toys. This toy based method will be used to estimate other systematic uncertainties in later sub-sections.

Additional case study is performed in Table 7.3. Branching ratio of $B \rightarrow D^{(*)}\eta\ell\nu$ $\pm 100\%$ and Asimov fit is performed. Because gap mode is separated from D^{**} decays, PDF shape is not changed by enlarging the BR by $+100\%$, then impact to the $R(D^{*})$ is small. When the BR is changed by -100% , PDF shape is changed largely and the impact is large. Table 7.4 shows the same study with different Asimov data, by setting branching ratio of gap mode to zero, which is indicated by sideband. Only fitted data is replaced with the different Asimov data, and PDF templates are nominal one. As expected, the impact to $R(D^{*})$ is small, because no gap mode is found in the fitted data.

7.2 Hadronic B decay branching fractions

The branching fractions of hadronic B decays are fluctuated following their errors in Table 4.5 to generate 1000 toys. Branching ratio of each decay mode is fluctuated independently by assuming Gaussian function. If the measured branching ratio is not available on PDG, they are fluctuated $\pm 100\%$ by uniform function. The branching fractions for each toy, PDF is regenerated and Asimov fit is performed as shown in Fig. 7.3. They are

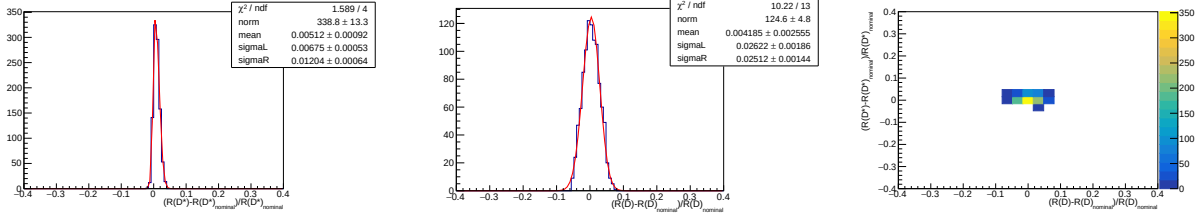


Figure 7.3: Comparison of Asimov fit results with nominal PDF and varied PDF with hadronic B decay branching fractions uncertainty.

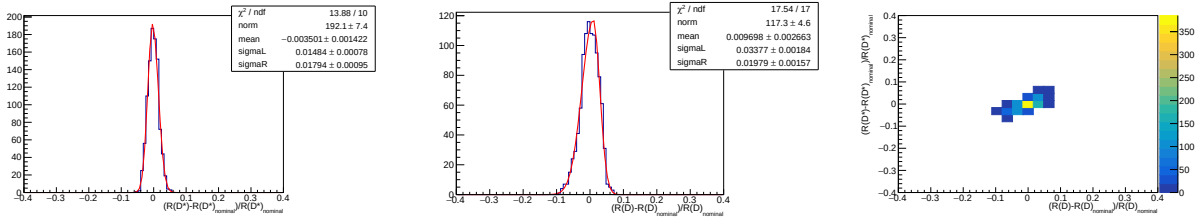


Figure 7.4: Comparison of Asimov fit results with nominal PDF and varied PDF with performance corrections uncertainty.

536 fitted by double-side Gaussian then systematic error is estimated.

537 7.3 FEI Efficiency correction

538 The number of reconstructed B_{tag} mode at q^2 side-band is different from MC expectation,
 539 as shown in Fig. 5.4. The MC is weighted by the data/MC ratio in each decay mode
 540 to regenerate PDF and signal efficiency. The difference of Asimov fit results between
 541 nominal PDF and regenerated PDF is taken as a systematic error.

542 7.4 Other efficiency corrections

543 Uncertainty of below correction in Sec.4.1 are estimated:

- 544 • hadron PID correction
- 545 • lepton PID correction
- 546 • slow $\pi^\pm$ efficiency
- 547 • γ efficiency
- 548 • π^0 efficiency (provided by neutral group locally)
- 549 • mode-dependent FEI efficiency

550 The correction factors are varied following their errors to generate 1000 toys. For each
 551 toy, PDF and efficiency are regenerated and Asimov fit is performed as shown in Fig. 7.4.
 552 They are fitted by double-side Gaussian then systematic error is estimated. Breakdown
 553 of each error is shown in Table 7.2.

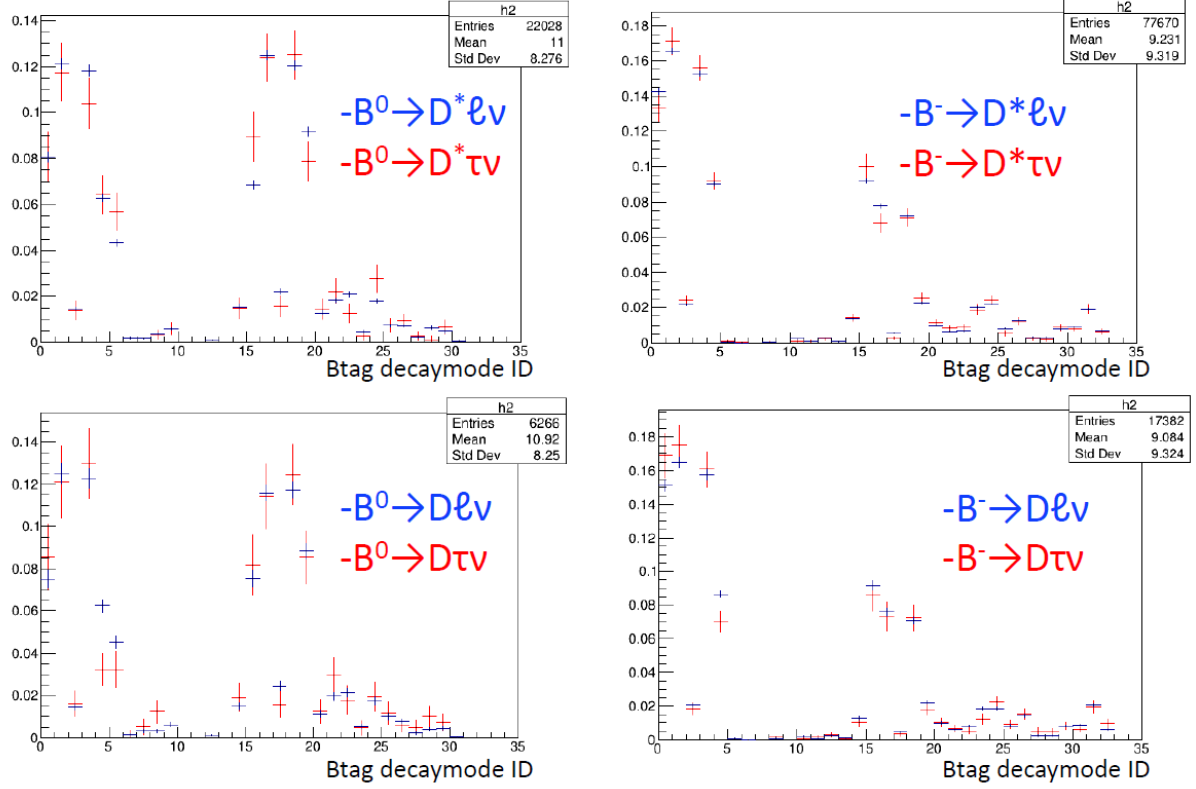


Figure 7.5: decay mode ID of B_{tag} in signal mode and normalization mode. The plots are area normalized. Error bar shows MC statistics error.

7.5 FEI Efficiency difference of signal and normalization mode

Figure 7.5 shows decay mode ID of B_{tag} by MC. The figures are area normalized to compare fraction of each decay mode. The fraction of ID=4, $B^0 \rightarrow D^- \pi^+ \pi^- \pi^0$ and $B^+ \rightarrow D^0 \pi^- \pi^+ \pi^0$, is different between the $B \rightarrow D\tau\nu$ and $B \rightarrow D\ell\nu$. Although several variables of B_{tag} and B_{sig} are checked, we do not understand the reason of the difference. In order to estimate possible systematic error conservatively, we performed Asimov fit study. We regenerate PDF and efficiency by applying a weight to scale efficiency of $B \rightarrow D\tau\nu$ at ID=4 to be same as that of $B \rightarrow D\ell\nu$, then Asimov fit is performed. The impact to $R(D^*)(R(D))$ is 0.1% (2.0%). These are assigned as systematic error.

7.6 Form factors correction

The uncertainty of form factor parameters of $B \rightarrow (D, D^*, D^{**})(\ell, \tau)\nu$ are fluctuated by Hammer, including the correlation of the parameters, to generate 1000 toy of PDF and signal efficiency. For each toy, Asimov fit is performed with the new PDF and efficiency, as shown in Fig. 7.6. They are fitted by double-side Gaussian then systematic error is estimated.

7.7 Continuum background

Continuum background is fixed in the fit. We changed weight of the continuum background $\pm 100\%$ uniformly to generate 1000 toys. Because their distribution is uniform, the systematic error is estimated by taking the maximum difference from 0.

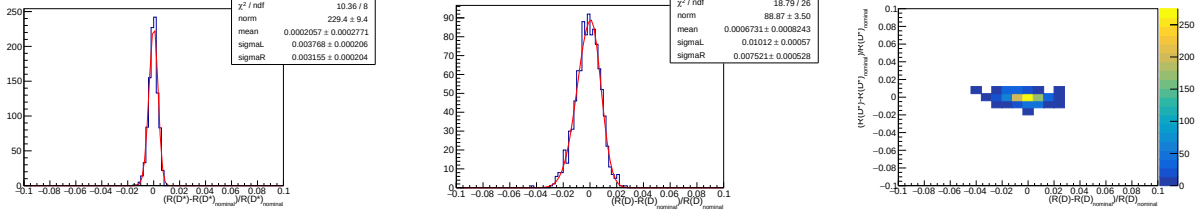


Figure 7.6: Comparison of Asimov fit results with nominal PDF and varied PDF with form factor uncertainty.

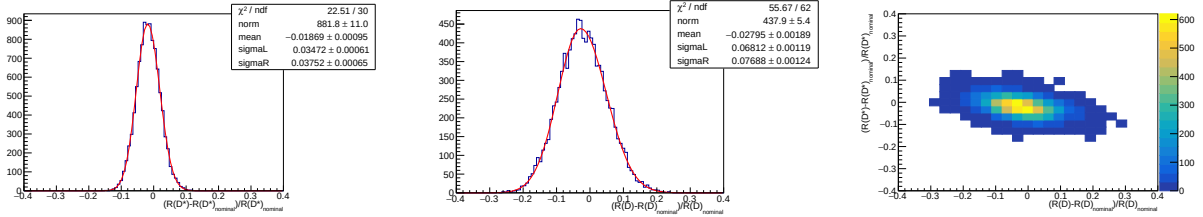


Figure 7.7: Comparison of Asimov fit results with nominal PDF and varied PDF with the MC statistics uncertainty.

7.8 MC statistics

MC statistics uncertainty is estimated by bootstrap method. MC event is re-sampled following poisson distribution. PDF and signal efficiency are regenerated with the re-sampled MC, and Asimov fit is performed. This is repeated 10000 times as shown in Fig. 7.7. They are fitted by double-side Gaussian then systematic error is estimated.

7.9 Fitter bias

As shown in Figure 6.11, fitter bias is seen depending on $R(D^*)$ and $R(D)$ input values. The maximum bias in the figure is taken as systematic error.

7.10 $\tau^- \rightarrow \ell^- \nu_\tau \bar{\nu}_\ell$ branching fractions

The branching fractions of $\tau^- \rightarrow \ell^- \nu_\tau \bar{\nu}_\ell$ is used for efficiency estimation. We assume $\text{BR}(\tau^- \rightarrow \ell^- \nu_\tau \bar{\nu}_\ell) = 0.1739 \pm 0.0004 (0.1782 \pm 0.0004)$ in e (μ) mode. In the fit, it is fixed to the central value. It is varied $\pm 1\sigma$ and Asimov fit is re-performed. The difference of Asimov fit result is negligible, 0.2%. Then no systematic error is assigned.

7.11 Fraction of correctly reconstructed $D^{(*)}$

The number of incorrectly reconstructed D can be estimated from D mass distribution, as shown in Figure 3.2-3.3. It is reliably estimated by MC, because the data agree with MC well and there is no peaking structure.

On the other hand, the number of incorrectly reconstructed D^* can not be estimated precisely from mass difference of $D^* - D$ as shown in Figure 3.4, because there is peaking structure in $D^{*+} \rightarrow \pi D^0$, $D^{*0} \rightarrow \pi^0 D^0$ and $D^{*0} \rightarrow \pi^0 D^0$. In order to estimate the

Table 7.5: Data/MC comparison of the number of selected events at q^2 side-band. The error shows statistical error of data. systematic error is not included in the error.

Category	Fraction of correctly reconstructed D^*					
	$D^{*+} \rightarrow D^0\pi^+$	$D^{*+} \rightarrow D^+\pi^0$	$D^{*0} \rightarrow D^0\pi^0$	$D^{*0} \rightarrow D^0\gamma$	D^0	D^+
before fit	0.929	0.853	0.850	0.500	0.907	0.919
after fit	1.000 ± 0.043	0.831 ± 0.137	1.000 ± 0.151	0.501 ± 0.070	0.865 ± 0.034	0.892 ± 0.065

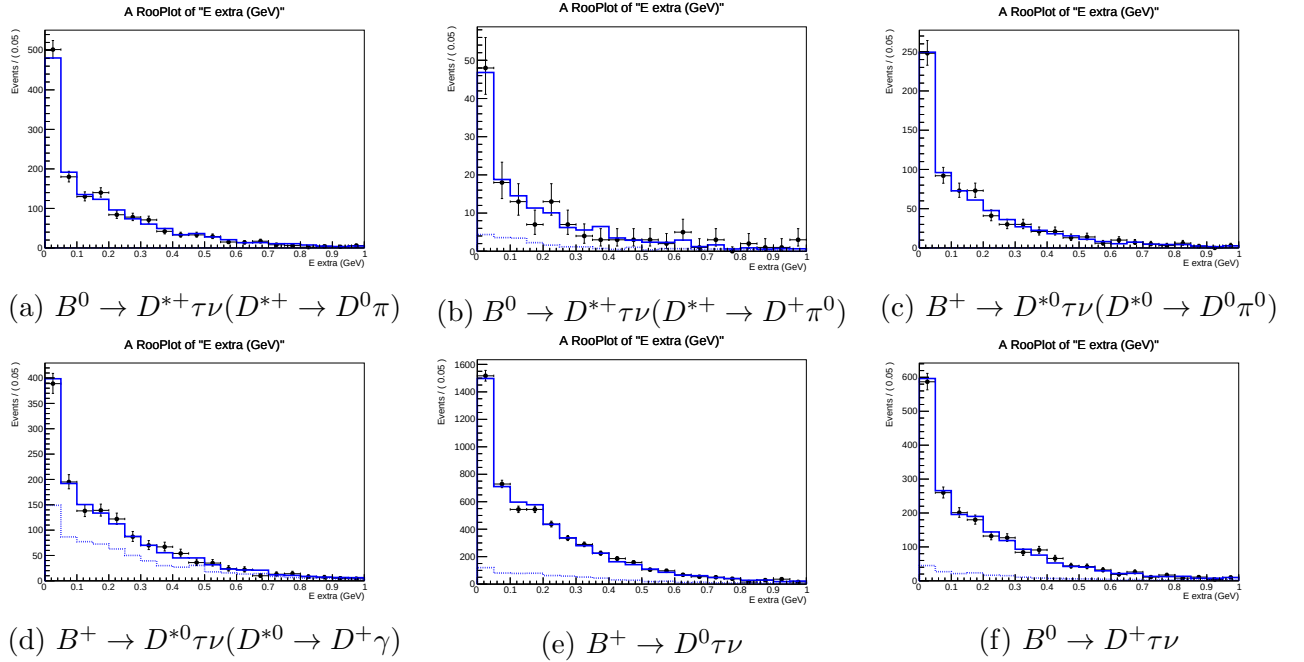


Figure 7.8: Distributions of the $E_{\text{extra}}^{\text{ECL}}$ in each $D^{(*)}$ decay mode at q^2 side-band after the fit. Straight (dotted) line is correctly (incorrectly) reconstructed signal.

fraction from sideband, $E_{\text{extra}}^{\text{ECL}}$ distribution is fitted at q^2 sideband. The PDF is categorized by correctly and incorrectly reconstructed $D^{(*)}$ as shown in Figure 5.9 and fraction of correctly reconstructed $D^{(*)}$ is floated as free parameter. Then fit is performed as shown in Figure 7.8 and Table 7.5. The fit results are consistent with MC prediction.

Finally, the effect to $R(D^{(*)})$ is estimated by Asimov fit. The fraction of correctly reconstructed $D^{(*)}$ is reweighted following the fit results at q^2 sideband and PDF is regenerated, then Asimov fit is performed. This is repeated 1000 times as shown in Fig. 7.9.

7.12 yield of D

As shown in Figure 5.1-5.2, the data yield of $D^+ \rightarrow K\pi\pi\pi^0$, $D^0 \rightarrow KK$, $D^+ \rightarrow K_S\pi\pi^0$ are about 20-30% different from MC expectation (more than 2σ of statistical error). The MC is weighted by the data/MC ratio to adjust the fraction of the three channels based on data, and new PDF and efficiency are generated. With the generated PDF, Asimov data is fitted and result is compared with nominal PDF. The impact to $R(D^*)$ is small, 0.1%, and no systematic uncertainty is assigned.

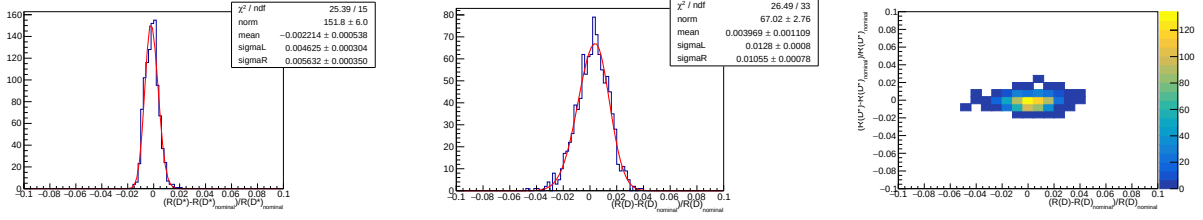


Figure 7.9: Comparison of Asimov fit results with nominal PDF and varied PDF with fraction of correctly reconstructed $D^{(*)}$ uncertainty.

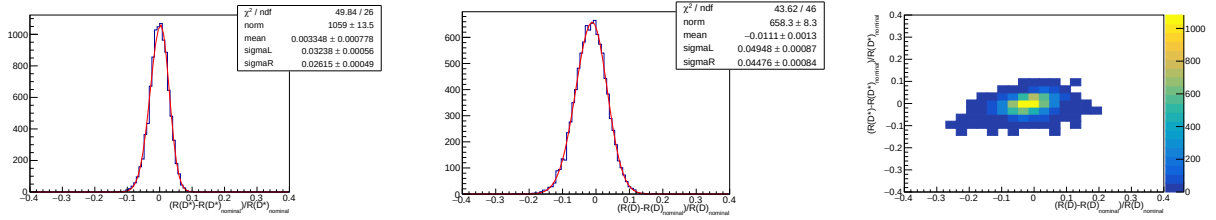


Figure 7.10: Comparison of Asimov fit results with nominal PDF and varied PDF with all systematic uncertainty, except for MC statistics, FEI efficiency correction, FEI efficiency difference of signal and normalization mode, and fitter bias.

7.13 M_{miss}^2 resolution

The uncertainty of M_{miss}^2 correction is taken into account as systematic error. The PDF shape is varied following the uncertainty in Table 5.4 and Asimov fit is performed.

7.14 Total uncertainty

In order to estimate total systematic uncertainty, PDF and efficiencies are fluctuated following all explained error sources, except for MC statistics, to generate 10000 toys. For each toy, PDF is regenerated and Asimov fit is performed as shown in Fig. 7.10. They are fitted by double-side Gaussian then systematic error is estimated. The uncertainties of MC statistics, FEI efficiency correction, FEI efficiency difference of signal and normalization mode, and fitter bias are not included in the toy. Their errors are added to total error by taking quadratic sum of them. Because not only sigma but also mean (bias) of double-sided gaussian is included in each systematic source in Table 7.1, quadratic sum of each uncertainty is different from the total uncertainty.

8 Results of Asimov Fit

The results of Asimov fit with statistics and systematic error are

$$R(D^*) = 0.254 \pm 0.022(\text{stat.}) \pm 0.013 - 0.021(\text{syst.}), \quad (8.1)$$

$$R(D) = 0.300 \pm 0.051(\text{stat.}) \pm 0.032 - 0.037(\text{syst.}), \quad (8.2)$$

$$\rho = -0.42(\text{stat.}) - 0.20(\text{syst.}). \quad (8.3)$$

The systematic error depends on amount of gap mode in the fitted Asimov data.

Table 9.1: Fitted parameters in the q^2 sideband

parameters	prefit	postfit with stat. error
$BR(B \rightarrow D^* \ell \nu)$	0.1013	0.1038 ± 0.0027
$BR(B \rightarrow D \ell \nu)$	0.0437	0.0436 ± 0.0012
f_{00}	0.4861 ± 0.0080	0.4861 ± 0.0080
f_{+-}	0.5113 ± 0.0108	0.5112 ± 0.0108
$N_{B\bar{B}}$	387.1 ± 5.6	387.2 ± 5.6

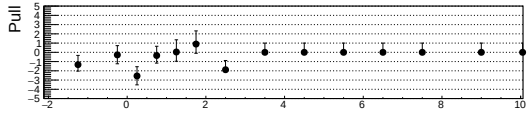
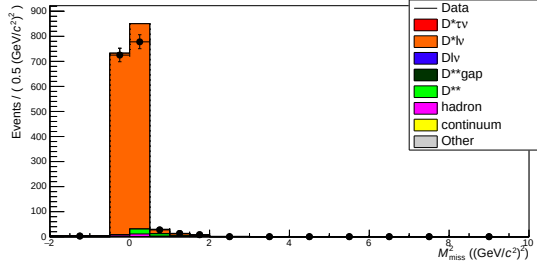
Table 9.2: Comparison of the measured $BR(B \rightarrow D^{(*)} \ell \nu)$ with this analysis and previous measurements. HFLAV 2022 is used for input of FEI efficiency calibration analysis of $B \rightarrow X_\ell \nu$. Sum of $BR(B^0 \rightarrow D^{(*)} \ell \nu)$ and $BR(B^+ \rightarrow D^{(*)} \ell \nu)$ are shown.

	pdg live	HFLAV 2022	this analysis q^2 sideband	this analysis selected events
$BR(B \rightarrow D^* \ell \nu)$	0.1013	0.1071	0.1038	0.1101
	± 0.0013 (stat.+sys.)	± 0.0026 (stat.+sys.)	± 0.0027 (stat.) ± 0.0070 (sys.)	± 0.0025 (stat.) ± 0.0070 (sys.)
$BR(B \rightarrow D \ell \nu)$	0.0436	0.0466	0.0436	0.0431
	± 0.0010 (stat.+sys.)	± 0.0013 (stat.+sys.)	± 0.0012 (stat.)	± 0.0013 (stat.)

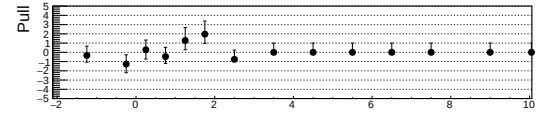
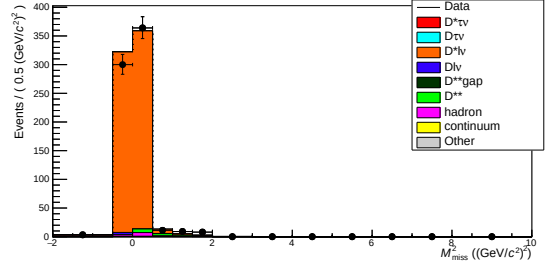
9 Data fit

9.1 Fit in q^2 sideband

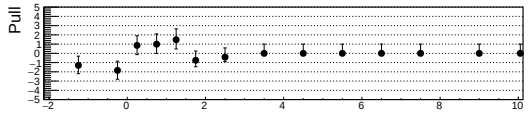
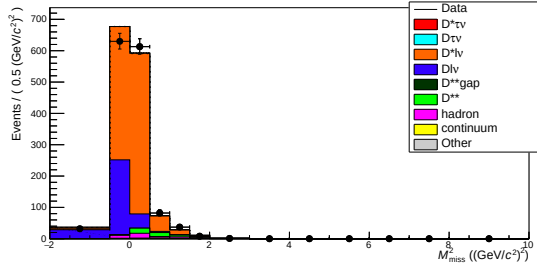
For consistency check, fit is performed in the q^2 sideband. Because amount of background is very small and fit is failed with the default fit configuration, the background parameters and $R(D^{(*)})$ are specially fixed. Figures 9.1-?? show postfit distributions of the fit variables. Table 9.1 shows the pre and post fit parameters. The measured $BR(B \rightarrow D^{(*)} \ell \nu)$ is consistent with the past measurement, as shown in Table 9.2. The $BR(B \rightarrow D^{(*)} \ell \nu)$ from HFLAV2022 is used as an input to the FEI calibration analysis of $B \rightarrow X_\ell \nu$. This assumption can affect our measured $BR(B \rightarrow D^{(*)} \ell \nu)$. Because the BR (absolute yield of the normalization mode) will be canceled for the $R(D^{(*)})$, such effect is not estimated quantitatively in this analysis.



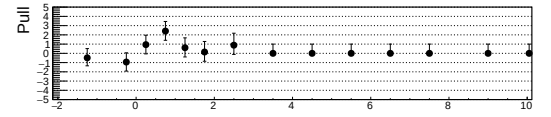
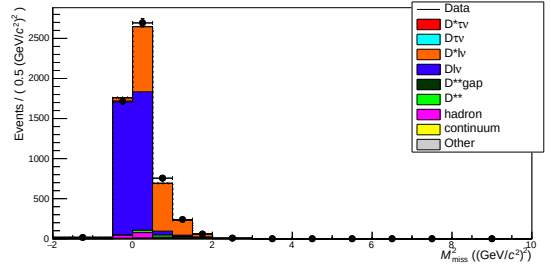
(a) $B^0 \rightarrow D^{*+} \tau \nu$



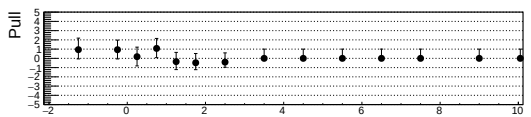
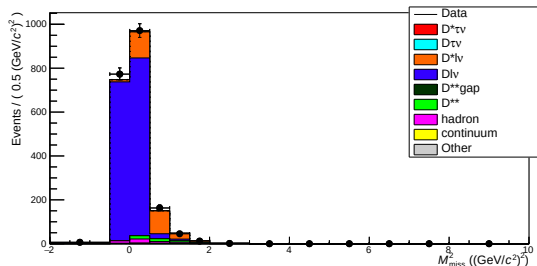
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

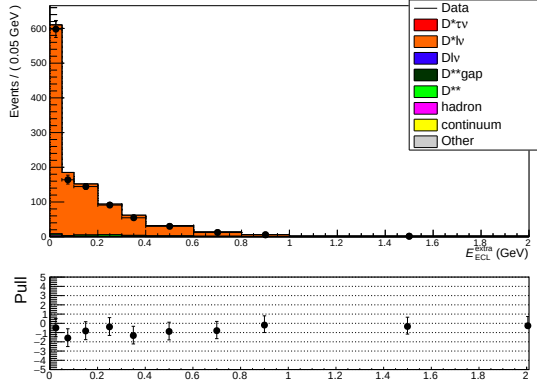


(d) $B^+ \rightarrow D^0 \tau \nu$

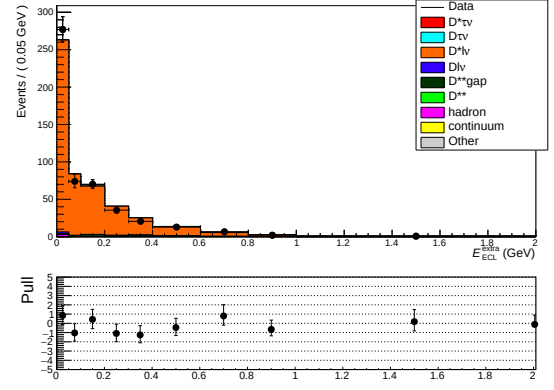


(e) $B^0 \rightarrow D^+ \tau \nu$

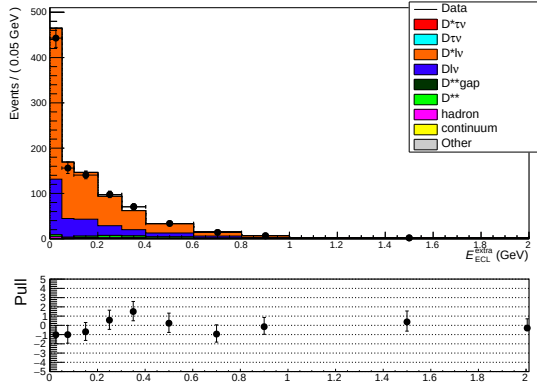
Figure 9.1: Post fit projection to the M_{miss}^2 in each sample in q^2 sideband.



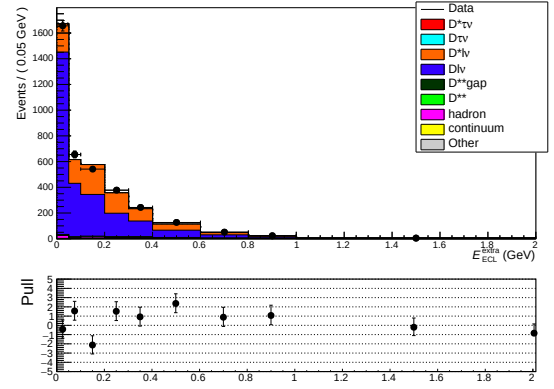
(a) $B^0 \rightarrow D^{*+} \tau \nu$



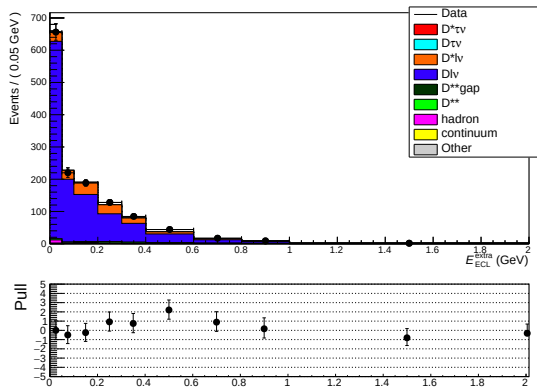
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$



(d) $B^+ \rightarrow D^0 \tau \nu$



(e) $B^0 \rightarrow D^+ \tau \nu$

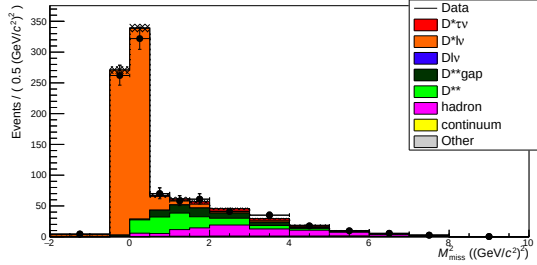
Figure 9.2: Post fit projection to the $E_{\text{extra}}^{\text{ECL}}$ in each sample in q^2 sideband.

Table 9.3: Fitted parameters in π^0 ROE sideband

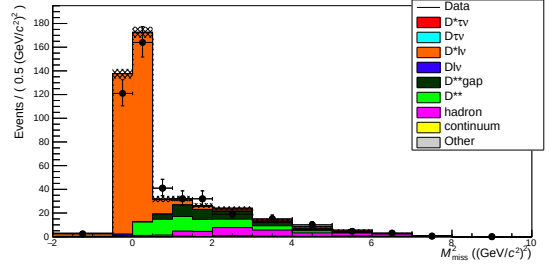
parameters	prefit	postfit with stat. error
$R(D^*)$	—	0.215 ± 0.037
$R(D)$	—	0.463 ± 0.131
$BR(B \rightarrow D^* \ell \nu)$	0.1013	0.1055 ± 0.0028
$BR(B \rightarrow D \ell \nu)$	0.0437	0.0431 ± 0.0013
$N(\text{gap mode}), B^0 \rightarrow D^{*+} \tau \nu$	148.08	72.0 ± 36.6
$N(\text{gap mode}), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \pi^0 D^0)$	89.259	35.1 ± 23.7
$N(\text{gap mode}), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \gamma D^0)$	342.39	-96.3 ± 71.0
$N(\text{gap mode}), B^+ \rightarrow D^0 \tau \nu$	1060.3	-60.8 ± 173
$N(\text{gap mode}), B^0 \rightarrow D^+ \tau \nu$	409.50	237.1 ± 107
$N(D^{**} \ell \nu), B^0 \rightarrow D^{*+} \tau \nu$	157.29	134.8 ± 39.3
$N(D^{**} \ell \nu), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \pi^0 D^0)$	71.475	74.0 ± 24.6
$N(D^{**} \ell \nu), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \gamma D^0)$	248.90	247.47 ± 70.5
$N(D^{**} \ell \nu), B^+ \rightarrow D^0 \tau \nu$	1130.6	1303.0 ± 185
$N(D^{**} \ell \nu), B^0 \rightarrow D^+ \tau \nu$	474.84	356.4 ± 105
$N(\text{hadronic } B), B^0 \rightarrow D^{*+} \tau \nu$	167.67	134.2 ± 22.5
$N(\text{hadronic } B), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \pi^0 D^0)$	87.558	50.413 ± 49.6
$N(\text{hadronic } B), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \gamma D^0)$	369.30	589.66 ± 39.6
$N(\text{hadronic } B), B^+ \rightarrow D^0 \tau \nu$	1210.9	1515.3 ± 89.6
$N(\text{hadronic } B), B^0 \rightarrow D^+ \tau \nu$	784.18	730.6 ± 49.6
f_{00}	0.4861 ± 0.0080	0.4861 ± 0.0080
f_{+-}	0.5113 ± 0.0108	0.5113 ± 0.0108
$N_{B\bar{B}}$	387.1 ± 5.6	387.1 ± 5.6

9.2 Fit in π^0 ROE sideband

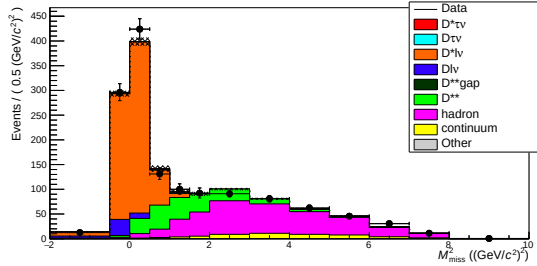
For consistency check, fit is performed in the π^0 ROE sideband. Figures 9.3-9.4 show the projected distributions of the fit variables at postfit. M_{miss}^2 and $E_{\text{extra}}^{\text{ECL}}$ cuts are removed from selection to unbox entire region. Table 9.3 shows the prefit and postfit parameters. The measured gap mode is lower than nominal value of the MC expectation, as discussed in Sec. 5.



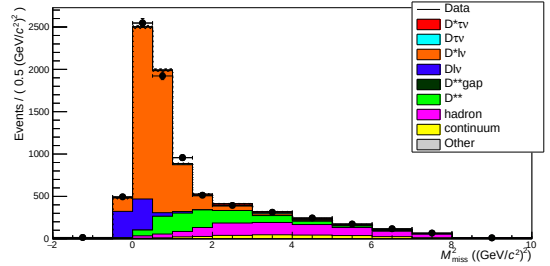
(a) $B^0 \rightarrow D^{*+} \tau \nu$



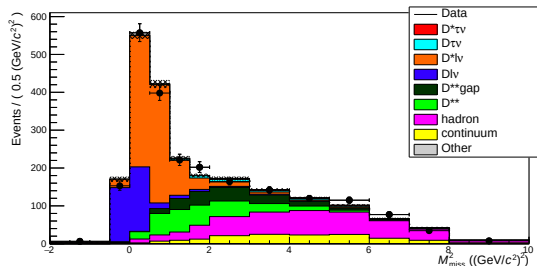
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

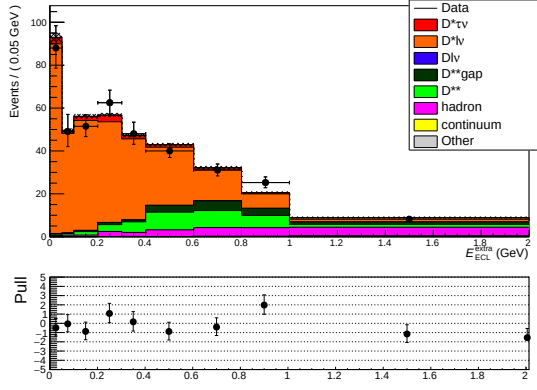


(d) $B^+ \rightarrow D^0 \tau \nu$

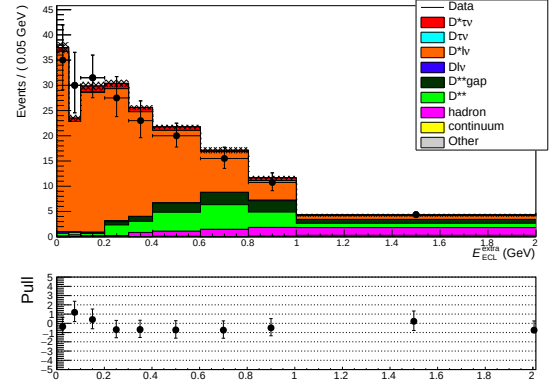


(e) $B^0 \rightarrow D^+ \tau \nu$

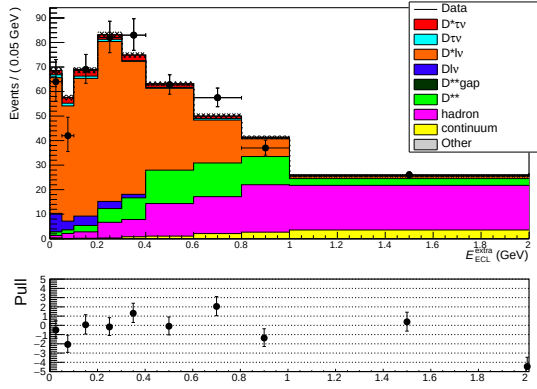
Figure 9.3: Post fit projection to the M_{miss}^2 in each sample in π^0 ROE sideband.



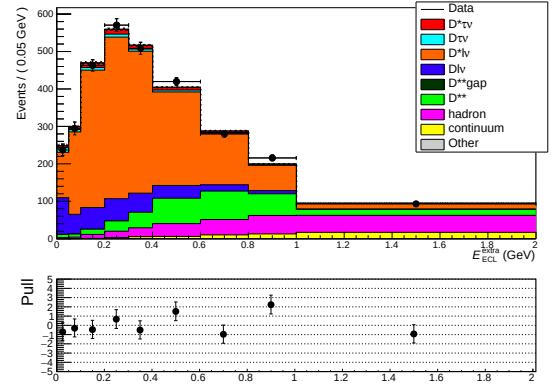
(a) $B^0 \rightarrow D^{*+} \tau \nu$



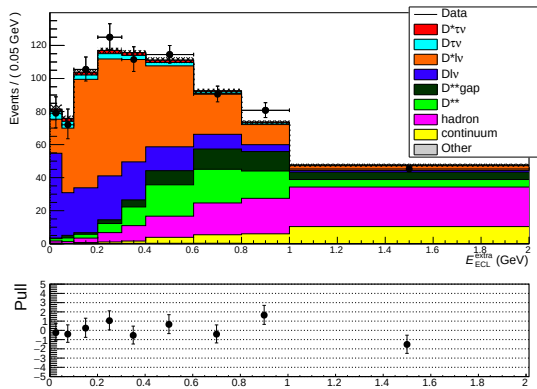
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$



(d) $B^+ \rightarrow D^0 \tau \nu$



(e) $B^0 \rightarrow D^+ \tau \nu$

Figure 9.4: Post fit projection to the $E_{\text{extra}}^{\text{ECL}}$ in each sample in π^0 ROE sideband.

Table 9.4: Fitted parameters to the selected events

parameters	prefit	postfit with stat. error
$R(D^*)$	—	0.242 ± 0.019
$R(D)$	—	0.439 ± 0.055
$BR(B \rightarrow D^* \ell \nu)$	0.1013	0.1101 ± 0.0025
$BR(B \rightarrow D \ell \nu)$	0.0437	0.0431 ± 0.0013
$N(\text{gap mode}), B^0 \rightarrow D^{*+} \tau \nu$	118.27	21.4 ± 24.8
$N(\text{gap mode}), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \pi^0 D^0)$	70.996	41.6 ± 15.1
$N(\text{gap mode}), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \gamma D^0)$	229.43	53.4 ± 49.6
$N(\text{gap mode}), B^+ \rightarrow D^0 \tau \nu$	776.79	240.3 ± 93.5
$N(\text{gap mode}), B^0 \rightarrow D^+ \tau \nu$	271.10	89.5 ± 46.4
$N(D^{**} \ell \nu), B^0 \rightarrow D^{*+} \tau \nu$	110.70	60.1 ± 31.2
$N(D^{**} \ell \nu), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \pi^0 D^0)$	46.964	28.9 ± 23.9
$N(D^{**} \ell \nu), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \gamma D^0)$	206.19	190.5 ± 58.3
$N(D^{**} \ell \nu), B^+ \rightarrow D^0 \tau \nu$	774.99	731.1 ± 134.0
$N(D^{**} \ell \nu), B^0 \rightarrow D^+ \tau \nu$	283.32	294.2 ± 63.5
$N(\text{hadronic } B), B^0 \rightarrow D^{*+} \tau \nu$	98.644	86.0 ± 25.3
$N(\text{hadronic } B), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \pi^0 D^0)$	41.766	3.5 ± 12.2
$N(\text{hadronic } B), B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow \gamma D^0)$	157.28	165.6 ± 38.2
$N(\text{hadronic } B), B^+ \rightarrow D^0 \tau \nu$	566.68	469.2 ± 94.6
$N(\text{hadronic } B), B^0 \rightarrow D^+ \tau \nu$	266.88	125.6 ± 25.3
f_{00}	0.4861 ± 0.0080	0.4861 ± 0.0080
f_{+-}	0.5113 ± 0.0108	0.5113 ± 0.0108
$N_{B\bar{B}}$	387.1 ± 5.6	387.1 ± 5.6

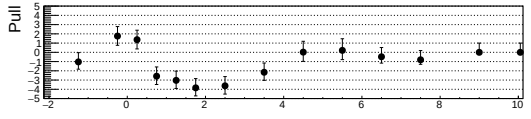
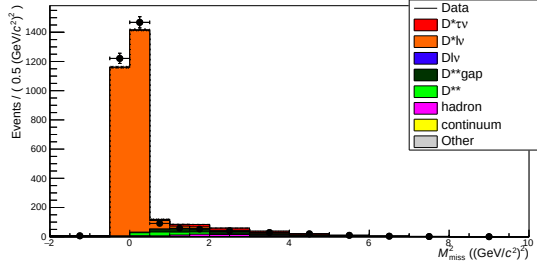
9.3 Data fit to selected events

The fit is performed to the selected events. Figures 9.5-9.8 (9.9-9.12) show projected distributions of the fit variables at the prefit (postfit). There is no local deviation more than 3σ in the postfit distributions. After the fitting, global $\chi^2/\text{ndf} = 1.15$. It indicates the data/MC agreement has no problem at the postfit. Table 9.4 shows the fit parameters.

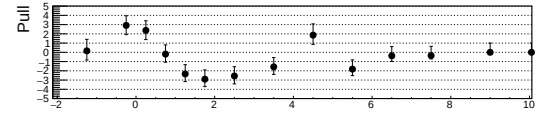
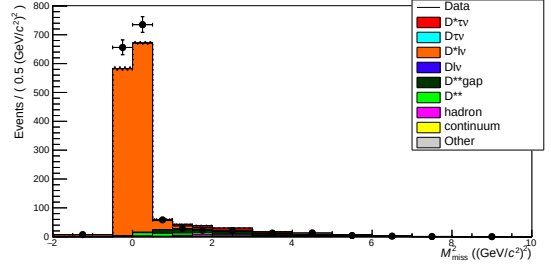
For consistency check, we compared the measured value of $BR(B \rightarrow D^* \ell \nu)$ and $BR(B \rightarrow D \ell \nu)$, as shown in Table 9.2. They agree each other within systematic error. The measured BR at q^2 sideband and selected events are 1.5σ different in each other, because some of systematics error is canceled and relative systematic error is reduced to 1.6%. In order to evaluate its impact on $R(D^*)$, sanity check is performed as in Sec. 9.3.1.

The measured gap mode is lower than nominal value of the MC expectation. It is consistent with the π^0 ROE sideband. Figure ?? show the comparison of the background parameters between the selected events and π^0 ROE sideband. In the plot, systematic error of continuum background is included, because amount of continuum background is large in the π^0 ROE sideband and it affects amount of the fitted yield of hadronic background. The background yields agree between the selected events and π^0 ROE sideband within the error.

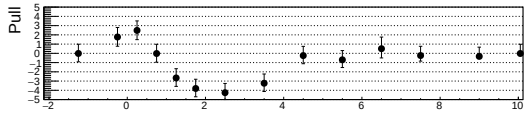
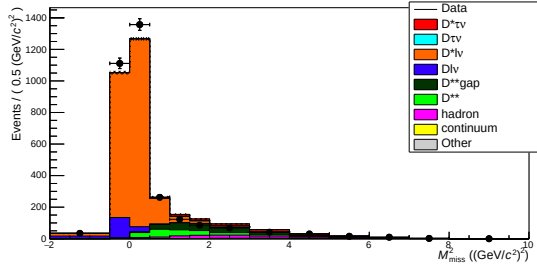
Table 9.5 shows the estimated systematic uncertainties on $R(D^{(*)})$ by using real data, instead of Asimov data in Sec. 7. Because amount of gap mode is smaller than the MC, systematic uncertainty of the gap mode branching fractions is smaller than Asimov data in Table 7.1.



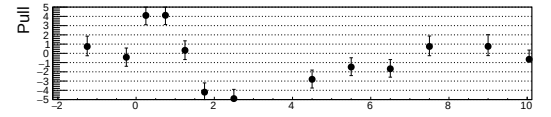
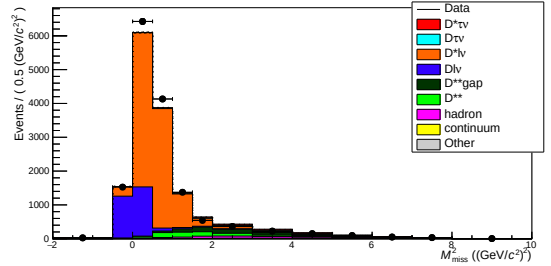
(a) $B^0 \rightarrow D^{*+} \tau \nu$



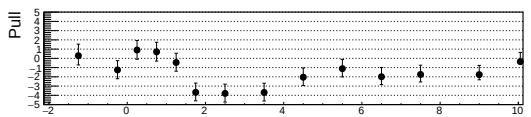
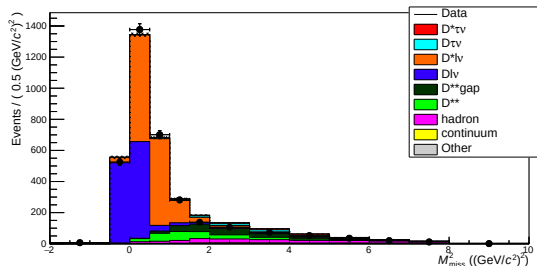
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

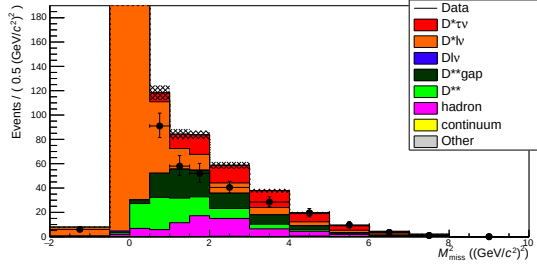


(d) $B^+ \rightarrow D^0 \tau \nu$

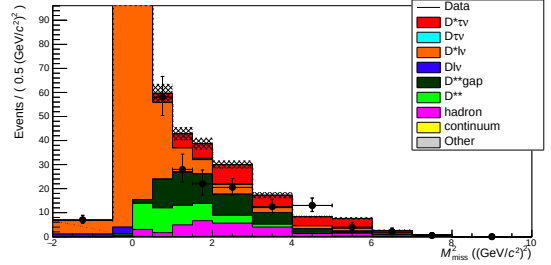


(e) $B^0 \rightarrow D^+ \tau \nu$

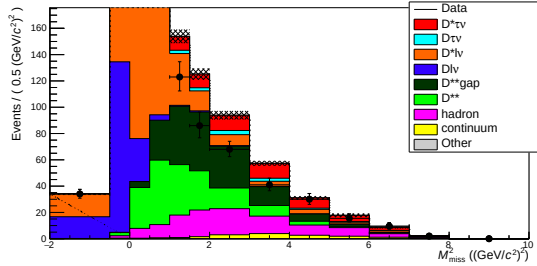
Figure 9.5: Pre fit projection to the M_{miss}^2 in each sample.



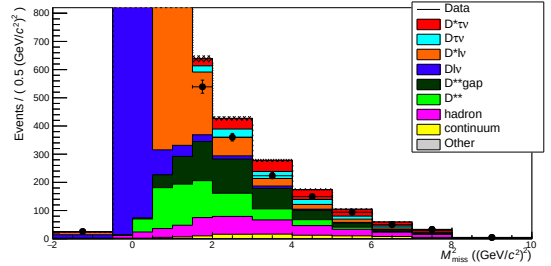
(a) $B^0 \rightarrow D^{*+} \tau \nu$



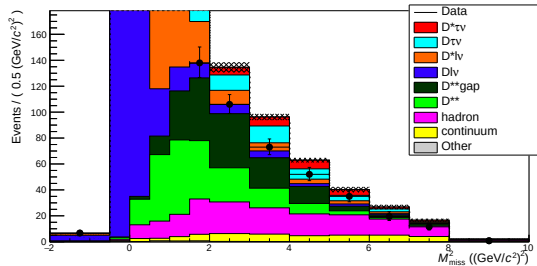
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

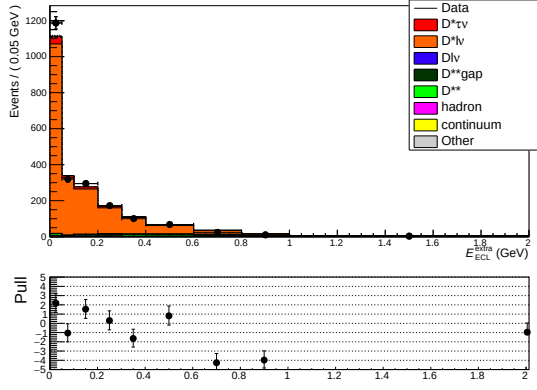


(d) $B^+ \rightarrow D^0 \tau \nu$

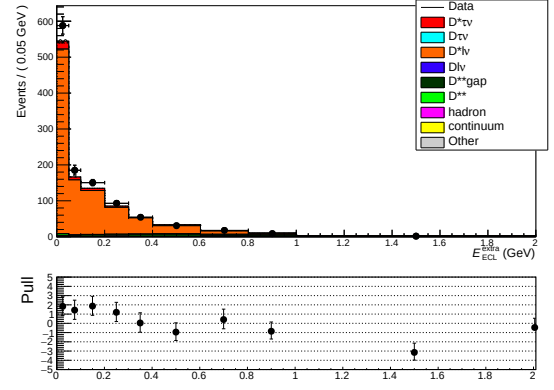


(e) $B^0 \rightarrow D^+ \tau \nu$

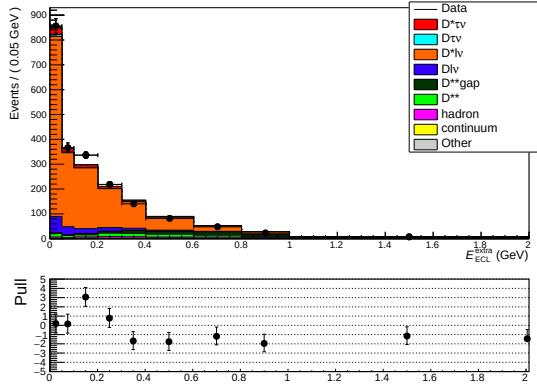
Figure 9.6: Pre fit projection to the M_{miss}^2 in each sample.



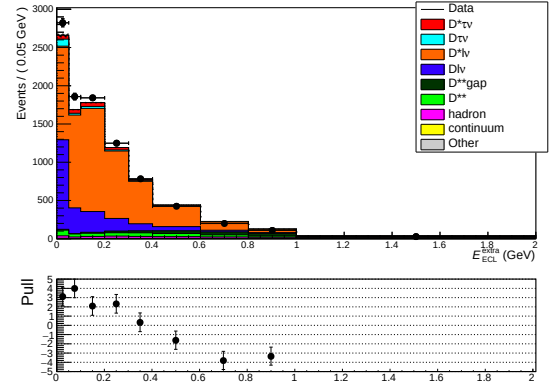
(a) $B^0 \rightarrow D^{*+} \tau \nu$



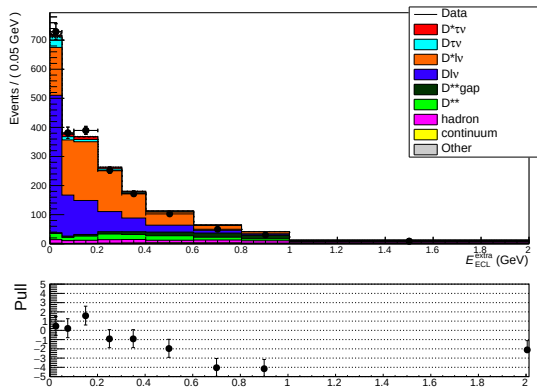
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

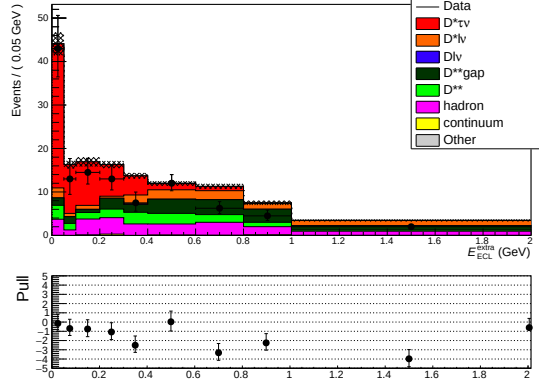


(d) $B^+ \rightarrow D^0 \tau \nu$

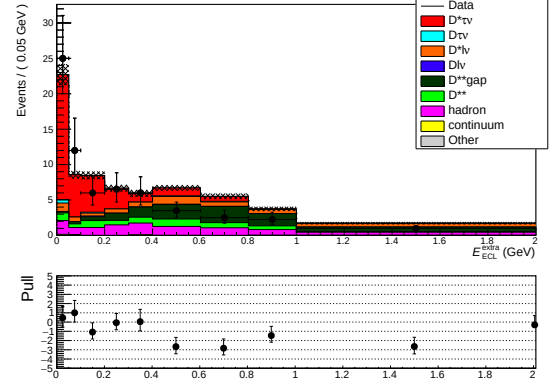


(e) $B^0 \rightarrow D^+ \tau \nu$

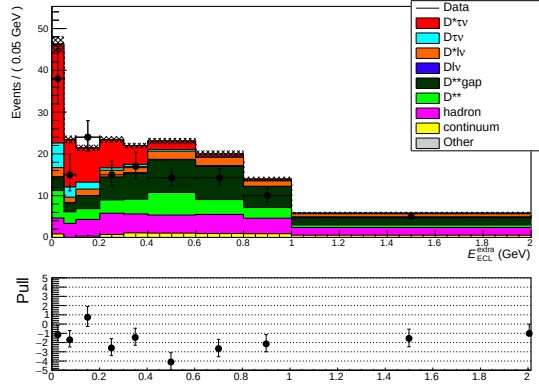
Figure 9.7: Pre fit projection to the E_{extra}^{ECL} in each sample.



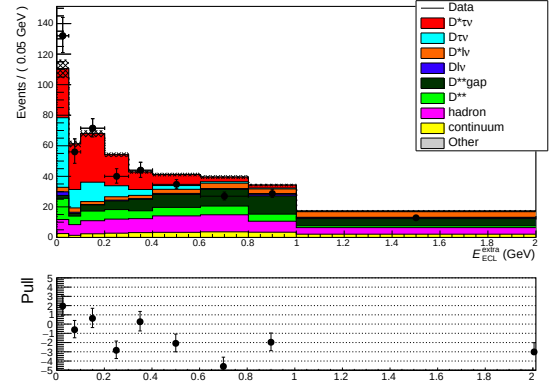
(a) $B^0 \rightarrow D^{*+} \tau \nu$



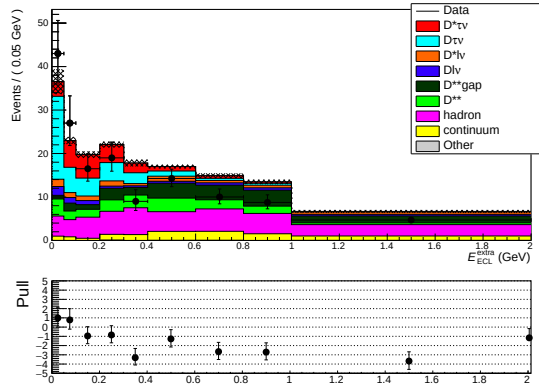
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

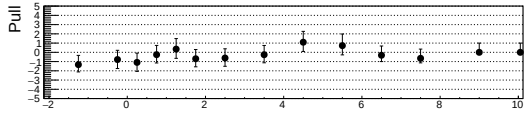
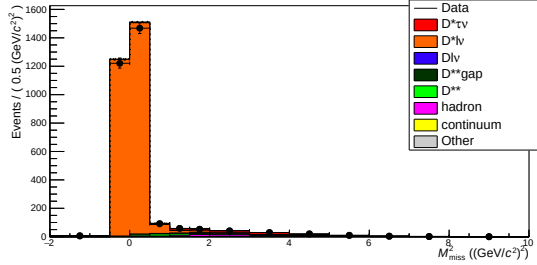


(d) $B^+ \rightarrow D^0 \tau \nu$

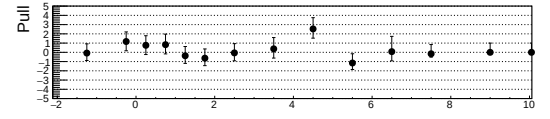
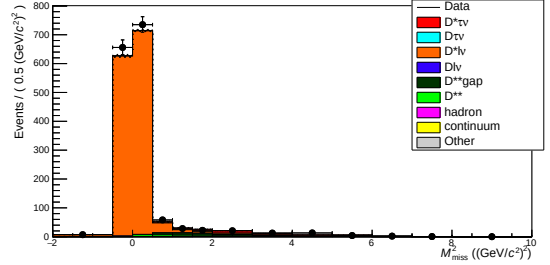


(e) $B^0 \rightarrow D^+ \tau \nu$

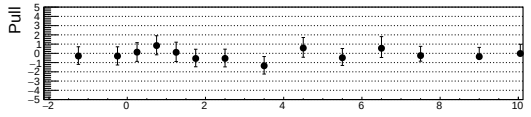
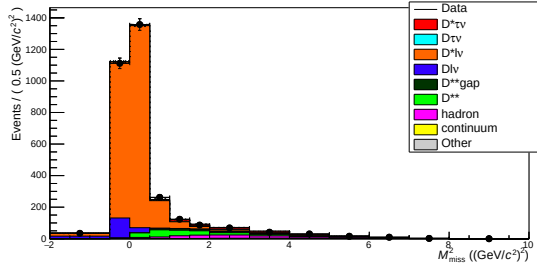
Figure 9.8: Pre fit projection to the $E_{\text{extra}}^{\text{ECL}}$ at $1.5 \text{ GeV}^2 < M_{\text{miss}}^2$ ($3.0 \text{ GeV}^2 < M_{\text{miss}}^2$) for $D^*(D)$ sample in each sample.



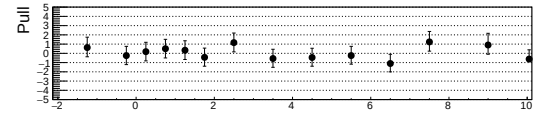
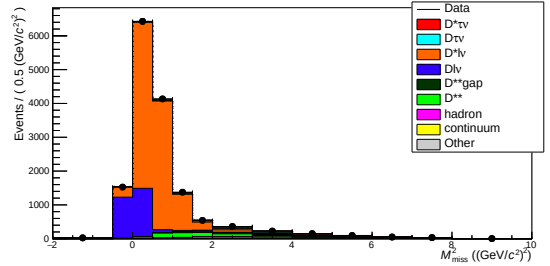
(a) $B^0 \rightarrow D^{*+} \tau \nu$



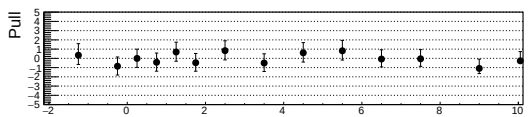
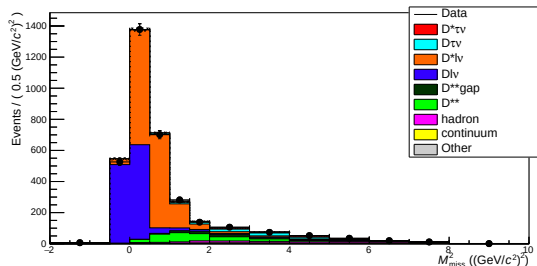
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

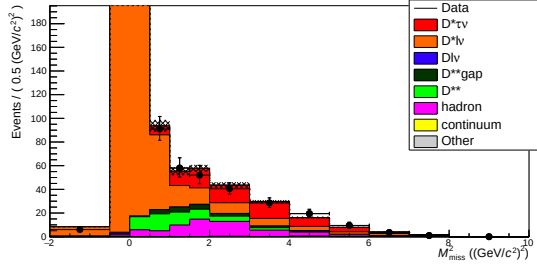


(d) $B^+ \rightarrow D^0 \tau \nu$

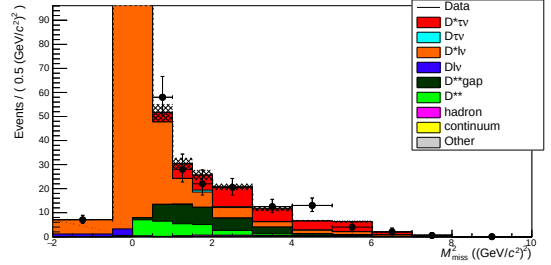


(e) $B^0 \rightarrow D^+ \tau \nu$

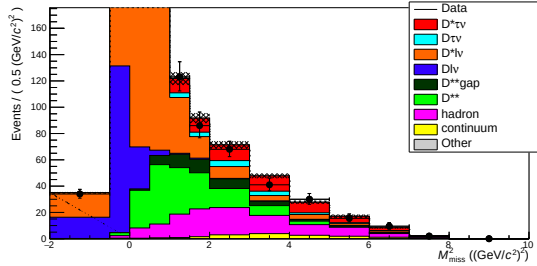
Figure 9.9: Post fit projection to the M_{miss}^2 in each sample.



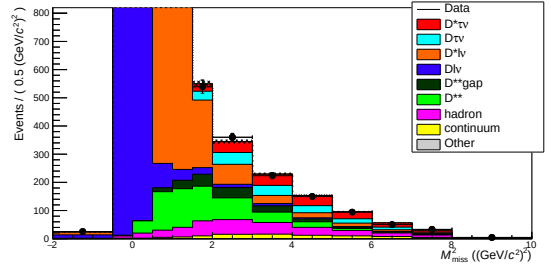
(a) $B^0 \rightarrow D^{*+} \tau \nu$



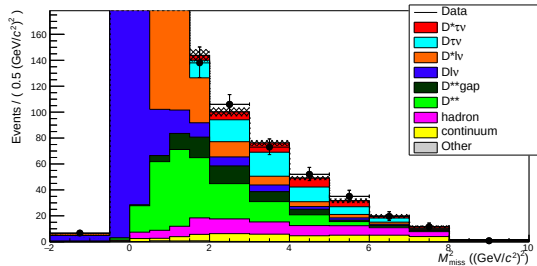
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

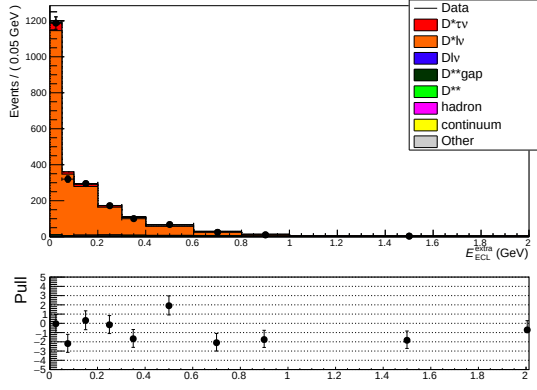


(d) $B^+ \rightarrow D^0 \tau \nu$

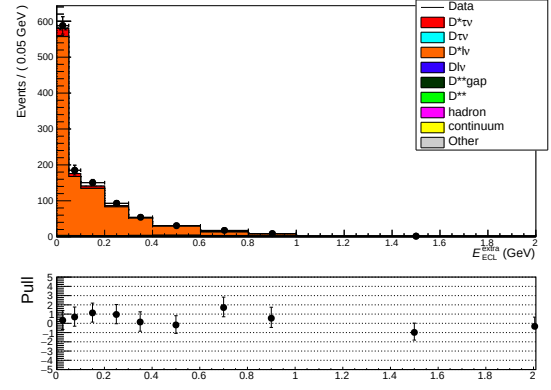


(e) $B^0 \rightarrow D^+ \tau \nu$

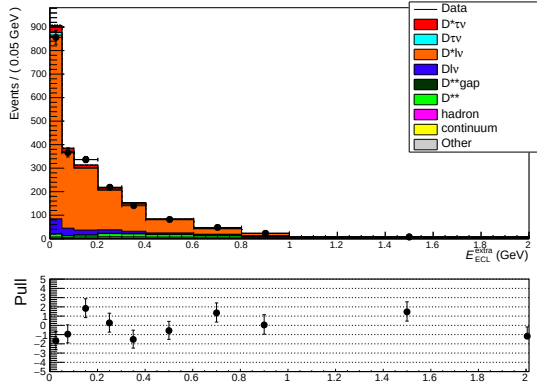
Figure 9.10: Post fit projection to the M_{miss}^2 in each sample. y axis is zoomed.



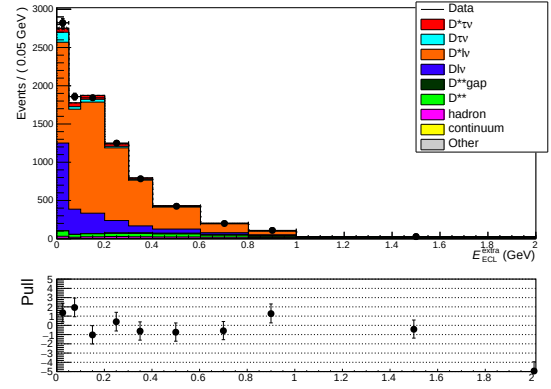
(a) $B^0 \rightarrow D^{*+} \tau \nu$



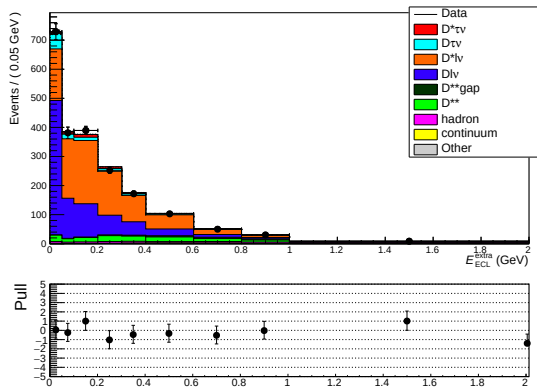
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$

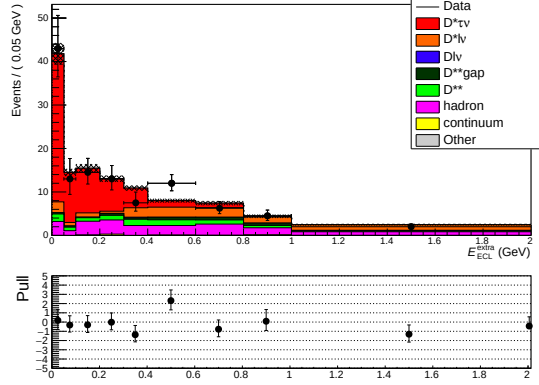


(d) $B^+ \rightarrow D^0 \tau \nu$

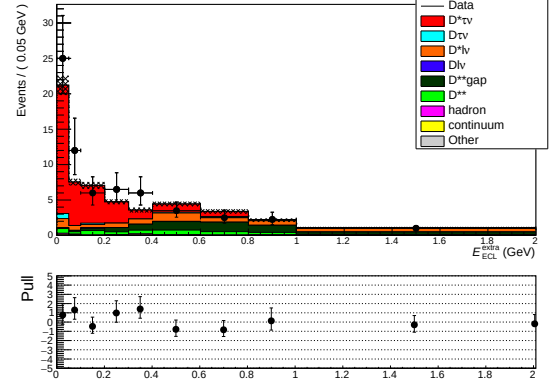


(e) $B^0 \rightarrow D^+ \tau \nu$

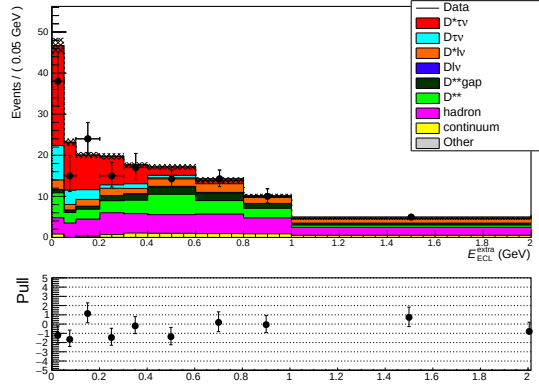
Figure 9.11: Post fit projection to the E_{extra}^{ECL} in each sample.



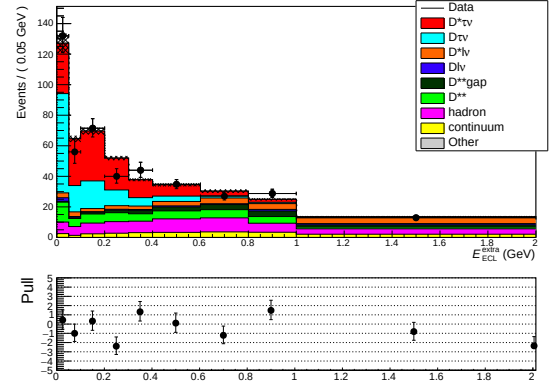
(a) $B^0 \rightarrow D^{*+} \tau \nu$



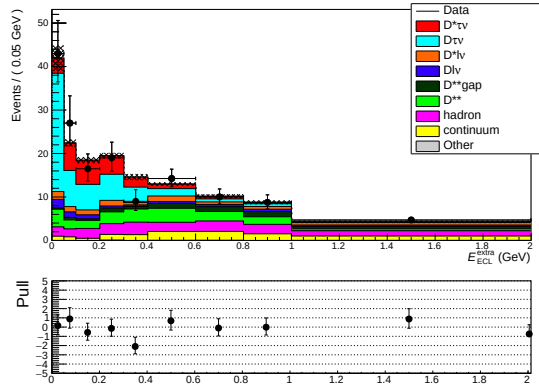
(b) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \pi^0)$



(c) $B^+ \rightarrow D^{*0} \tau \nu (D^{*0} \rightarrow D^0 \gamma)$



(d) $B^+ \rightarrow D^0 \tau \nu$



(e) $B^0 \rightarrow D^+ \tau \nu$

Figure 9.12: Post fit projection to the $E_{\text{extra}}^{\text{ECL}}$ at $1.5 \text{ GeV}^2 < M_{\text{miss}}^2$ ($3.0 \text{ GeV}^2 < M_{\text{miss}}^2$) for $D^*(D)$ sample in each sample.

Table 9.5: Summary of the systematic uncertainties.

Error source	$R(D^*)$	$R(D)$	ρ
$\bar{B} \rightarrow D^{**}\ell^-\bar{\nu}_\ell$ branching fractions	0.3%	1.3%	0.25
MC statistics	4.8%	8.4%	-0.44
gap mode branching fractions $\pm 100\%$	2.6%	2.6%	0.0
gap mode branching fractions	0.5%	0.3%	-0.06
Hadronic B decay branching fractions	1.6%	1.5%	-0.26
Continuum background	2.4%	2.1%	0.93
Slow π^0, γ efficiency	2.2%	2.4%	0.99
Other efficiency corrections	0.7%	1.4%	0.92
FEI efficiency of data	0.9%	1.8%	-1.0
FEI efficiency of $B \rightarrow D\tau\nu$	0.1%	1.8%	1.0
M_{miss}^2 resolution	0.5%	0.8%	0.48
Fraction of fake $D^{(*)}$	0.5%	1.2%	0.00
Fitter bias	0.3%	1.2%	0.00
Form factors correction	0.5%	0.9%	-0.70
Total systematic uncertainty	6.7%	10.2%	-0.20

9.3.1 Sanity check

For sanity check, data sample is split as follows:

- D^0 mode and D^+ mode
- electron mode and muon mode
- data in exp7-18 and exp20-26
- add q^2 sideband to the fit

In each case, fit is re-performed and fitted values of $R(D^{(*)})$ are checked as shown in Table 9.6. They agree each other within statistical error. The $R(D)$ is different by 1.37 σ between electron and muon mode.

Table 9.6: Comparison of $R(D^{(*)})$ by splitting the data set for sanity check. Only statistical error is shown.

Sample	$R(D^*)$	$R(D)$	ρ
Nominal	0.242 ± 0.019	0.439 ± 0.055	-0.40
D^0 mode only	0.246 ± 0.021	0.411 ± 0.069	-0.43
D^+ mode only	0.214 ± 0.068	0.484 ± 0.927	-0.41
electron mode only	0.265 ± 0.024	0.360 ± 0.065	-0.40
muon mode only	0.218 ± 0.029	0.505 ± 0.084	-0.41
exp7-18 only	0.245 ± 0.027	0.408 ± 0.074	-0.39
exp20-26 only	0.247 ± 0.029	0.480 ± 0.079	-0.41
add q^2 sideband	0.246 ± 0.020	0.403 ± 0.052	-0.41

673 **10 Results**

674 The results of data fit are

$$675 \qquad R(D^*) = 0.242 \pm 0.019(\text{stat.}) \pm 0.016(\text{syst.}), \qquad (10.1)$$

$$676 \qquad R(D) = 0.439 \pm 0.055(\text{stat.}) \pm 0.045(\text{syst.}), \qquad (10.2)$$

$$677 \qquad \rho = -0.40(\text{stat.}) - 0.20(\text{syst.}). \qquad (10.3)$$

678 **11 Comparison with past measurements**

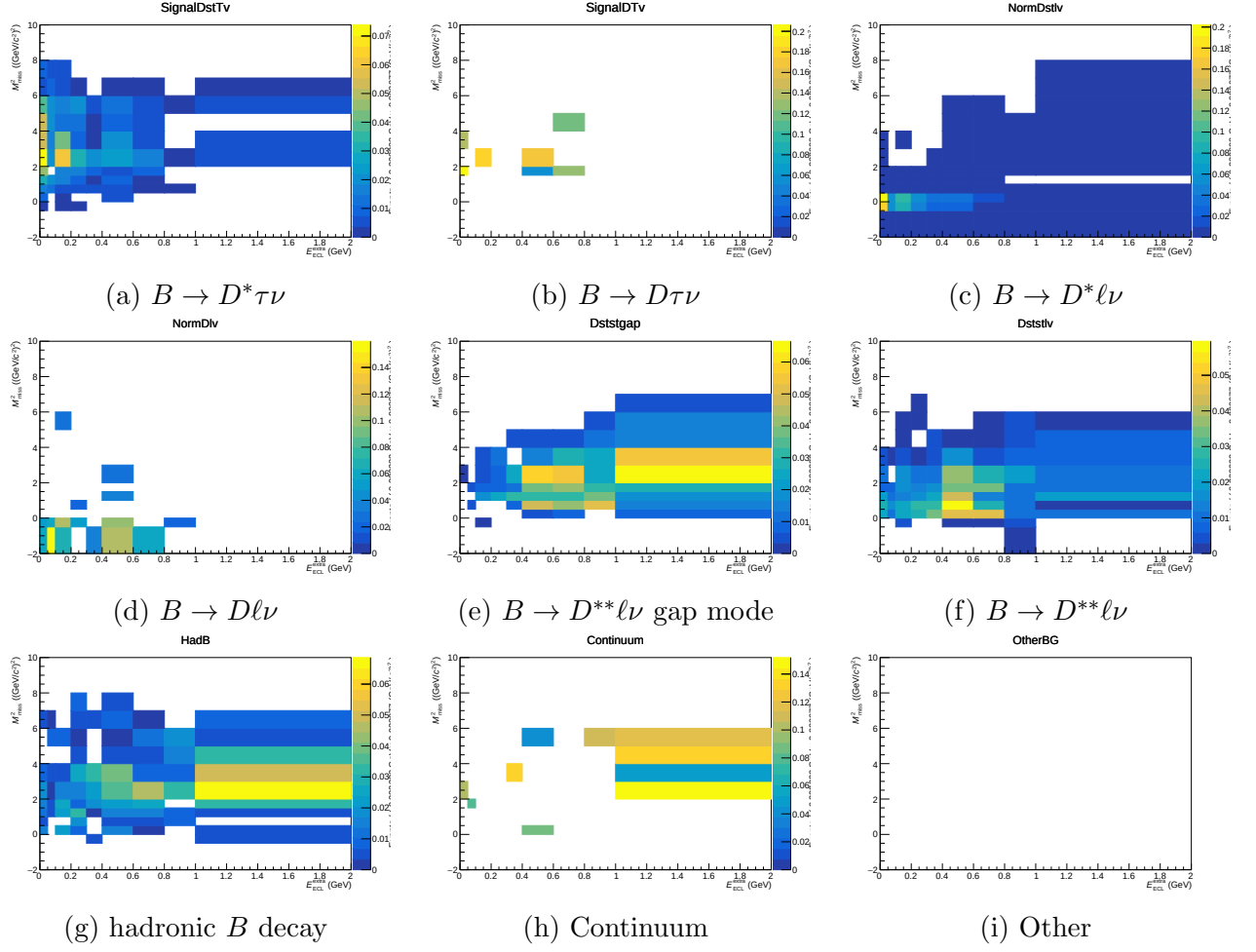


Figure 1.1: PDF of each category in $B^+ \rightarrow D^{*0} \ell^+$ ($D^{*0} \rightarrow D^0 \pi^0$) sample. z-axis is probability.

Appendix A PDF

Figure 1.1–1.4 show the PDF in each sample and category.

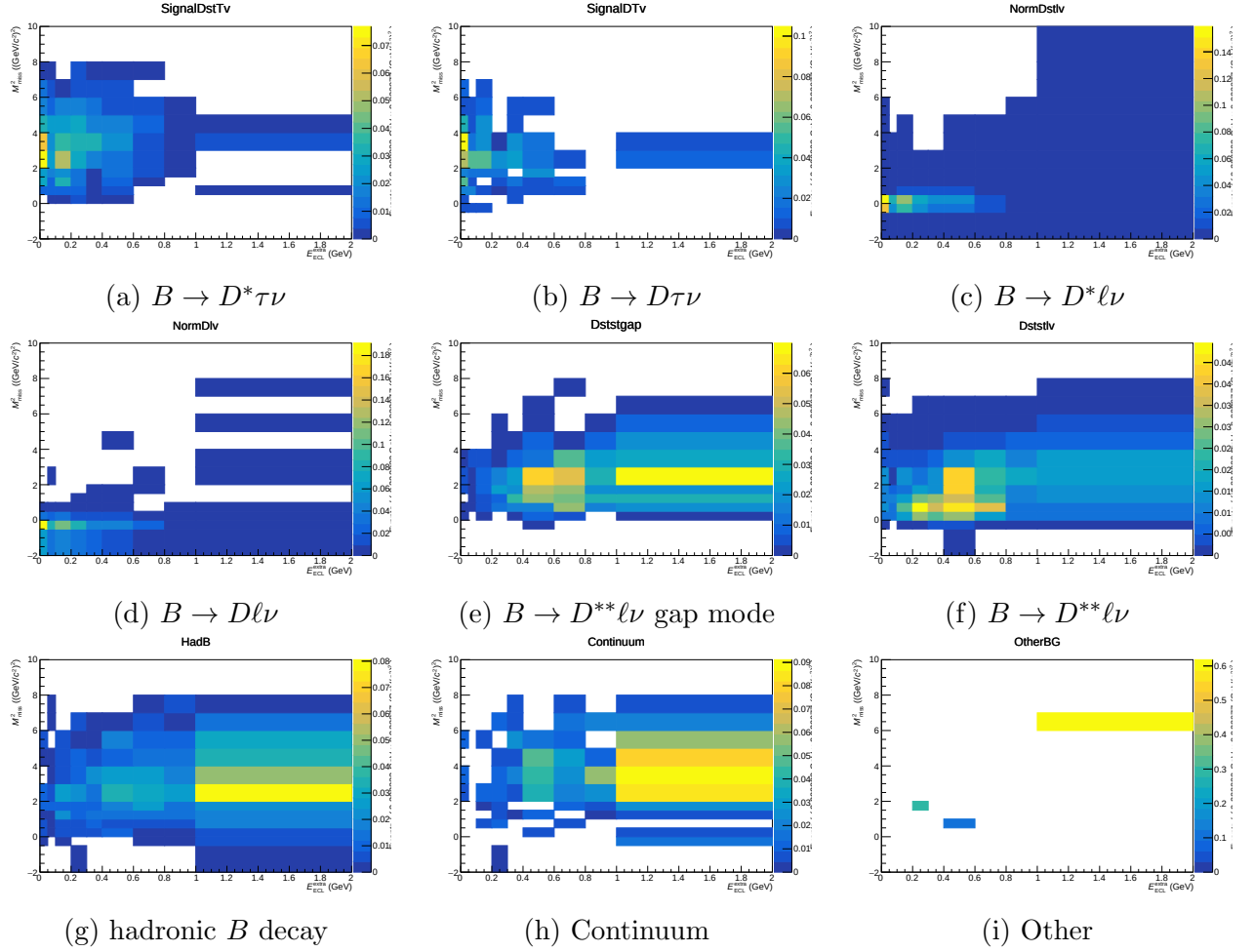


Figure 1.2: PDF of each category in $B^+ \rightarrow D^{*0} \ell^+$ ($D^{*0} \rightarrow D^0 \gamma$) sample. z-axis is probability.

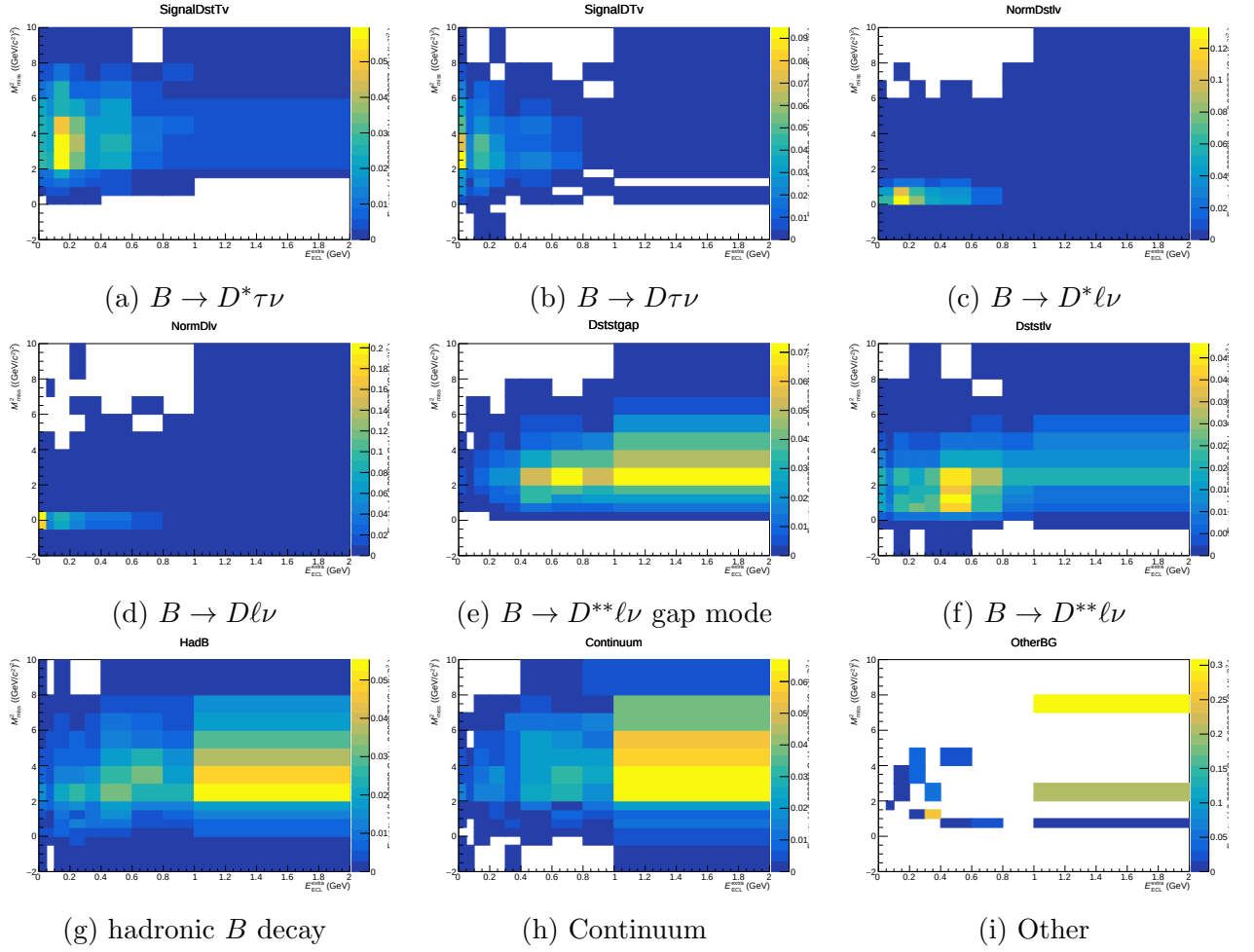


Figure 1.3: PDF of each category in $B^+ \rightarrow D^0 \ell^+$ sample. z-axis is probability.

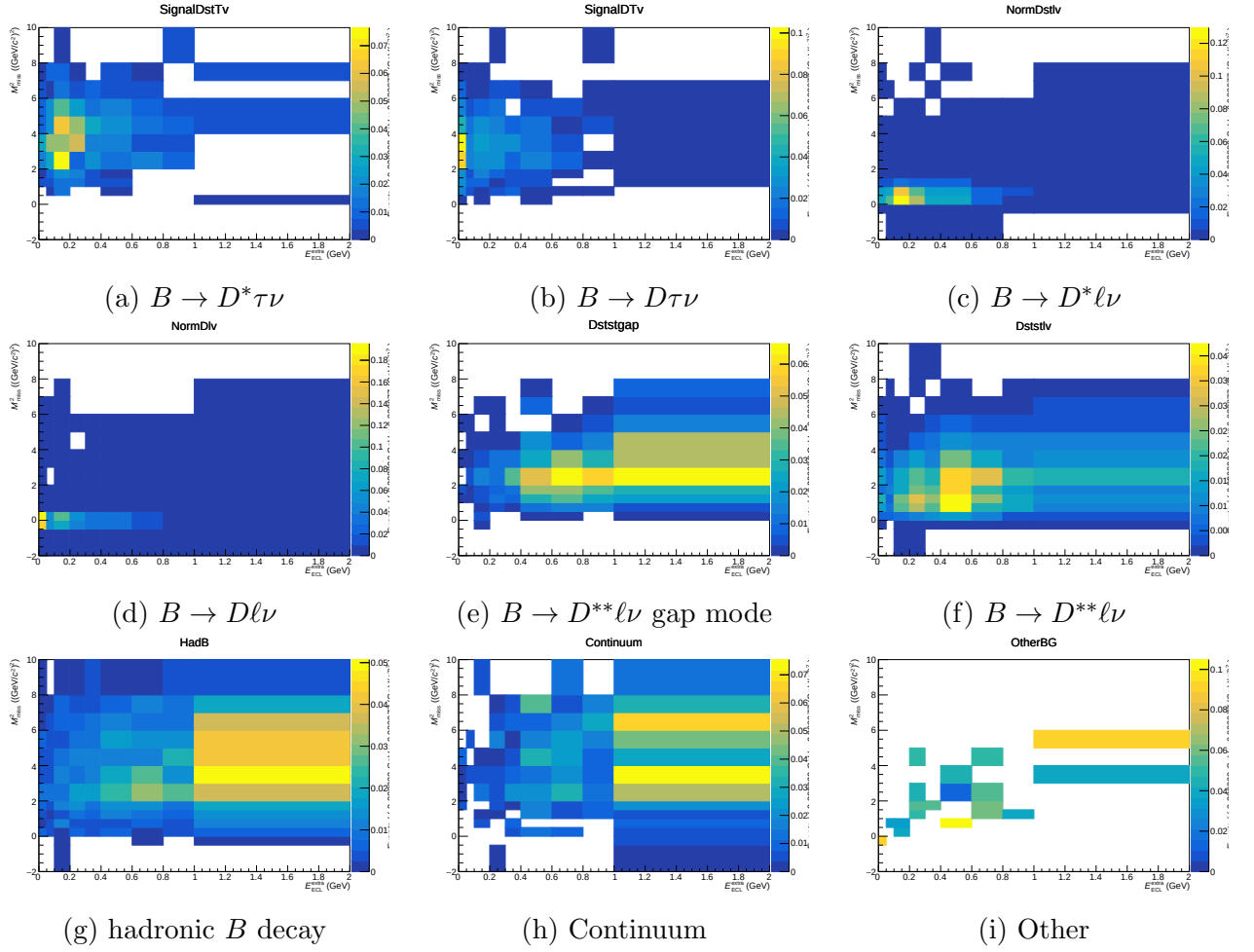


Figure 1.4: PDF of each category in $B^0 \rightarrow D^- \ell^+$ sample. z-axis is probability.

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